



COMSATS University Islamabad, Lahore Campus

Software Project Management Lab Manual



Lab 1

Introduction: Introduction to Software Project Management and its Tools

Problem Statement: In this 1st lab we will learn different type of **software projects** and their **constraints**. What are different **tools** being used professionally for software project management. In the end we will **install** MS Project 2013.

Software Project Management

Software project management refers to the branch of project management dedicated to the **planning, scheduling, resource allocation, execution, tracking** and **delivery** of software projects.

Software Projects

- Mobile Patient Tracker App.
- Online Learning Management Systems / e-learning system.
- Fingerprint voting system.
- Home automation system.
- Game development for mobile users.
- Ecommerce store.



Project Constraints

Project are executed in constraints. **PMBOK's 5th ed** recognizes many constraints:

- Scope: What is the project trying to accomplish?
- Time: How long should it take to complete?
- Cost: What should it cost?
- Quality: What are the quality specifications to be delivered?
- Resource: What man, material, equipment is required?
- Risk: What are the potential risks?

Project Management Tools

Tools	Features
Asana	Combines elements of project management, file storage, and collaboration and helps to manage projects across a team without email.
JIRA	It is a cross-platform issue and bug tracking software with advanced project management capabilities and features.
Trello	Is known for visualizing project tasks on a cardboard-like dashboard that's great for managing short and quick everyday assignments.
Zoho Projects	is one of the best-known tools for simple project management , that can help teams streamline their upcoming work and tasks.
MS Project	Helps to track the information about Project goals, cost, deadlines, and resources.

In this course we will work using MS Project. We might put light on other tools if we get time.



Lab Task:

Install MS Project 2013 into your laptops.

Link : <https://getintopc.com/software/development/microsoft-project-2013-free-download-6366710/>

Link to the video to follow the steps: <https://www.youtube.com/watch?v=02ncvGsp9yw>

Lab 2

Introduction: Getting Started with MS Project

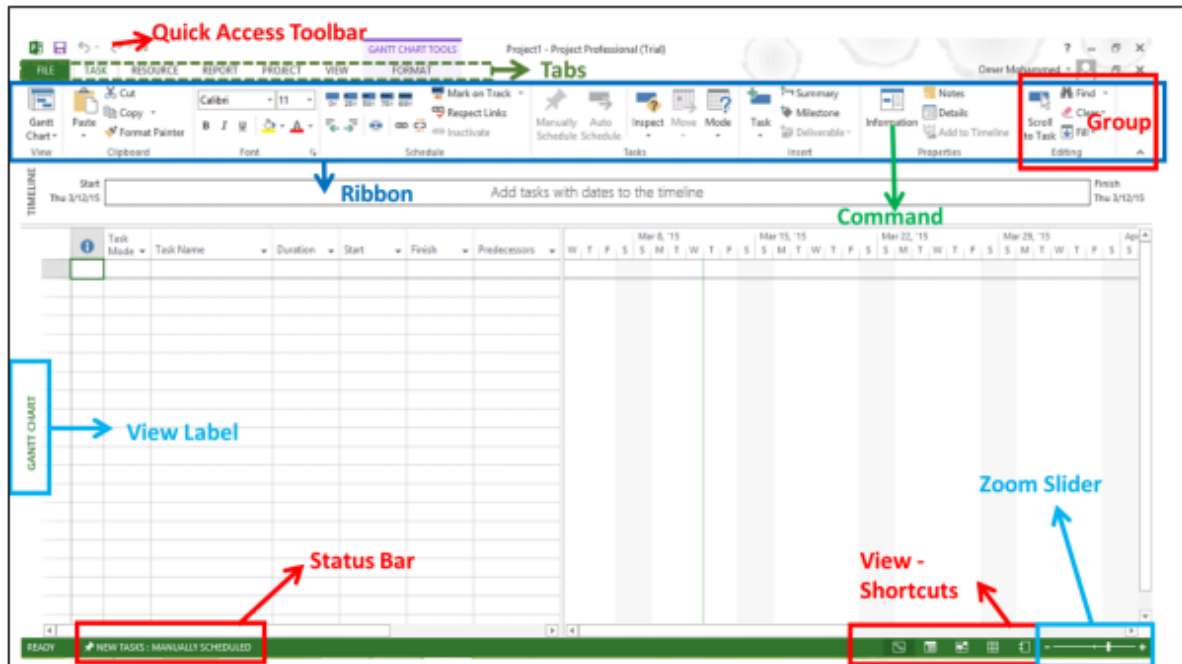
Problem statement:

- Learning how to get started with MS Project 2013
- Understanding User Interface of MS Project
- Learning and implementing different views in MS Project.

Windows 7 – Click on Start menu > All Programs > click Microsoft Office > and then click Project 2013.

Windows 8 – On the Start screen > tap or click Project 2013.

Windows 10 – Click on Start menu > All apps > Microsoft Office > Project 2013.



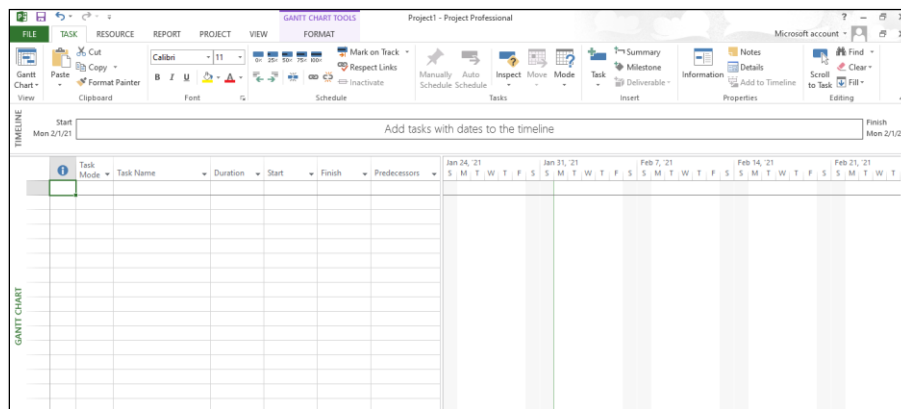
Major part of this interface are:

- **Quick Access Toolbar:** A customizable area where you can add the frequently used commands.
- **Tabs on the Ribbon, Groups:** With the release of Microsoft Office 2007 came the "Fluent User Interface" or "Fluent UI", which replaced menus and customizable toolbars with a single "Office menu", a miniature toolbar known as "quick-access toolbar" and what came to be known as the ribbon having multiple tabs, each holding a toolbar bearing buttons and occasionally other controls. Toolbar controls have heterogeneous sizes and are classified in visually distinguishable Groups. Groups are collections of related commands. Each tab is divided into multiple groups.
- **Commands:** The specific features you use to perform actions in Project. Each tab contains several commands. If you point at a command, you will see a description in a tooltip.
- **View Label:** This appears along the left edge of the active view. Active view is the one you can see in the main window at a given point in time. Project includes lots of views like Gantt Chart view, Network Diagram view, Task Usage view, etc. The View label just tells you about the view you are using currently. Project can display a single view or multiple views in separate panes.
- **View Shortcuts:** This lets you switch between frequently used views in Project.
- **Zoom Slider:** Simply zooms the active view in or out.
- **Status bar:** Displays details like the scheduling mode of new tasks (manual or automatic) and details of filter applied to the active view.

Learning About Different Views in MS Project

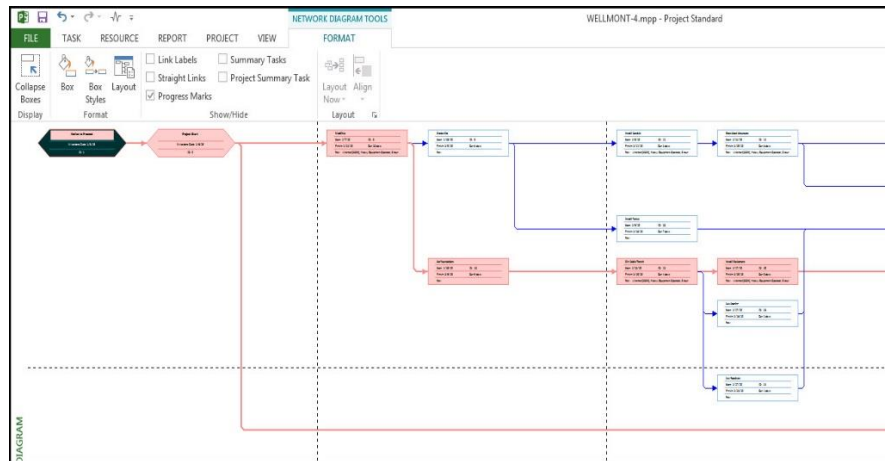
Project uses three types of views: task views, resource views, and assignment views. Let's explore them.

- **Gantt Chart View:** It is the default view of MS Project. It lists the tasks in your project, and illustrates their relationship to one another and the schedule using Gantt bars.

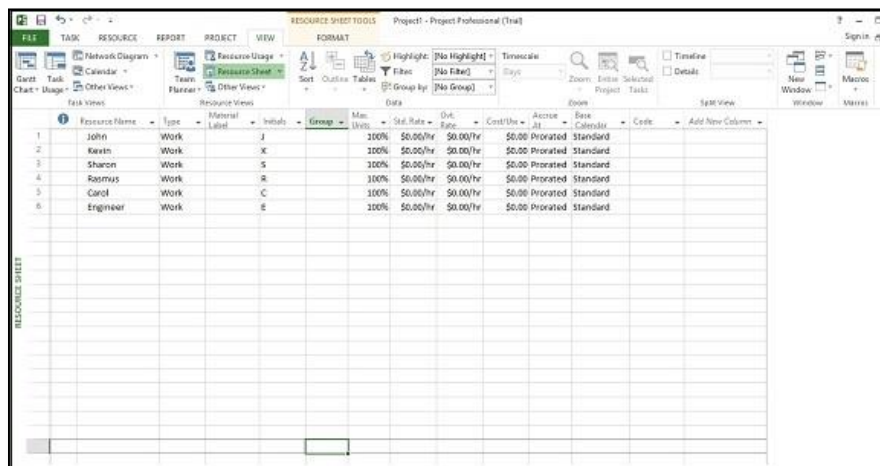


- **Network Diagram View:** A Network Diagram is a graphical way to view tasks, dependencies, and the critical path of your project.

The Network Diagram view was called the PERT Chart in earlier versions of Project. This view shows the dependencies between tasks in a graphical manner. Gantt chart is primarily meant to view the schedule time line, whereas Network diagram to view the all type of dependencies in the project.



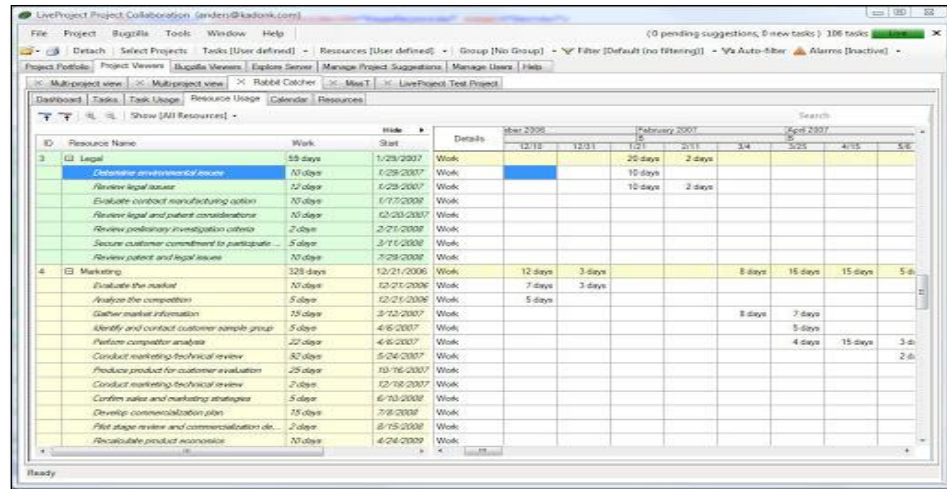
- **Resource Sheet View:** is a view within Project that allows the user to view, create, and edit resources and resource information.



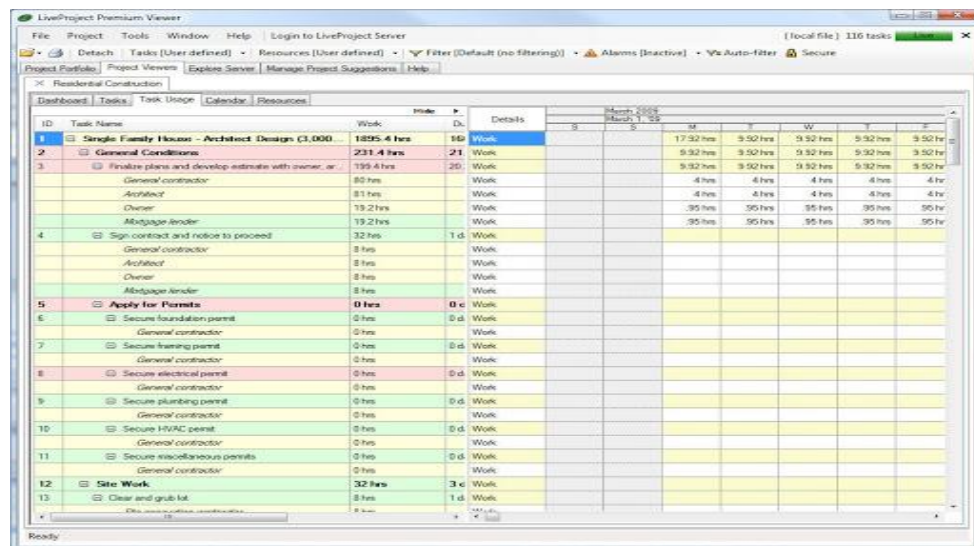
The screenshot shows the 'Resource Sheet' view in Microsoft Project. The interface includes a ribbon with tabs for FILE, TASK, RESOURCE, REPORT, PROJECT, VIEW, and FORMAT. The 'RESOURCE' tab is active, showing options for Resource Usage, Resource Sheet, and Resource Views. The main area displays a table of resources and their assigned tasks. The table has columns for Resource Name, Type, Material Label, Initials, Group, Max Units, Std. Rate, Rate, Cost, Accrue, Base, and Cedit. The resources listed are John, Kevin, Sharon, Rasmus, Carol, and Engineer, all of type 'Work'.

Resource Name	Type	Material Label	Initials	Group	Max Units	Std. Rate	Rate	Cost	Accrue	Base	Cedit
1 John	Work	J			100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
2 Kevin	Work	K			100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
3 Sharon	Work	S			100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
4 Rasmus	Work	R			100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
5 Carol	Work	C			100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
6 Engineer	Work	E			100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	

- **Resource usage view:** Resource Usage view is a functionality that lists for each resource the assigned tasks and the total amount of work that the resource is scheduled to perform on each task. Basically it groups the tasks against each resource.



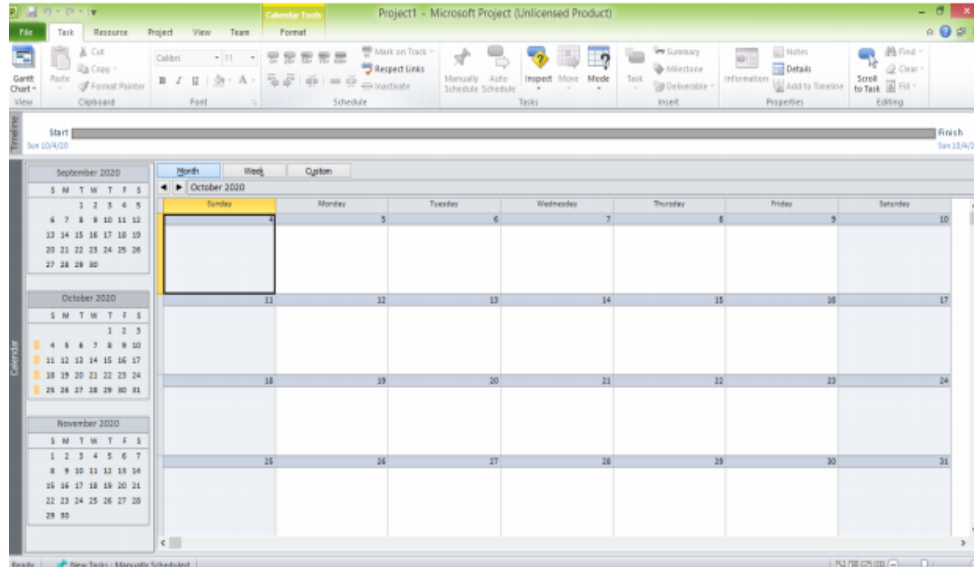
- **Task Usage View:** shows details about each task that which task is assigned to whom and working schedule of each resource.



- **Calendar view:** It assists you in showing which tasks are scheduled for which days in a calendar layout.



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Lab 3 & 4

Project Plan

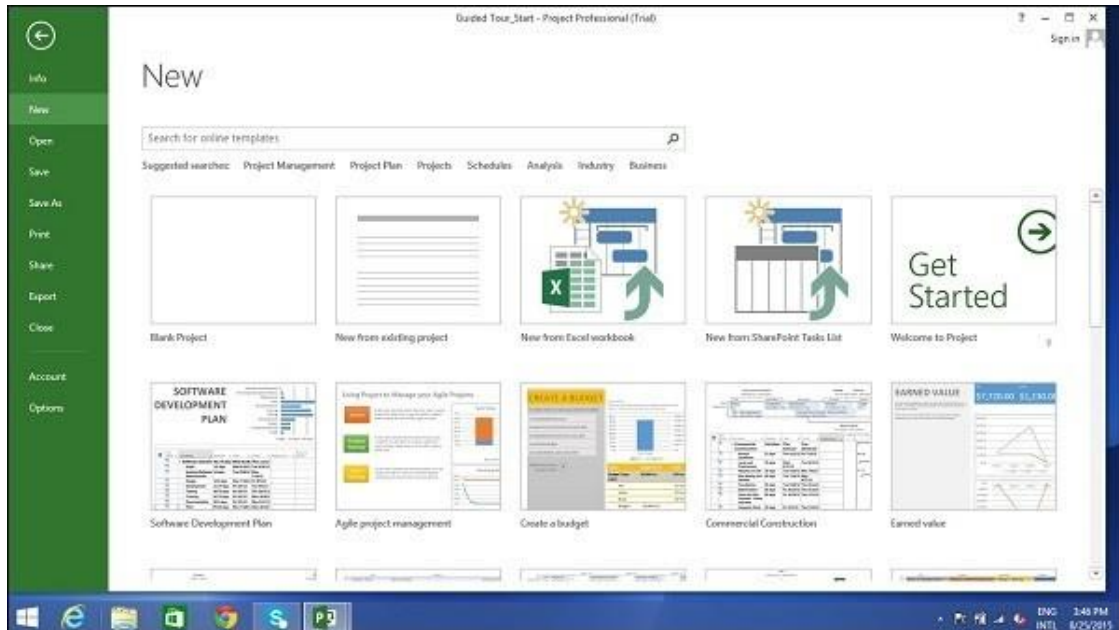
Introduction: Create A New Project Plan

Problem Statement:

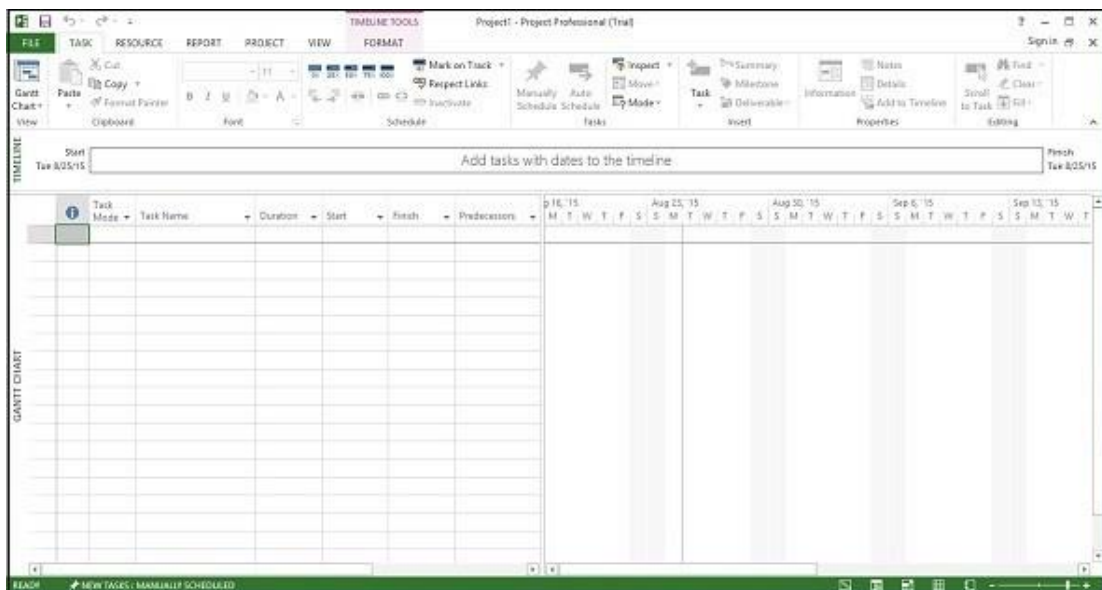
- Learn how to create a new project plan
- Learn how to set calendar
- Add exception to the calendar
- Set resource calendar
- Add working and non-working days.
- Learn to build task list
- Learn what is a duration and different ways to enter duration in MS Project
- Learn how to change default time dimensions
- Learn what is elapsed duration

Create Blank Project

MS Project 2013 will display a list of options. In the list of available templates, click Blank Project.



Project sets the plan's start date to current date, a thin green vertical line in the chart portion of the Gantt Chart



View indicates this current date.

Project Information

Let us change the project start date and add some more information.

Step 1: Start Date

Click Project tab → Project Information.

A dialog box appears. In the start date box, type the date, or click the down arrow to display the calendar, select any date of your choice.

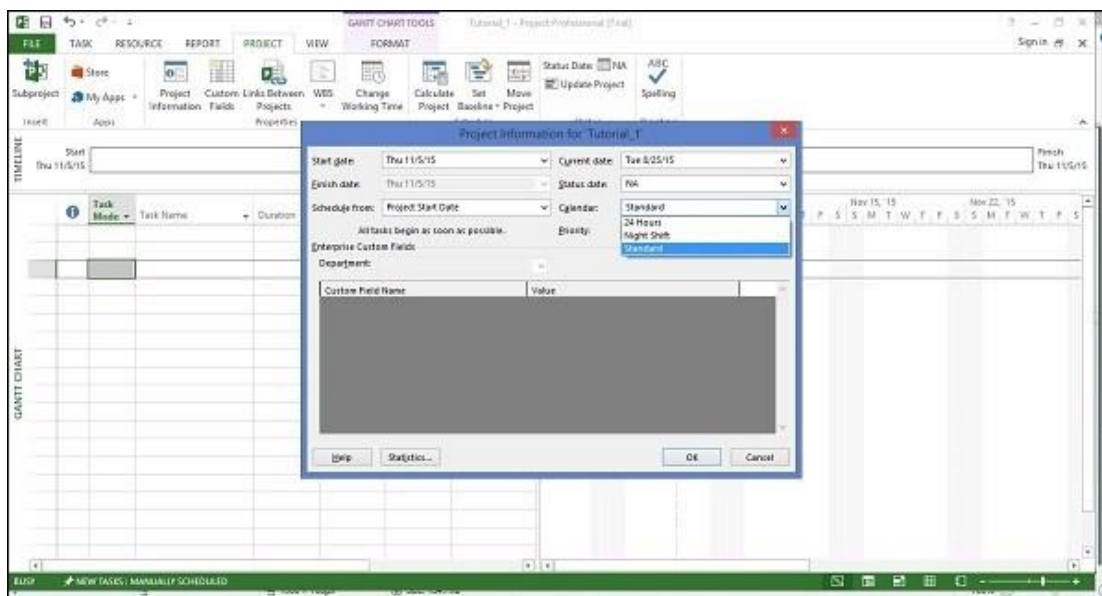
Click OK to accept the start date.

Step 2: Set Up Calendar

Click Project tab → Project Information.

Click the arrow on the Current Date dropdown box. A list appears containing three base calendars.

- 24 Hour – A calendar with no non-working time.
- NightShift – Covers 11 PM to 8 AM, night shifts covering all nights from Monday to Friday, with one hour breaks.
- Standard – Regular working hours, Monday to Friday between 8 AM to 5 PM, with one hour breaks.



Select “Standard Calendar” as your project Calendar.

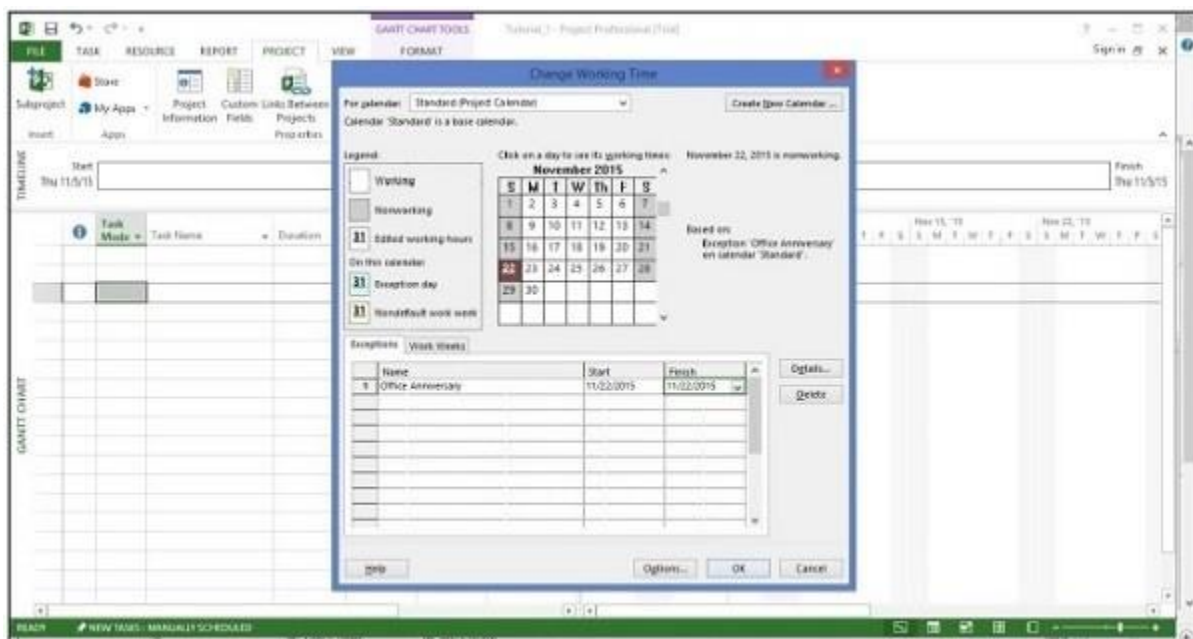
Step 3: Adding Exceptions to Calendar

Exceptions are used to modify a Project calendar to have a non-standard workday or a non- working day. You can also allot unique working hours for a particular resource as well.

Here is an example to create a non-working day, which could be because of a holiday or office celebrations or events other than the standard office work effort.

Click Project tab → Change Working Time.

Change Working Time dialog box appears. Under Exceptions Tab click on the Name Field, enter event as “Office Anniversary”. In the Start field enter the date, and then enter the same date in the Finish field. This date is now scheduled as a non-working day for the project. You can also verify the changed color indicated in the calendar within the dialog box as below. Click Ok to close.





Step 4: Setting up Resource Calendar

Just like you can change a Standard Base Calendar, you can change the work and non-working time for each resource. You can modify the resource calendar to accommodate vacation time, training time, etc.

Also remember, Resource Calendar can only be applied to work resources and not to material and cost resources.

By default, when we create the resources in a plan, the resource calendar matches the Standard base calendar. And any changes you make to the Project Calendar, gets reflected automatically in resource calendars, except when you create an exception in the resource calendar. In that case even if you update the project calendar, the exception in resource calendar is not affected.

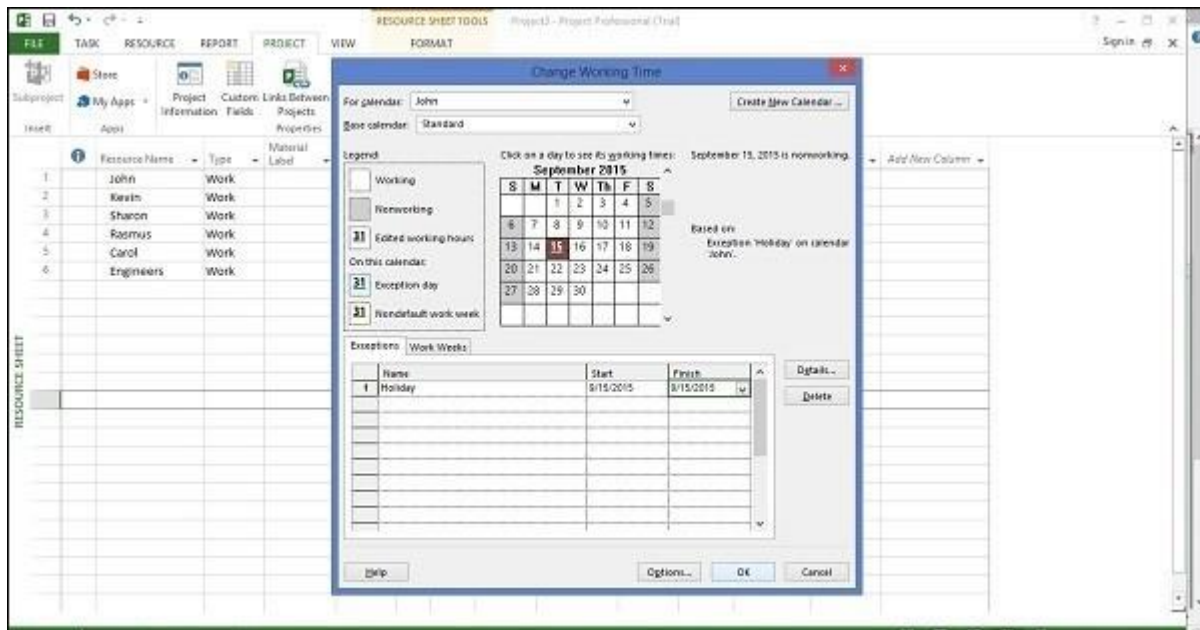
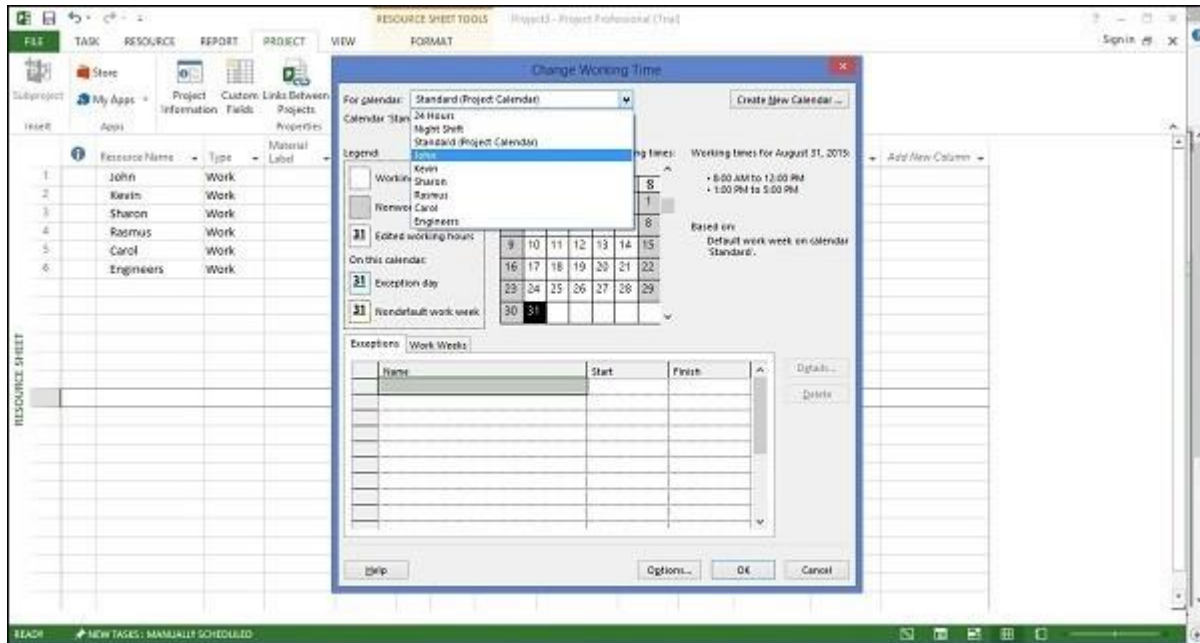
Click Project tab → **Click Change Working Time Change Working**

Time dialog box appears.

Click the down arrow **for** the “**For Calendar**” drop-down box.

Select the resource **for** whom you want to create an exception. **In** example below I have chosen John

Under Exceptions Tab click on the Name Field, enter event as “Personal holiday”. In the Start field enter the date and then enter the same date in the Finish field.





Step 5: Change Working times for Each Resource

Click Project tab → **Click Change Working Time**.

The **Change Working Time** dialog box appears.

Click the down arrow **for** the **“For Calendar”** dropdown box.

Select the resource **for** whom you want to change work schedule.

In the following screen you can see we have chosen **John**.

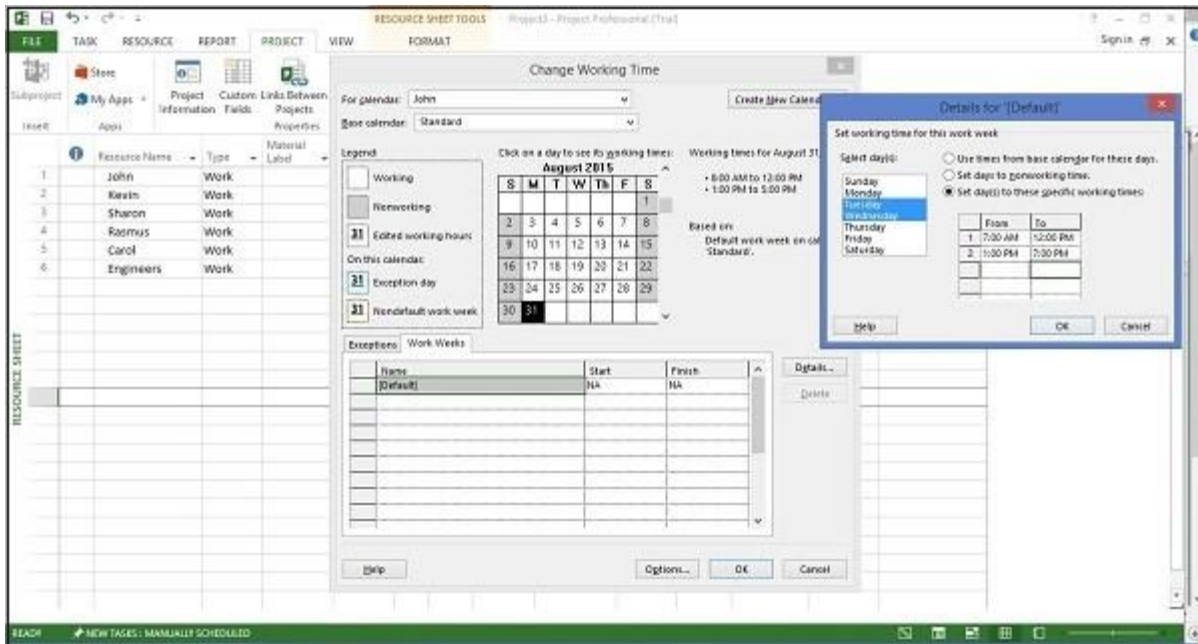
Click **“Work Weeks”** tab.

Double-click the **[default]** cell below the **Name** column heading.

Under **“Selected Day(s)”** choose any day you want to change the work schedule.

We have chosen **Tuesday and Wednesday**.

Click Set day(s) to these specific working times. **Change** the time.



Step 6: Create Non-working Days

Click Project tab → Click Change Working Time.

The Change Working Time dialog box appears.

Click the down arrow for the “For Calendar” dropdown box.

Select the resource for whom you want to change work schedule. We have chosen John again.

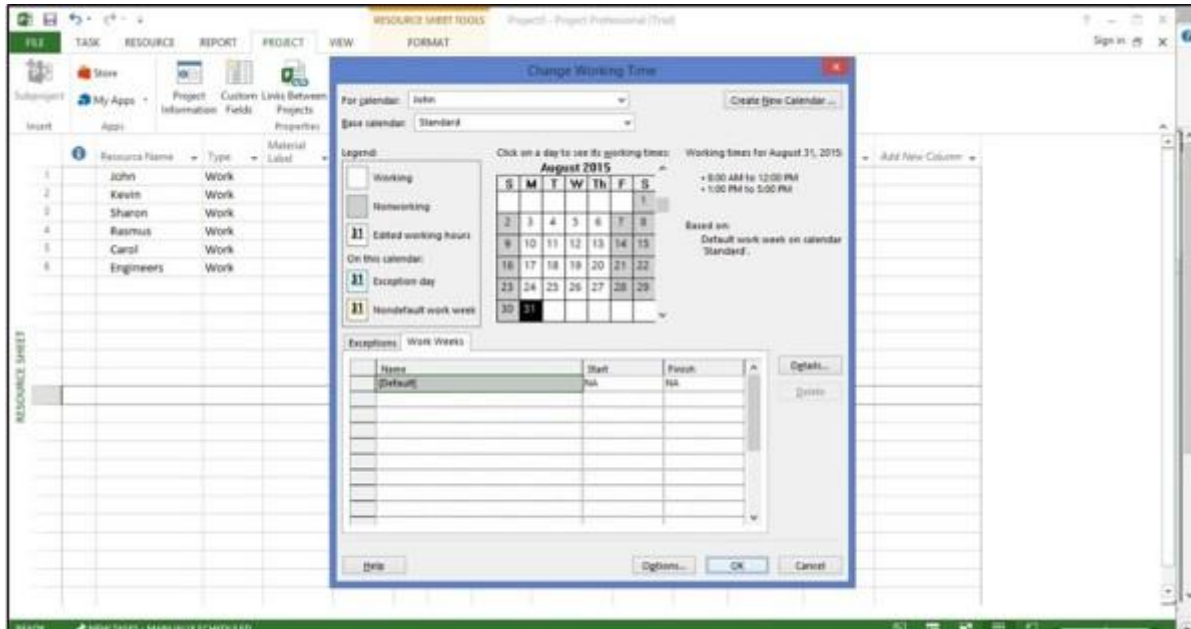
Click “Work Weeks” tab.

Double-click the [default] cell below the Name column heading.

Under “Selected Day(s)” choose any day you want to change the work schedule.

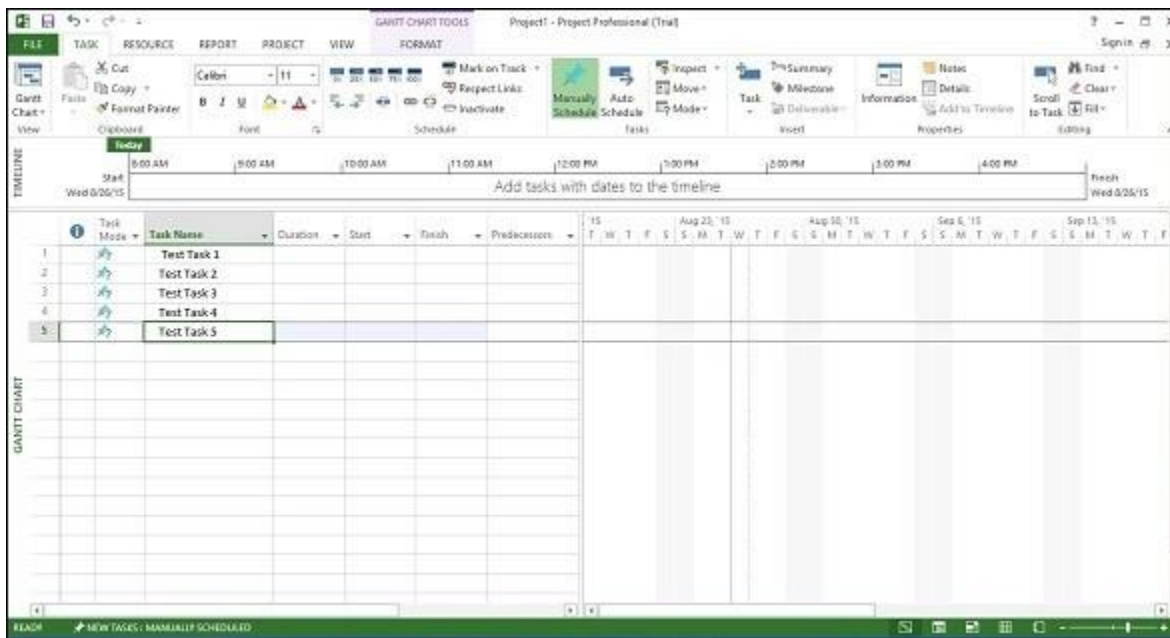
Click any day (we have chosen Friday) and use the radio button “Set days to nonworking time”.

Click OK to close the Dialog box. You will now see all Fridays are greyed out in the calendar.



Enter Task

This is simple. In Gantt Chart View, just click a cell directly below the Task Name column. Enter the task name. In the following screen, we have entered 5 different tasks.





Enter Duration

A duration of the task is the estimated amount of time it will take to complete a task. As a project manager you can estimate a task duration using expert judgment, historical information, analogous estimates or parametric estimates.

You can enter task duration in terms of different dimensional units of time, namely minutes, hours, days, weeks, and months. You can use abbreviations for simplicity and ease as shown in the following table.

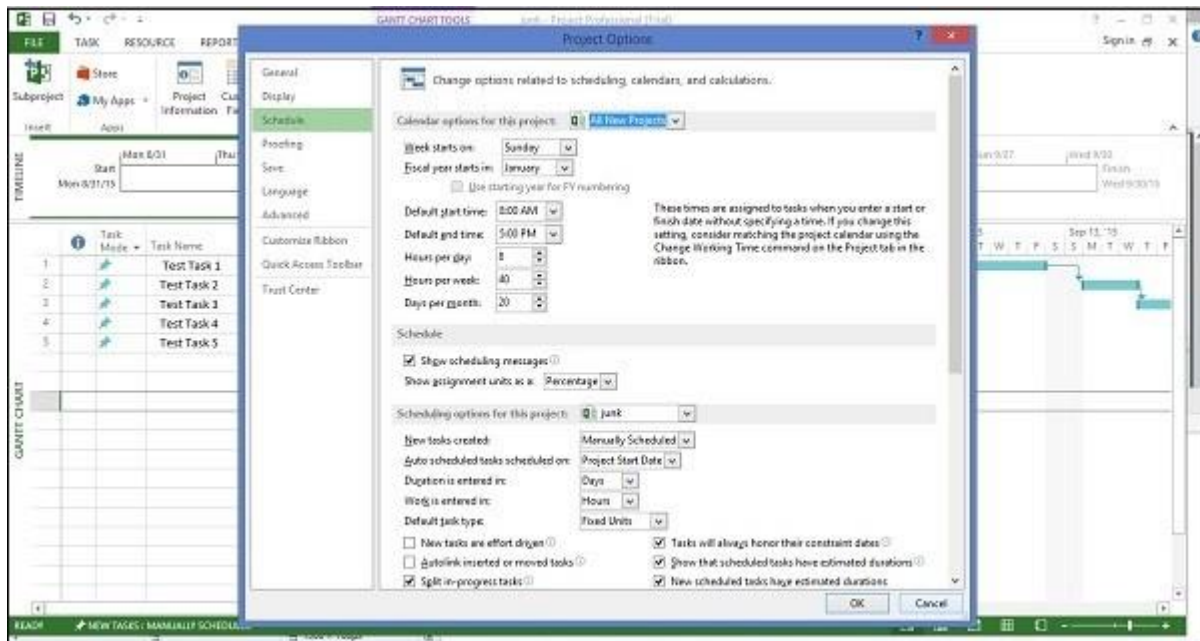
Value you want to enter	Abbreviation	Appearance
45 minutes	45 m	45 mins
2 hours	2h	2 hrs
3 days	3d	3 days
6 weeks	6w	6 weeks
2 months	2mo	2 mons

Remember, Project default values depend on your work hours. So 1 day is not equivalent to 24 hours but has 8 hours of work for the day. Of course, you can change these defaults anytime you want.

Value entered	Value	Project default Value
1 minute	60 seconds	60 seconds
1 hour	60 minutes	60 minutes
1 day	24 hours	8 hours (1 workday)
1 week	7 days	40 hours (5 workdays)
1 month	28 to 31 days	160 hours (20 workdays)

Change Default Time Dimensions

Click Project tab → Properties Group → Click Change Working Time → Click Options. You can apply this to all projects or a specific project that you are working on currently.

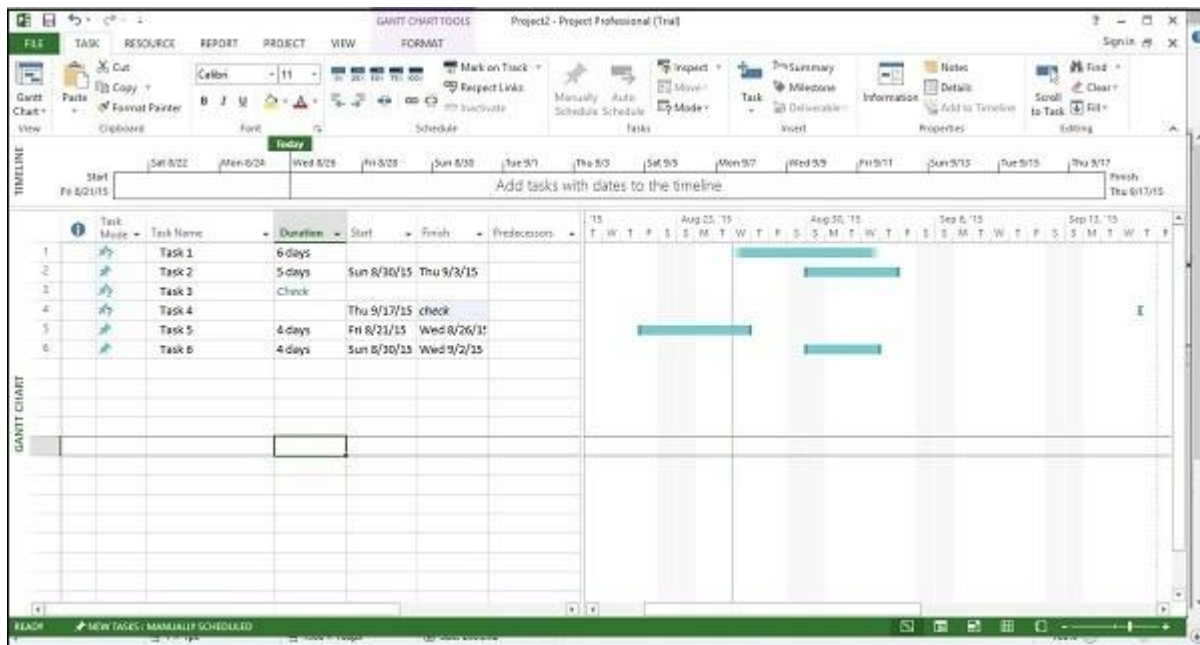


One of the neat tricks MS Project possesses is, it considers duration of the task in workday sense. So if you have a non-working day in between, it accommodates this and ensures a task that takes 16 hours to complete to end on the 3rd day. In other words, if you have a task that needs 16 hours to complete starting on Monday 8:00 AM (if this is the time your work day starts, and 8 hours being total work hours in a day), and Tuesday being a holiday, the task will logically end on the evening of Wednesday.

Tip – With manually scheduled tasks, if you are not sure about a task duration, you can just enter text such as “Check with Manager/Engineer” to come back to this later.

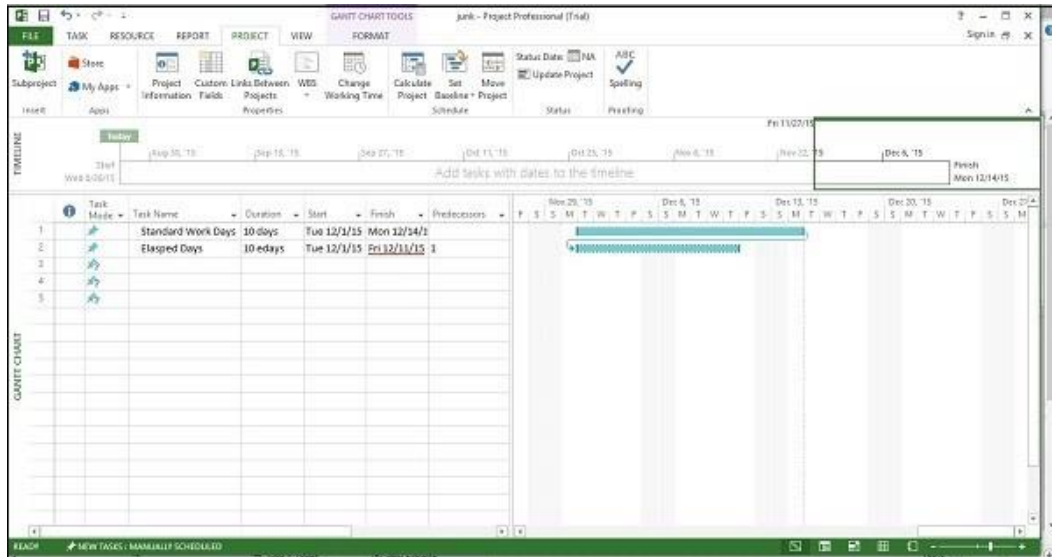
Enter Task Duration

- This is simple in Gantt Chart View, click the cell below Duration column heading. Enter the duration. (Task 1 in the following screenshot)
- You can also enter Start and Finish date and MS Project will calculate the duration on its own. (Task 2 in the following screenshot)
- You can enter text as well when you don't have a duration metric currently. (Task 3 and Task 4 in the following screenshot)



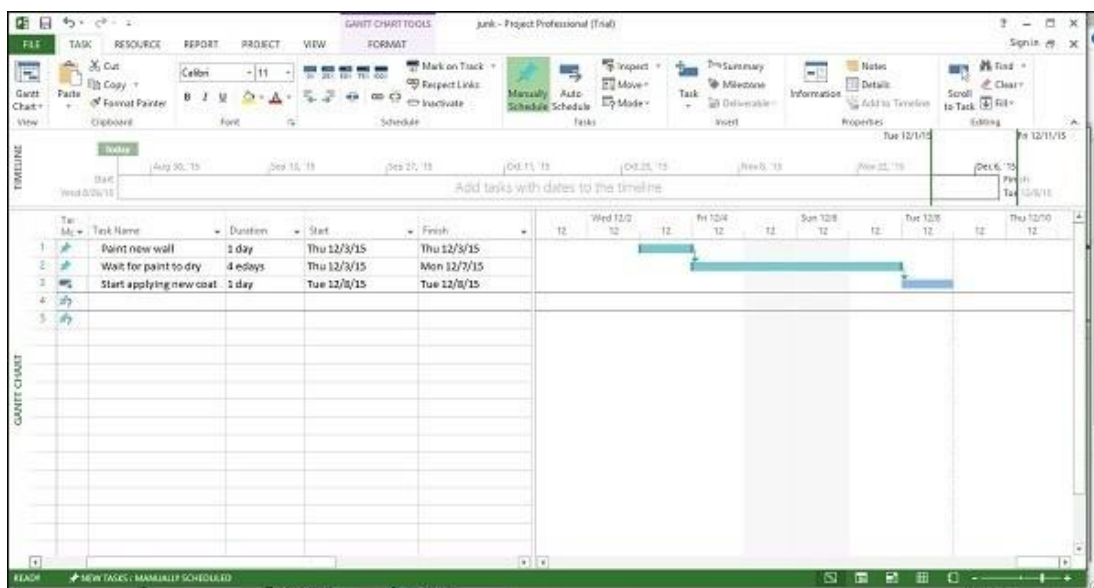
Elapsed Duration

Elapsed Duration is the time that elapses while some event is occurring which does not require any resources. Elapsed duration for a task can be used in instances where a task will go on round-the-clock without any stoppage. A normal workday has 8 hours, and an elapsed day duration will have 24 hours. The task also continues over non-working (holidays and vacations) and working days.



You can enter elapsed duration by preceding any duration abbreviation with an “e”. So 1ew is seven 24-hour days.

For example, when you are ‘Waiting for the paint to dry’. And it takes 4 days for this to happen. It does not need a resource or a work effort, and all you are doing is waiting for it to dry. You can use 4ed as the time duration, which signifies 4 elapsed days, the paint can dry regardless of whether it is a weekend or if it falls on a holiday. Here in this example, the drying occurs over 24 hours over the weekend.





Lab Task

Implement above explained steps on the Gantt chart view created in the previous lab, set up your calendar add exception, set up resource calendar and add working and non-working days.

Lab 5

Build Tasks

Introduction: Learn to build task lists and links between tasks.

Problem Statement:

- Learn to create milestones
- Learn different types of task dependencies and how to implement them
- Learn how to create respect links for your tasks.



Create Milestones

In Project Management, Milestones are specific points in a project timeline. They are used as major progress points to manage project success and stakeholder expectations. They are primarily used for review, inputs and budgets.

Mathematically, a milestone is a task of zero duration. And they can be put where there is a logical conclusion of a phase of work, or at deadlines imposed by the project plan.

- Simply put, it's a reference point that marks a major event
- Milestones do not impact project duration

There are two ways you can insert a milestone.

Method 1: Inserting a Milestone

Click name of the Task which you want to insert a Milestone

Click Task tab → Insert group → Click Milestone.

MS Project names the new task as <New Milestone> with zero-day duration.

Click on <New Milestone> to change its name.

You can see the milestone appear with a rhombus symbol in the Gantt Chart View on the right.

Method 2: Converting a Task to a Milestone

Click on any particular task or type in a new task under the **Task Name** Heading.

Under **Duration** heading type in “0 days “.

MS Project converts it to a Miles

Link Tasks

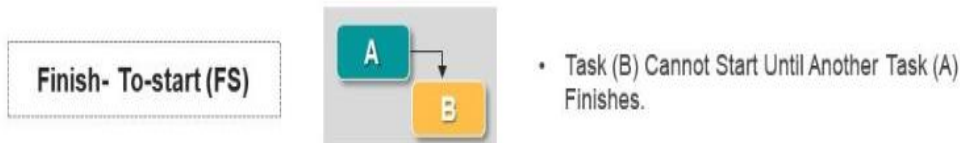
Once you have a list of tasks ready to accomplish your project objectives, you need to link them with their task relationships called dependencies. For example, Task 2 can start once Task 1 has

finished. These dependencies are called Links.

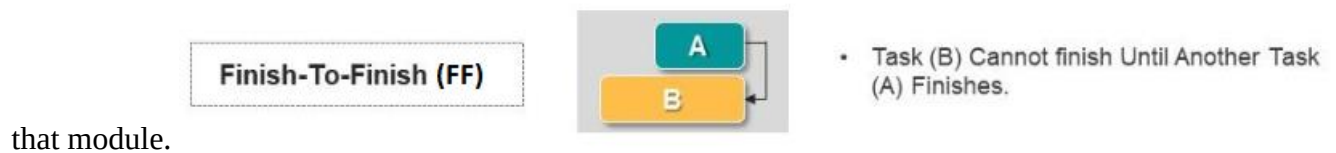
- In MS Project, the first task is called a predecessor because it precedes tasks that depend on it.
- The following task is called the successor because it succeeds, or follows tasks on which it is dependent.

There are only four types of task dependencies, here we present them with examples.

- Finish to Start (FS) – Finish the first floor before starting to build the second floor. *Most used.*



- Finish to Finish (FF) – Testing for S/W module will finish only once you have finished writing code for



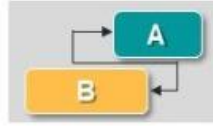
that module.

- Start To Start (SS) – When doing a survey, we would seek survey responses but will also start tabulating the responses. One does not have to finish collecting survey response before starting the tabulation.



- Start to Finish (SF) – Exam preparation will end when exam begins. *Least used.*
 - Duty of morning guard cannot finish until duty of evening guard starts.

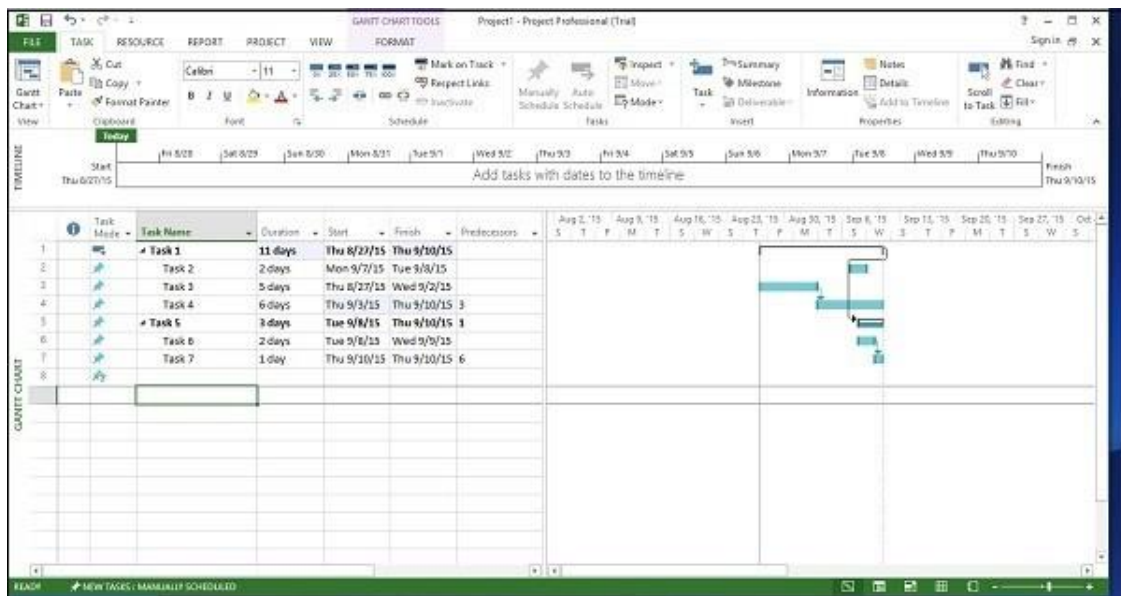
Start-To-Finish (Sf)



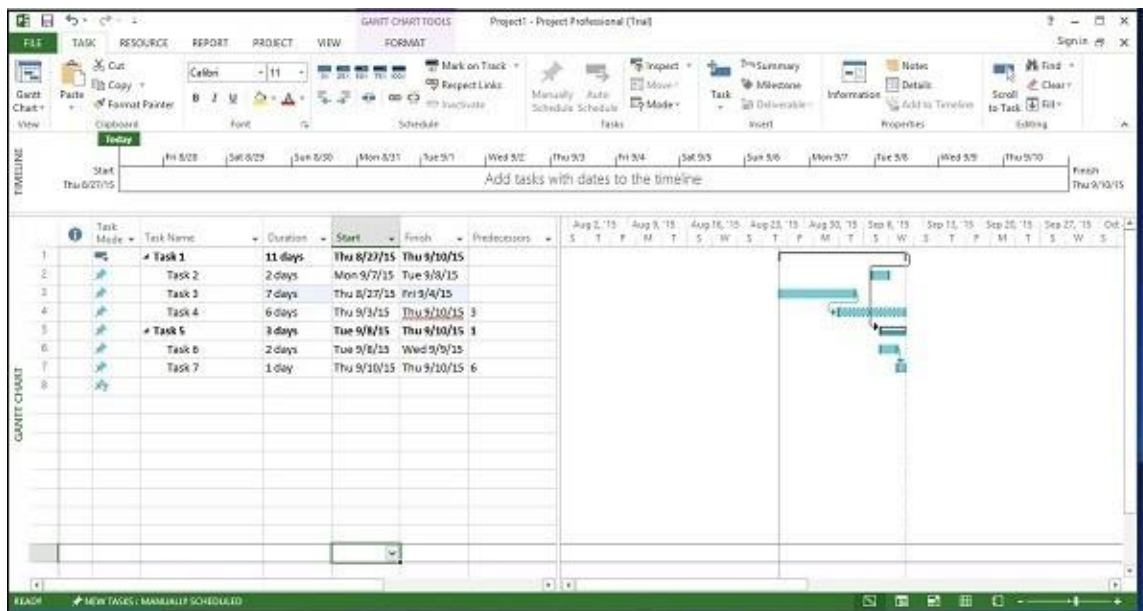
- Task (B) Cannot finish Until Another Task (A) Starts.

Respect Links

If you are in Manually Scheduled mode, any change in duration of the predecessor task will not reflect on Start date of Task 4. For example, Task 4 starts on 9/3/20 which is the next day of Finish date of Task 3.

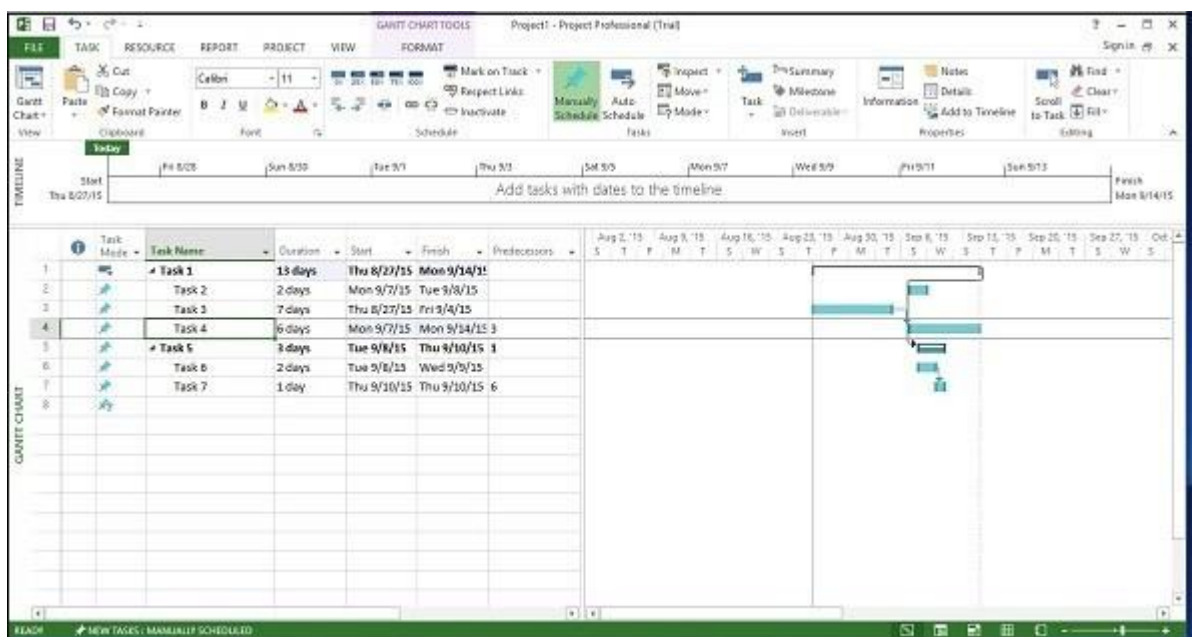


Now when we change the Duration of Task 3 from 5 to 7 days, the start date is not automatically updated for Task 4 in Manual Scheduling.



You can force MS Project to respect the link (dependency) by doing the following –

- Select task 4
- Click Task tab → Schedule group → Respect Links.





Lab Tasks

Create milestones for your Project.

Try out different dependency links, implement wherever you think is appropriate.

Create Respect links.

Lab 6 & 7

Setup Resources

Introduction: set up resources in MS Project.

Problem Statement:

- Learn what are different types of resources and how to enter their names
- Learn what is max capacity
- Learn resources as group and as part time
- Learn how to add notes to resources
- Learn how to set up cost



Resource Types

- Work resources – People and equipment to complete the tasks.
- Cost resources – Financial cost associated with a task. Travel expenses, food expenses, etc.
- Material resources – Consumables used as project proceeds. For example, paint being used while painting a wall, office stationery

Note – Be aware of the crucial difference between People and Equipment resources. People resources will have limited work hours, say 6, 8 or 12 hours. Equipment resources have different working capacities of 2, 8 or 24 hours and could have maintenance breaks as well. Also note, that it is possible multiple people resources might be using one equipment resource, or one equipment might be accomplishing multiple tasks.

Task1-Enter Work Resource Names

You can enter resource names according to your convenience.

Resource	Example
Work resource as an identified person	John, Kevin
Work resource as a job function or group	Engineer, Coordinator, Typist
Work resource as an equipment	Computers, cpu, hard-drive

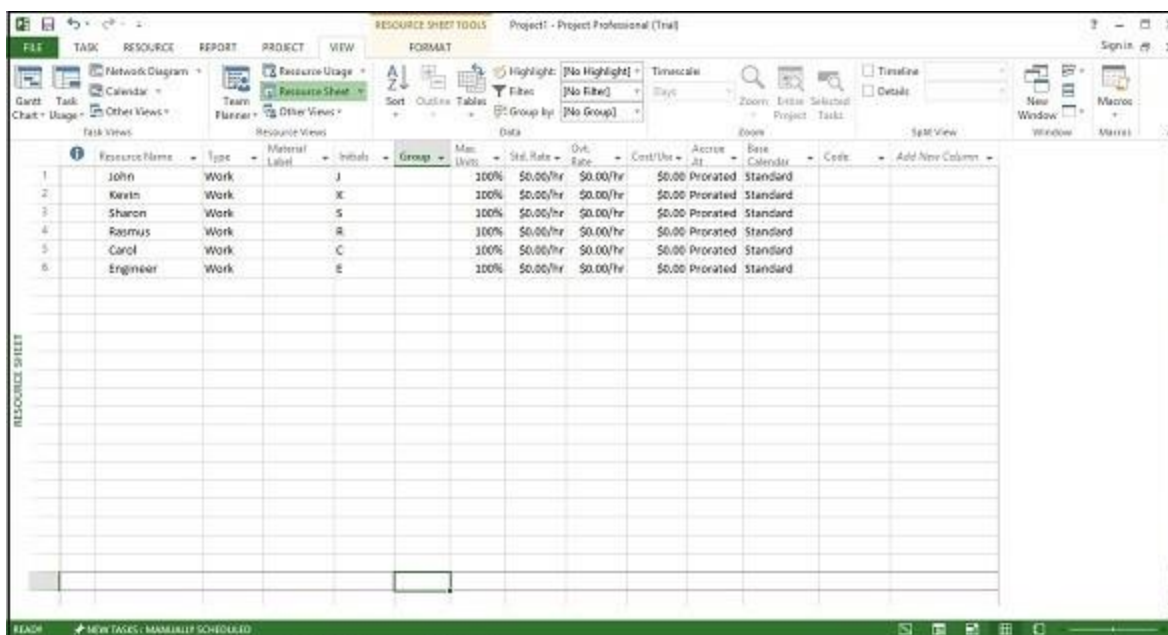
Steps involved

Click View tab → Resource Views group → Click Resource Sheet.

Click the cell directly below the Resource Name heading column.

Enter Resources as an individual person, job function or group.

By default, the Max Units field is set to 100%.



The screenshot shows the 'Resource Sheet' view in Microsoft Project. The 'Resource Name' column lists five resources: John, Kevin, Sharon, Rasmus, and Carol, all of type 'Work'. The 'Max Units' column for all resources is set to 100%. The 'Status' column shows 'Prorated' for all resources. The 'Base Calendar' is set to 'Standard' for all resources. The 'Code' column is empty for all resources. The 'Add New Column' button is visible on the right side of the table.

Resource Name	Type	Material Label	Initials	Group	Max Units	Std. Rate	Ovt. Rate	Cost/Use	Accres. At	Base Calendar	Code	Add New Column
1 John	Work		J		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		
2 Kevin	Work		K		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		
3 Sharon	Work		S		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		
4 Rasmus	Work		R		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		
5 Carol	Work		C		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		
6 Engineer	Work		E		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		



Resource Max Capacity

Max Units field represents the maximum capacity of a resource to work on assigned tasks. 100% stands for 100 percent of resource's working time is available for work on task assigned. The resource is available full-time on each workday. If the resource gets allocated to task or tasks that would require more than his/its work hours, the resource is over allocated and MS Project will indicate this in red formatting.

If a resource does not represent an individual person but a job function, where a group of people with the same skill set can work on the task, we can enter larger Max Units to represent the number of people in the group. So 400% would indicate, 4 individual people working full-time every workday.

Work Resources as a Group

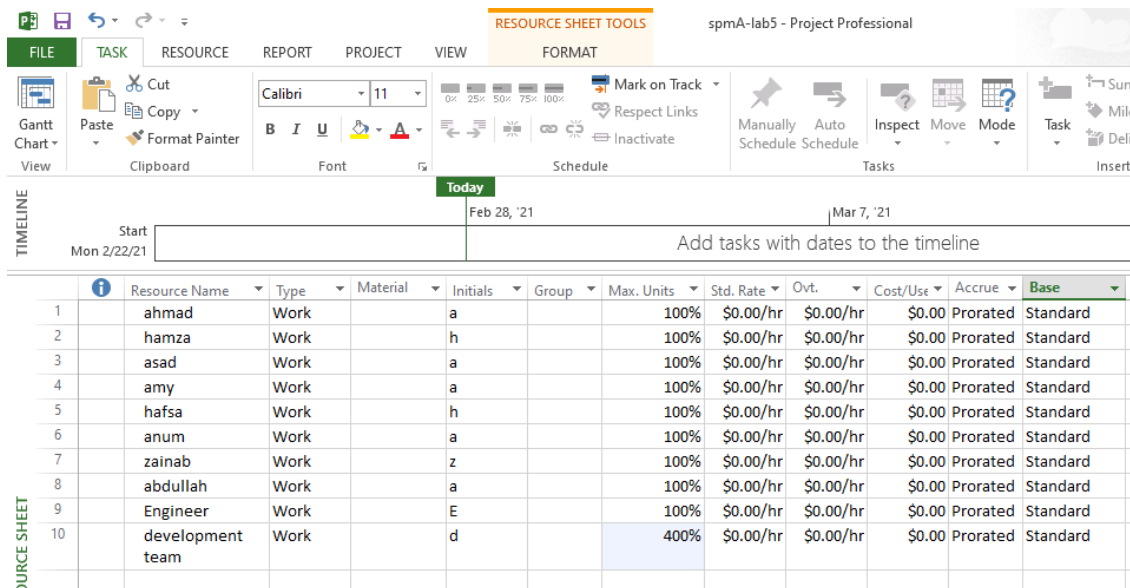
Click View tab → Resource Views group → Click Resource Sheet

Click the cell directly below Resource Name heading column

Enter Resources as group, here we take an example of Engineers.

Click the Max. Units field for the Engineers resource.

Type or select 400%. Press Enter.



	Resource Name	Type	Material	Initials	Group	Max. Units	Std. Rate	Ovt.	Cost/Use	Accrue	Base
1	ahmad	Work		a		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
2	hamza	Work		h		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
3	asad	Work		a		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
4	amy	Work		a		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
5	hafsa	Work		h		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
6	anum	Work		a		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
7	zainab	Work		z		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
8	abdullah	Work		a		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
9	Engineer	Work		E		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
10	development team	Work		d		400%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard

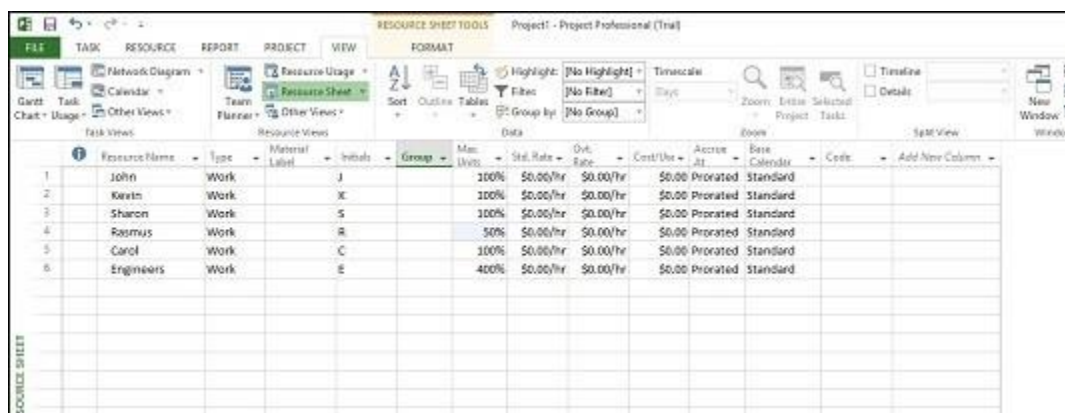
Work Resource as Part-time

Entering a value less than 100% in Max.Units would mean you expect the resource capacity to be lower than a full-time resource. So 50% would mean the individual works for half of the normal full capacity, so if a normal work week is 40 hours, this equals 20-hour capacity.

Click View tab → Resource Views group → Click Resource Sheet.

Click the cell directly below Resource Name heading column.

Enter Resource as an individual or job function. Here let's take an example. For Ramsus I have entered 50%.



	Resource Name	Type	Material	Initials	Group	Max. Units	Std. Rate	Ovt.	Cost/Use	Accrue	Base
1	John	Work		J		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
2	Karen	Work		K		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
3	Sharon	Work		S		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
4	Ramsus	Work		R		50%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
5	Carol	Work		C		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
6	Engineers	Work		E		400%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard



Task2-Enter Resource Cost

You can enter standard rates and costs per use for work and material resources. You can also enter overtime rates for work resources. Standard rates are calculated on per hour basis. Costs per use on the other hand are costs that do not vary with task. Cost per use is a set fee used up to complete a task. There are three types of resources – work, material, and cost.

- Work resources – People and equipment to complete the tasks.
- Cost resources – Financial cost associated with a task. Travel expenses, food expenses, etc.
- Material resources – Consumables used as project proceeds. Like paint being used while painting a wall.

Cost resources do not use pay rates. Remember cost per use and cost resources are two different things. Cost resources are financial cost associated with a task, like travel expenses, food expenses, etc. The cost value of cost resource is only assigned when you assign cost resource to a task.

Project calculates the cost of a task by using this formula –

$$\text{Cost of Task} = \text{Work Value (in number of hours)} \times \text{Resource's Pay Rate.}$$

To enter standard and overtime pay rates for work resources –

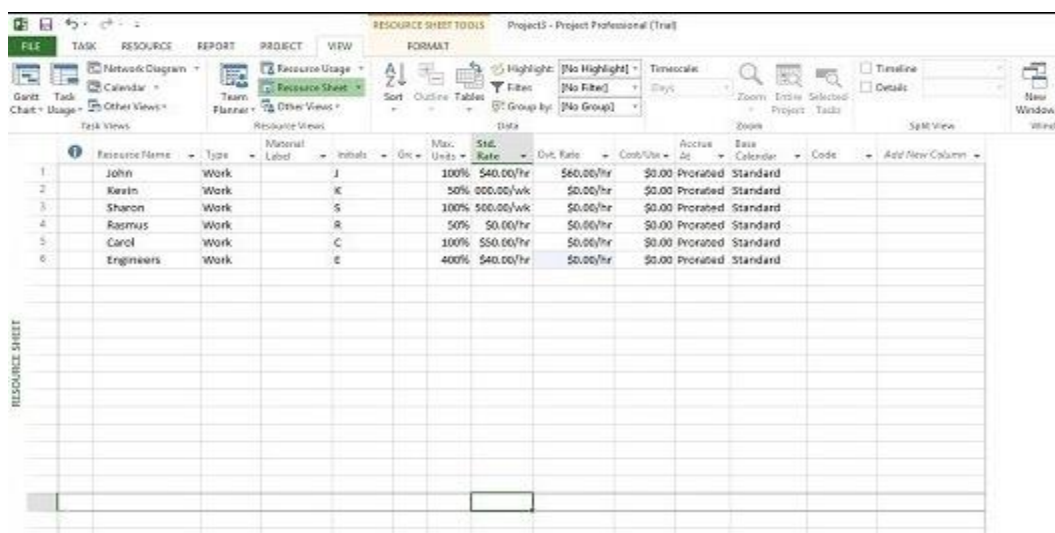
Click View tab → Resource Views group → **Resource Sheet**.

Click the cell directly below Resource **Name** heading column to **create** Resources.

Click the **Std. Rate field** for each resource to costs in **hourly (default), daily, weekly, monthly and yearly rates**.

In the following example, the resource Rasmus is left at zero. This is useful when you don't have to track rate-based costs for some resources.

Click the Ovt. Rate field to enter overtime rates.



The screenshot shows the 'Resource Sheet' view in Microsoft Project. The 'Resource Name' column lists John, Karen, Sharon, Rasmus, Carol, and Engineers. The 'Type' column is set to 'Work' for all. The 'Initials' column contains J, K, S, R, C, and E. The 'Max. Units' column shows 100%, 50%, 100%, 50%, 100%, and 400% respectively. The 'Std. Rate' column shows \$40.00/hr, \$00.00/wk, \$00.00/wk, \$0.00/hr, \$50.00/hr, and \$40.00/hr. The 'Ovt. Rate' column shows \$60.00/hr, \$0.00/hr, \$0.00/hr, \$0.00/hr, \$0.00/hr, and \$0.00/hr. The 'Cost/Use' column shows \$0.00, \$0.00, \$0.00, \$0.00, \$0.00, and \$0.00. The 'Accrue At' column shows 'Prorated' for all. The 'Base Calendar' column shows 'Standard' for all. The 'Code' column is empty.

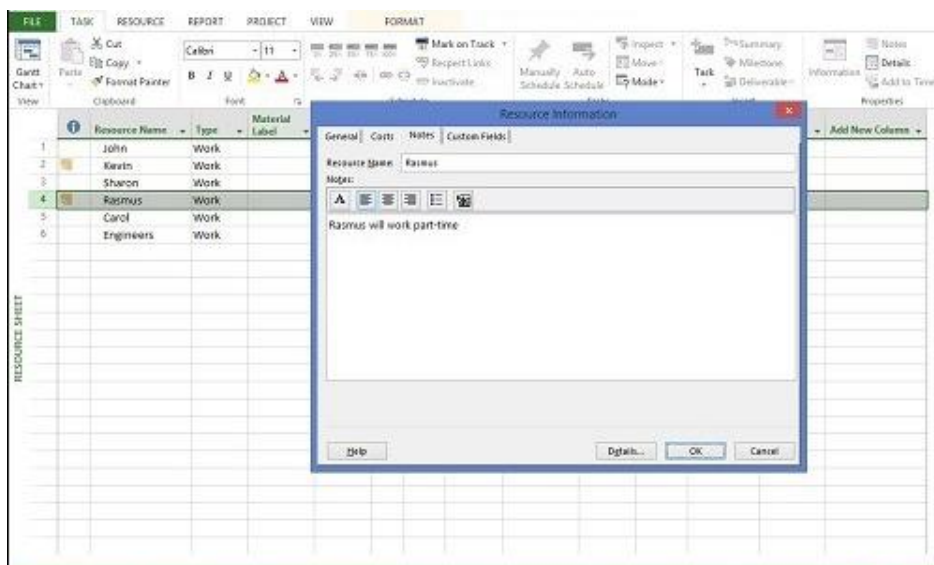
Resource Name	Type	Material Label	Initials	Grp	Max. Units	Std. Rate	Ovt. Rate	Cost/Use	Accrue At	Base Calendar	Code
John	Work		J		100%	\$40.00/hr	\$60.00/hr	\$0.00	Prorated	Standard	
Karen	Work		K		50%	\$00.00/wk	\$0.00/hr	\$0.00	Prorated	Standard	
Sharon	Work		S		100%	\$00.00/wk	\$0.00/hr	\$0.00	Prorated	Standard	
Rasmus	Work		R		50%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
Carol	Work		C		100%	\$50.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
Engineers	Work		E		400%	\$40.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	

Add Notes to Resources

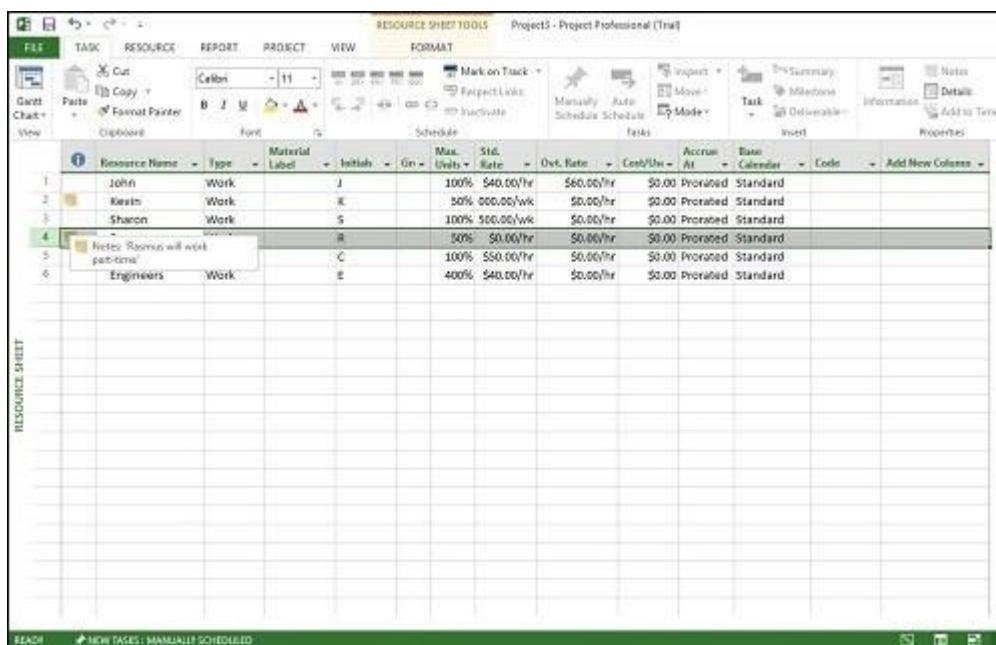
Click View tab → Resource Views group → Resource Sheet.

Double-click the Resource, a Resource Information dialog box opens.

Click on Notes tab. Here let's enter a note for Rasmus as "Rasmus will work part time"
Click OK.



A note icon now appears to the left of Rasmus' name in the Resource Sheet view. Hovering over it will make the note appear.



Resource Name	Type	Material Label	Initials	Gr	Max. Units	Std. Rate	Ovt. Rate	Cost/Unit	Accrual At	Base Calendar	Code	Add New Columns
John	Work		J		100%	\$40.00/hr	\$60.00/hr	\$0.00	Prorated	Standard		
Kevin	Work		K		50%	\$00.00/wk	\$0.00/hr	\$0.00	Prorated	Standard		
Sharon	Work		S		100%	\$00.00/wk	\$0.00/hr	\$0.00	Prorated	Standard		
Rasmus	Work		R		50%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		
Carol	Work		C		100%	\$50.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		
Engineers	Work		E		400%	\$40.00/hr	\$0.00/hr	\$0.00	Prorated	Standard		

Lab Task

Case: It's a project where a company is moving to a new office space, and we are in charge of planning the move, including getting office furniture and all office stuff shipped to the new location. And we want to do this without interfering with daily business. People should be able to do their work, then on the weekend all items will be transferred, so that your colleagues can start working from the new office building the following Monday.

Key project data: ▪ **Goal:** Office relocation ▪ **Timeline:** March – October ▪ **Project phases:** Project preparation, Selection of offices, Space design and furnishing, Physical relocation

- Build task list
- Enter task links
- Create sub tasks
- Set up base calendar and resource calendar
- Create milestones

Lab 8 & 9

Assign resources to tasks

Introduction: assign resources to tasks

Problem Statement:

- Learn how to set cost and material resources
- Learn different methods of how to assign material & cost resources to tasks
- Learn how to assign multiple resources to a single task.
- Learn how to assign cost resources to tasks.

Set Up Cost Resources

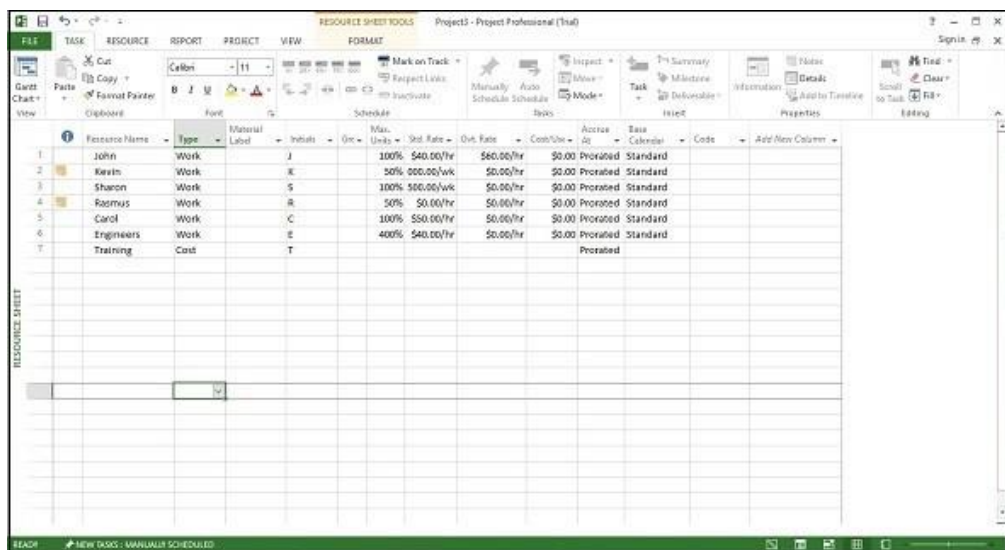
You can use a cost resource to represent a financial cost associated with a task in a plan. Common types of cost resources might include categories of expenses you want to track in a plan for accounting or financial reporting purposes, such as travel, entertainment, or training. **Cost resources do no work and have no effect on the scheduling of a task. The Max. Units, Standard and Overtime pay rate, and Cost per Use fields do not apply to cost resources.** After you assign a cost resource to a task and specify the cost amount per task, you can then see the cumulative costs for that type of cost resource, such as total travel costs in a project. The way in which cost resources generate cost values differs from that of work resources. When you assign a work resource to a task, the work resource can generate a cost based on a pay rate. **However, you enter the cost value of a cost resource only when you assign it to a task.** You do this in the task.

Click View tab → Resource Views group → Resource Sheet.

Click the empty cell in the Resource Name column.

Type Training and press the Tab Key.

In the Type field, click the down arrow to select Cost.



	Resource Name	Type	Material Label	Initials	Max. Units	Std. Rate	Ovr. Rate	Cost/Use	Accrue At	Base Calendar	Code
1	John	Work	J		100%	\$40.00/hr	\$40.00/hr	\$0.00	Prorated	Standard	
2	Kevin	Work	K		50%	\$20.00/wk	\$0.00/hr	\$0.00	Prorated	Standard	
3	Sharon	Work	S		100%	\$50.00/wk	\$0.00/hr	\$0.00	Prorated	Standard	
4	Rasmus	Work	R		50%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
5	Carol	Work	C		100%	\$50.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
6	Engineers	Work	E		400%	\$40.00/hr	\$0.00/hr	\$0.00	Prorated	Standard	
7	Training	Cost	T						Prorated		



Once the task and resource list are complete, resources need to be assigned to tasks in order to work on them. With MS Project you can track task progress, resource and tasks costs.

Assign Cost Resource to Tasks

Click View Tab → Gantt Chart View → Task Name column.

Double-click the Task Name. Task Information dialog box opens.

Click the Resources tab.

Click the cell below the Resource Name column. Select the resource from the dropdown list.

In the following example below, let's choose "Travel" as cost resource and enter the cost at \$800.

The screenshot shows the Microsoft Project interface. The 'Task Information' dialog box is open, with the 'Resources' tab selected. The 'Resources' list includes 'Travel' (cost \$800.00), 'Tansy' (cost \$0.00), and 'Tanya' (cost \$3,760.00). The 'Travel' resource is selected. The background shows a Gantt chart view of a project with tasks like 'Assign_Resources' and 'Phase 1'.



Assign Material Resource to Task

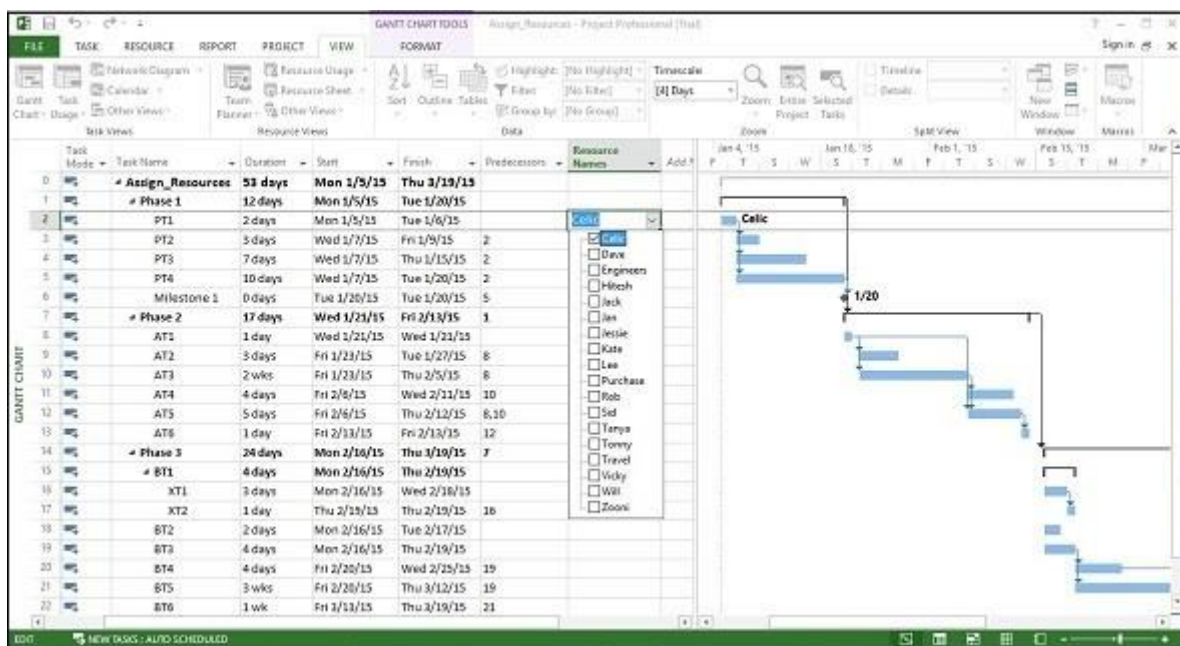
Method 1

Click View Tab → Gantt Chart View → Resource Name column.

Click the box below the Resource Name column for the task you need the resource to be assigned.

From the dropdown, choose the resource name. In the following screenshot as an example. For Task 1 "PT1", we have chosen the resource "Celic".

You can also select multiple resources to work on a single task.





Method 2

Click Resource tab → Under Assignments group → Assign Resources.

In the Assign Resources dialog box, click the resource name you like to assign.

Here let's choose "Hitesh". Now click the Assign button.

You can also select multiple resources to work on a single task.

The screenshot displays the Microsoft Project interface with the 'Assign Resources' dialog box open. The dialog box is titled 'Assign Resources' and shows the task 'PT2' selected. Below the task name, there is a section for 'Resource list options' which includes a table of resources from the 'Assign_Resources' project. The table has columns for 'Resource Name', 'R/D', 'Units', and 'Cost'. The resources listed are Celic, Dave, Engineers, Hitesh, Jack, Jah, Jesse, Kabe, and Lee. The 'Assign' button is highlighted in the dialog box. The background shows a Gantt chart with various tasks and their durations.

Resource Name	R/D	Units	Cost
Celic			
Dave			
Engineers			
Hitesh			
Jack			
Jah			
Jesse			
Kabe			
Lee			

Method 3

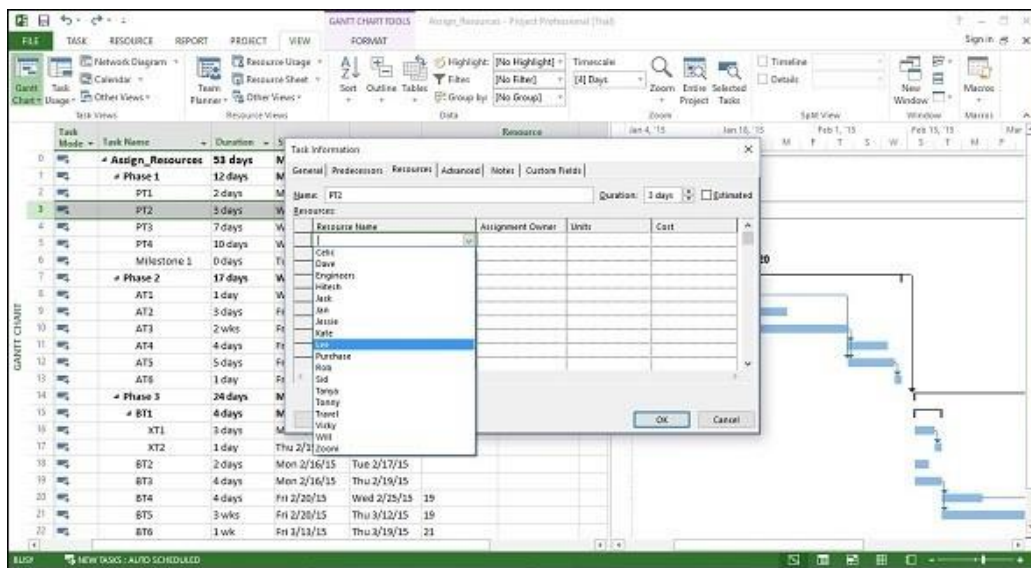
Click View Tab → Gantt Chart → Task Name column.

Double-click the Task Name. Task Information dialog box opens.

Click the Resources tab.

Click the cell below the Resource Name column. Select the resource from the dropdown list.

You can also select multiple resources to work on a single task.



Method 4

Click View Tab → Split View group → Details → Task Form.

The window is split in two, Gantt Chart view and Task Form view below it.



In the Task Form view, click under the Resource Name column and select the resource.

You can also select multiple resources to work on a single task.

The screenshot displays the Microsoft Project software interface in the 'Task Form' view. The 'Assign Resources' dialog box is open, showing a list of tasks and resources. The 'Resource Name' column is highlighted, and the 'Resource' dropdown menu is open, showing 'Celic' and 'Lee' as available resources. The 'Task Name' column shows tasks like 'Phase 1', 'Phase 2', and 'Milestone 1'. The 'Duration' column shows durations like '12 days', '3 days', and '7 days'. The 'Start' and 'Finish' columns show dates. The 'Predecessors' column shows task dependencies. The 'Resource Name' column shows the assigned resource for each task.

Task Name	Duration	Start	Finish	Predecessors	Resource Name
Assign Resources	53 days	Mon 1/5/15	Thu 3/19/15		
Phase 1	12 days	Mon 1/5/15	Tue 1/20/15		
PT1	2 days	Mon 1/5/15	Tue 1/6/15		Celic
PT2	3 days	Wed 1/7/15	Fri 1/9/15	2	Lee
PT3	7 days	Wed 1/7/15	Thu 1/15/15	2	
PT4	10 days	Wed 1/7/15	Tue 1/20/15	2	
Milestone 1	0 days	Tue 1/20/15	Tue 1/20/15	5	
Phase 2	17 days	Wed 1/21/15	Fri 2/13/15	1	
AT1	1 day	Wed 1/21/15	Wed 1/21/15		
AT2	3 days	Fri 1/23/15	Tue 1/27/15	8	



Lab Task

Define work, material and cost resources in your FYP gantt chart and assign them to tasks.

Lab 10

Plan Duration Cost and Time

Introduction: Plan duration cost and time.

Problem Statement:

- Learn how to check plan's cost
- Learn how to check plan's work
- Learn how to check project statistics

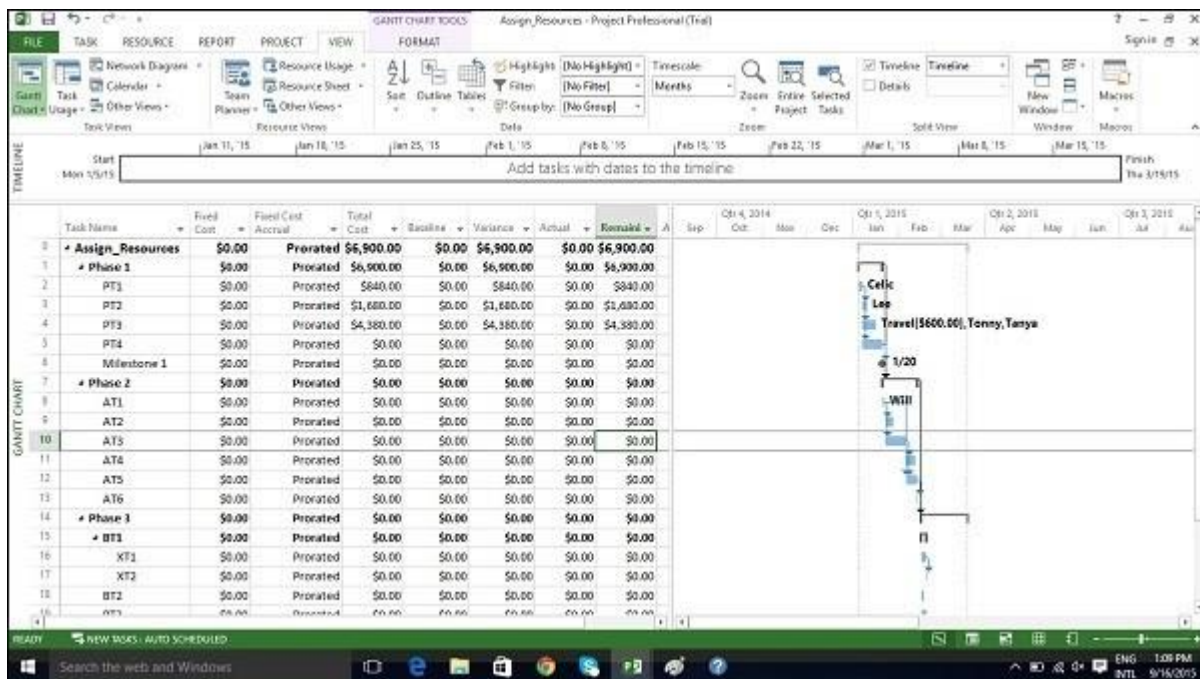


MS Project- Plan Duration Cost & Time

Check Plan's Cost

Click View tab → Data group → Tables → Cost.

Cost for each task gets rolled up into summary tasks, and then ultimately to project summary task.

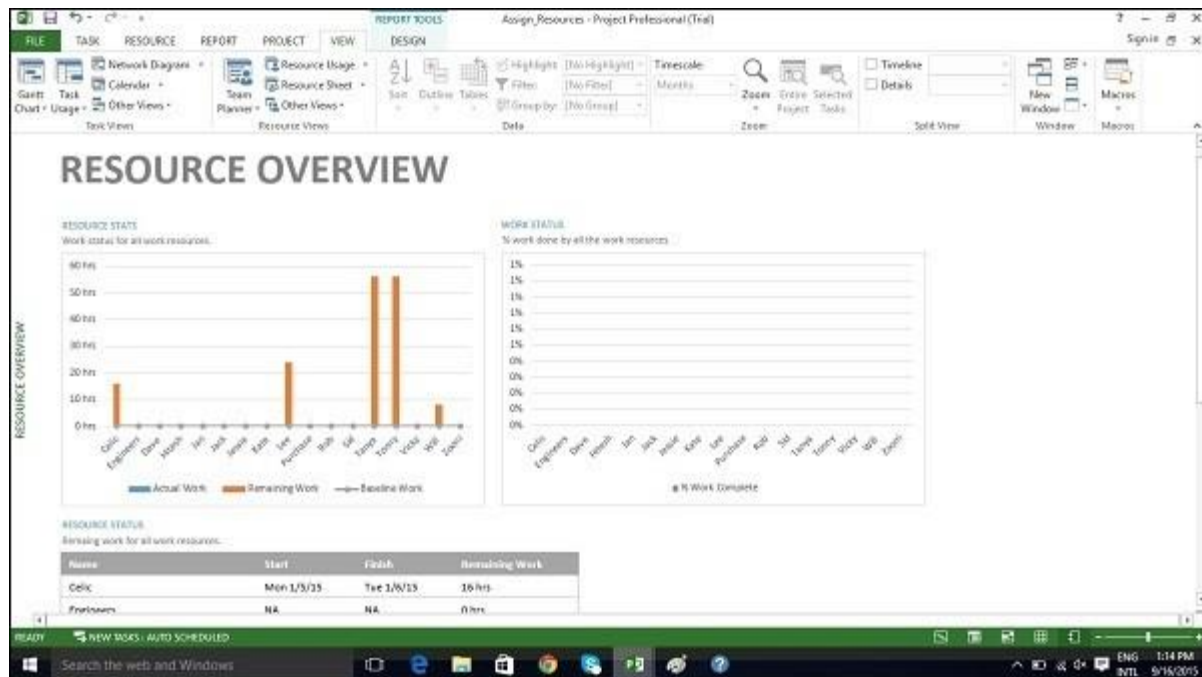


Check Plan's Work

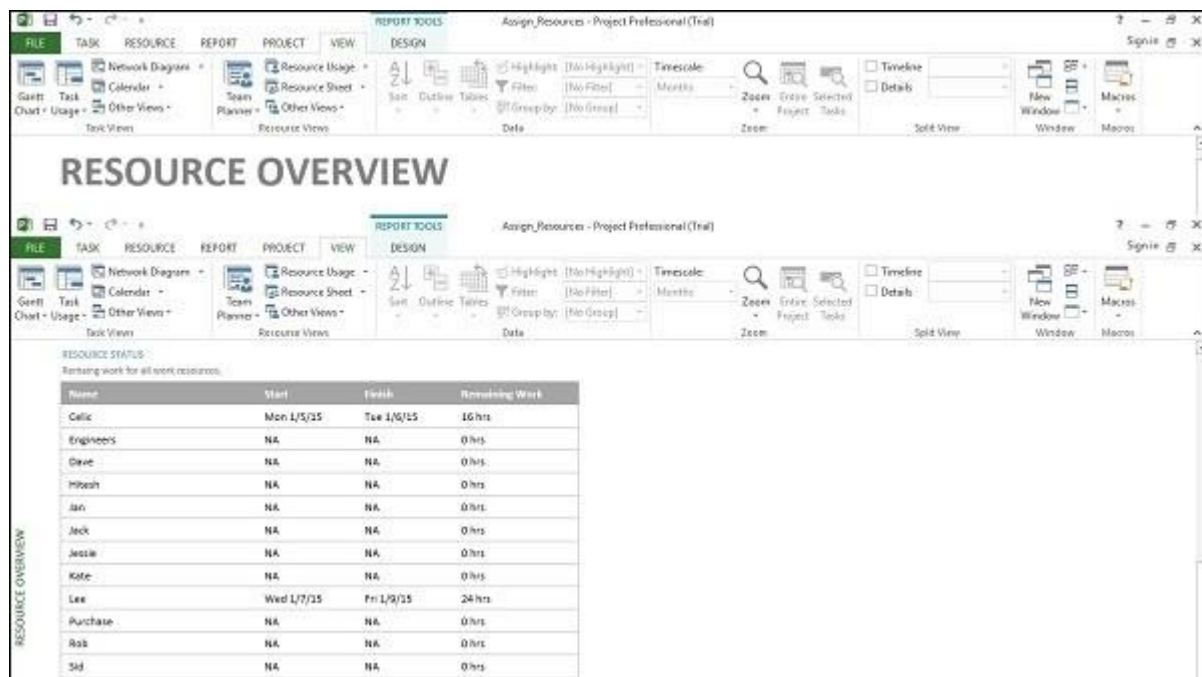
Click Report Tab → View Reports group → click Resources → click Resource overview



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In Resource status table which appears at the bottom, you will get a summary of resource's earliest start dates and latest finish dates as well as remaining work.





Check Project Statistics

Click Project Tab → Properties group → Project Information → in the new dialog box click Statistics...

Lab Task

You've been asked to build a software to support a low-cost video editing system. The system accepts videotape as input, stores the video on disk, and then allows the user to do a wide range of edits to the digitized video. The result can then be output to a tape.

Estimated duration 1 month

Staff required:

Project manager \$30/hr.

Emma (senior software engineer) \$20/hr.

Austin (junior developer) \$10/hr.

Harry (junior developer) \$10/hr.

Andrew (junior developer) \$10/hr.

Albert (senior graphic designer/ video editor) \$20/hr.

Frank (junior graphic designer/ video editor) \$10/hr.

Jane (Quality assurance engineer) \$10/hr.

Create complete project plan, build tasks, subtasks, create milestones, mention predecessors, write proper costing and duration. Also show your plan's cost.



Lab 11

Introduction: Get better hands on practice with MS Project

Problem Statement:

- Learn to divide a project proposal into tasks
- Learn to create summary tasks and their subsequent sub tasks.
- Learn how to manage resources
- Learn how to add costing

Online Learning Management System

You are a Project Manager in a renowned software company in Lahore. You have been asked to create an Online Learning Management System. The point of this project is to build up an online app that gives instructors and mentors the capacity to communicate easily with their Students. The project will bring about a framework with **dashboards, courses, attendance, grades, report cards, discussions, participation, tasks(quizzes/assignment/exams)**. The application will likewise permit guardians to remain associated with what their youngsters are doing in the university, to see their advancement, on schedule/late entries and so forth.

The application will have three panels:

- **Admin**
- **Teacher**
- **Student**
- **Parent**

Duration of this Project is 5 months starting from march 2021 till July 2021.

You have a team of software developers consisting of 2 junior developers lead by a senior developer.

There are 2 in house designers; Ahmad and Ali and a senior QA engineer Rabia.

Ali is on leave from 11th to 18th may.

Create a project plan for the Learning management system. Add task dependencies, milestones, resources and costs for each resource. Setup the calendar where required.



Lab 12

Introduction: Advanced Scheduling with MS Project 2013

Problem Statement:

- Learn about recurring tasks
- Learn how to view critical path
- Total float and free float



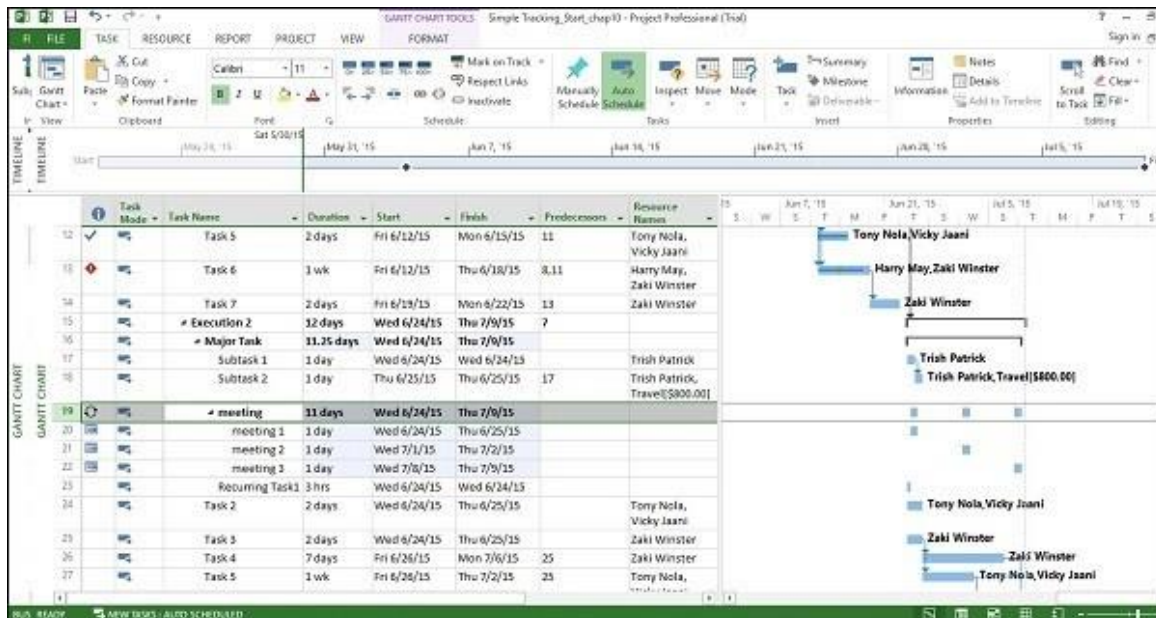
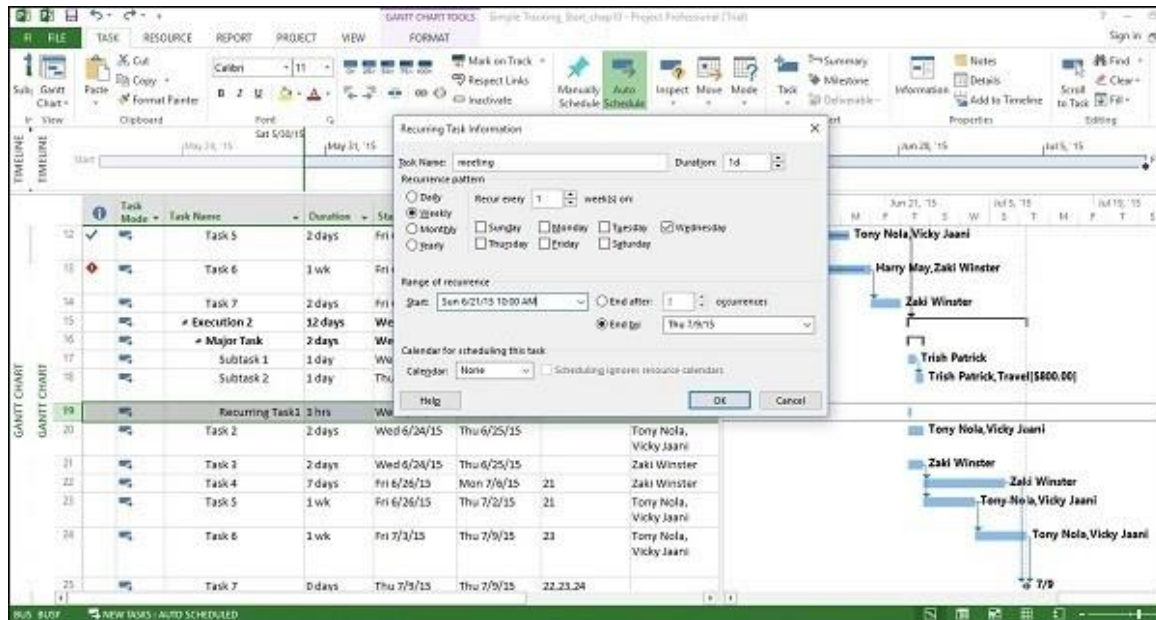
Enter a Recurring Task

Status meetings, status reports, inspection dates can recur with a particular frequency. In MS Project 2013, you can specify recurring tasks without having to assign tasks each time separately. You can also assign resources to these task.

In Gantt Chart View → Task Tab → Insert group → dropdown box for Task → Recurring Task.

Enter Task Name and choose Recurrence pattern.

You can also choose a specific time for the task to start as well. By default Project schedules a recurring task to start on plan's default start time. You can add time value in the Start box for Recurring Task Information dialog box to change this. In the following figure, start time of 10:00 AM is entered.



View Critical Path

Critical Path is the succession of connected tasks that will take the longest to complete. The word “critical” does not mean that the tasks are complex or important or need to be closely monitored, but the focus is on term schedule that will affect the project finish date.

- If you want to shorten the duration of a project, you should first start with activities/tasks on the



critical path.

- Critical path can be a single sequence of tasks (a single critical path) or there can be more than 1 critical paths for a single project.
- While schedule changes are made, it is also likely that the critical path will change from time to time.
- If a task in critical path is late, the project is late.

One needs to always focus on the Critical Path first, when one wants to apply fast-tracking or crashing to shorten the project duration.

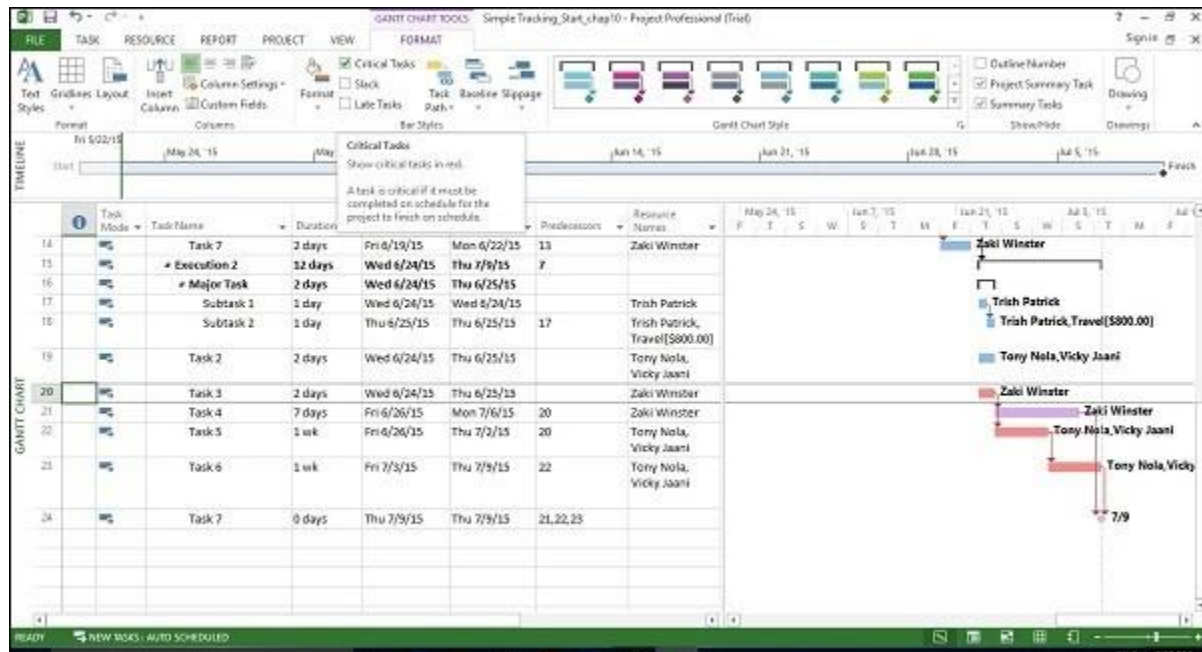
Slack or Float are key to understanding Critical path. There are two types of Float –

- **Free Float** – It is the amount of time a task can be delayed without delaying another task.
- **Total Float** – It is the amount of time a task can be delayed without delaying the completion of the project.

A task is on the critical path if its total float is less than or equal to a certain threshold—by default, if it's float is zero days (you can adjust this threshold if you want). In contrast, noncritical tasks have float, meaning that they can start or finish earlier or later within their float time without affecting the completion date of the plan.

All task bars in the critical path, in the Gantt Chart View on the right, will turn Red in color.

In Gantt Chart view → Format Tab → Bar Styles Group → Check the Critical Tasks box ON.



Lab Task

- 1-Create a Recurring task for your FYP project plan.
- 2-Create columns for free slack and total slack. Identify the critical task in you FYP project plan.



Lab 14 & 15

Resource Over Allocation

Introduction: Advanced Scheduling with MS Project 2013

Problem Statement:

- Learn resource allocations
- Resolve resource over allocation

Check Resource Allocations

Relationship between a resource's capacity and task assignments is called allocation.

This can be defined by 3 states –

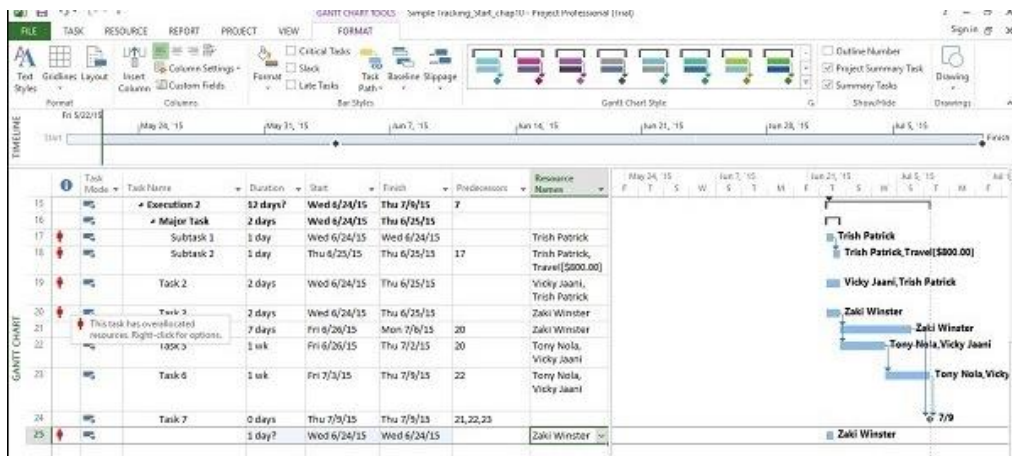
- Under allocated – An Engineer who works for 40 hours a week, has work assigned for only 20 hours.
- Fully allocated – A skilled worker who works for 40 hours a week, is assigned 40 hours of work in that week.
- Over allocated – A carpenter is assigned 65 hours of work, when he only has a 40 hour work week.

In Gantt Chart View

Click View Tab → Task Views group → Gantt Chart view.

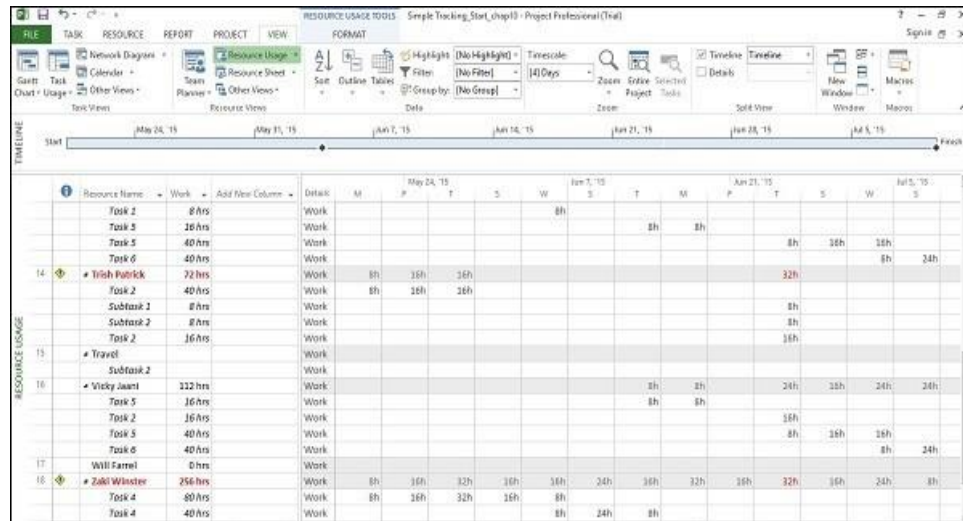
Gantt Chart View displays some limited resource information, as shown in the following screenshot.

It summarizes whether there may be a problem by the red over allocated icon in the indicator column.



Click View Tab → Resource Views group → Resource Usage view.

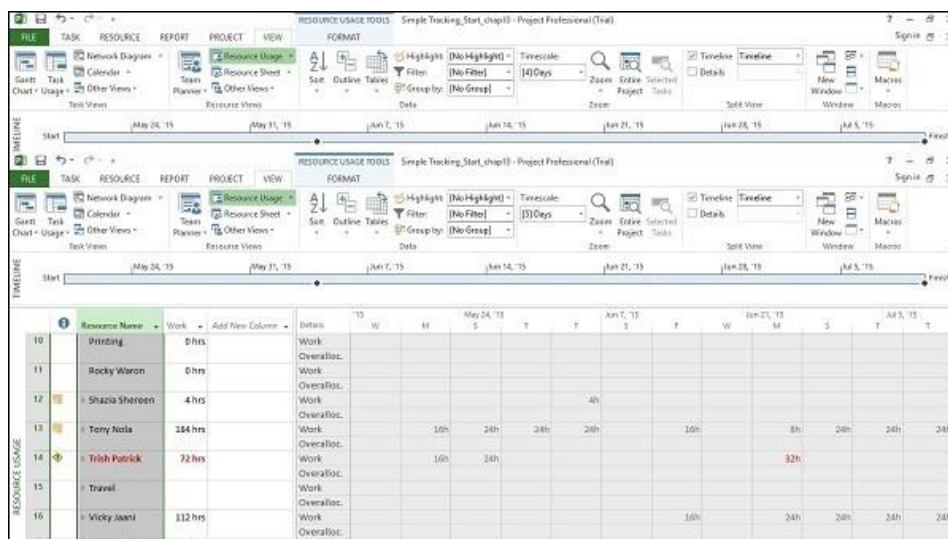
The Resource Usage view displays resources and all tasks assigned to them underneath the ResourceName. The left-hand side of the screen lists the Resources and the Task Name together with columns of total information for the resource or assignment. The right-hand side shows a time-phased view.



You can also collapse the outline in the table to see total work per resource overtime.

Click on Resource Name column heading.

Click View Tab → Data group → Outline → Hide Subtasks.



Resolve Resource Over Allocation

One would need to either change the scope (reduce the amount of work), assign more resources, or accept a longer schedule to resolve over allocation.

This can be achieved by using some of the following techniques –

Adjust Schedule

By changing its lead or lag time when the resource has more tasks assigned than can be completed during a given time period. If you add delay that is less than or equal to the amount of slack on the task, you will not affect the finish date of the project.

By default when you link tasks, they are assigned a “Finish to Start” relationship. In this relationship,

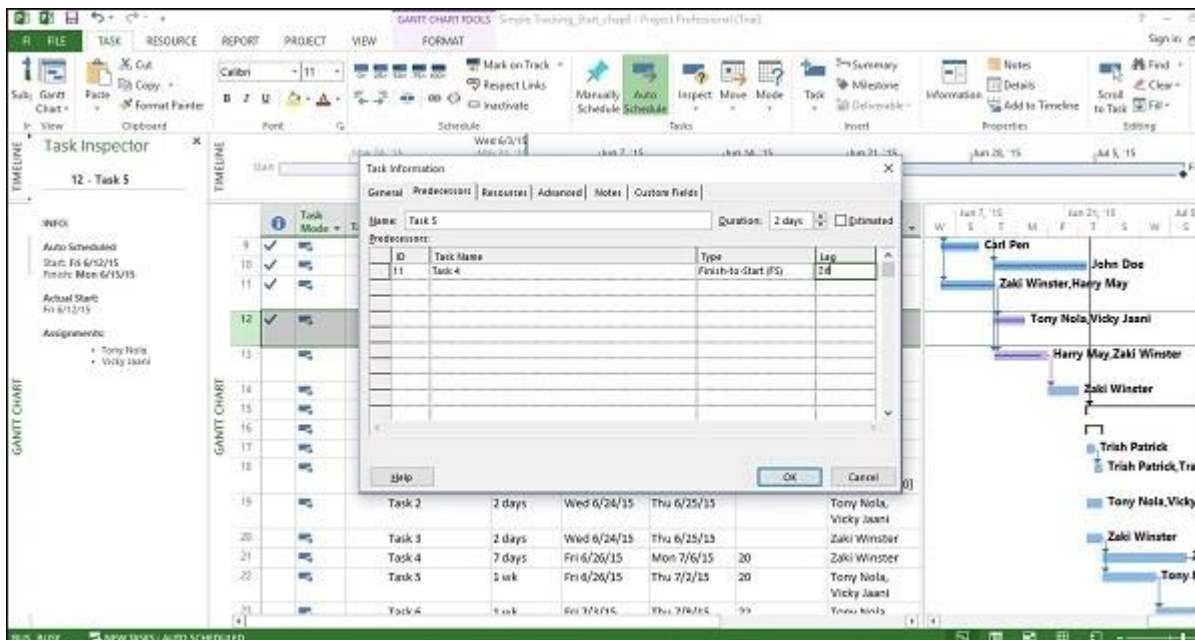
- **Lead** – Lead time causes successor task to begin before its predecessor tasks ends.
- **Lag** – Lag time causes successor task to start after its predecessor task ends.

Click Task Tab → double-click the required Task under Task Name column → Task Information dialog box opens → Predecessors Tab.

Under Lag heading column, enter the lag in terms of hours, days, weeks, or years.

You can also apply lag or lead as a percentage. If you enter 50% for the selected Task which is 6 days long, the task is delayed by 3 days after the predecessor ends.

Lag is entered as positive units and lead in negative units (example, -3d or -50%).





Substitute Resources or Add Additional Resources

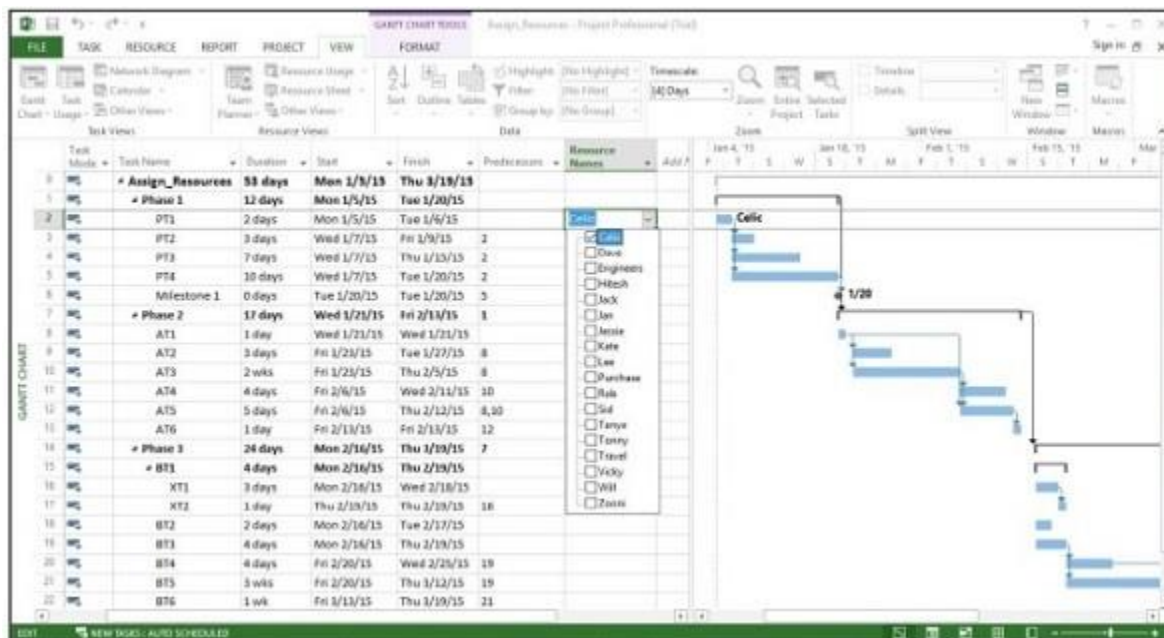
You can manually allot some other resource to the task.

Click View Tab → Gantt Chart View → Resource Name column.

Click the box below the Resource Name column for the task you need the resource to be assigned.

From the dropdown, choose the resource name. In the following example, for Task 1 "PT1", we have chosen the resource "Celic".

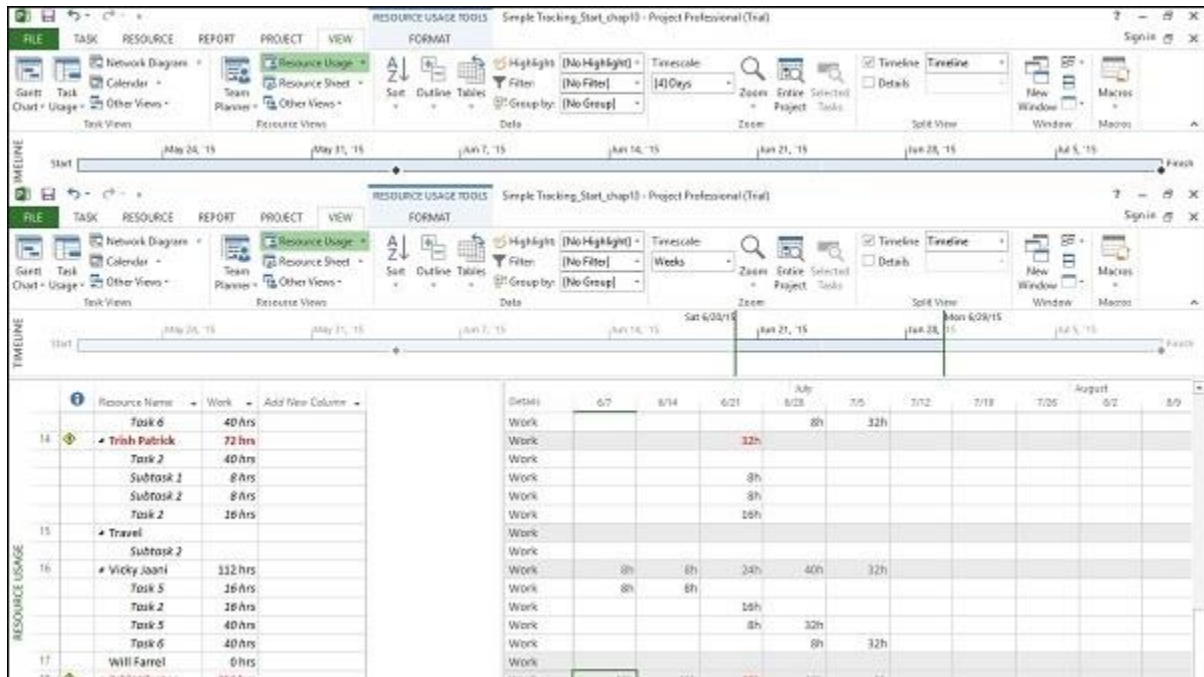
You can also select multiple resources to work on a single task.



Reduce Assigned Work

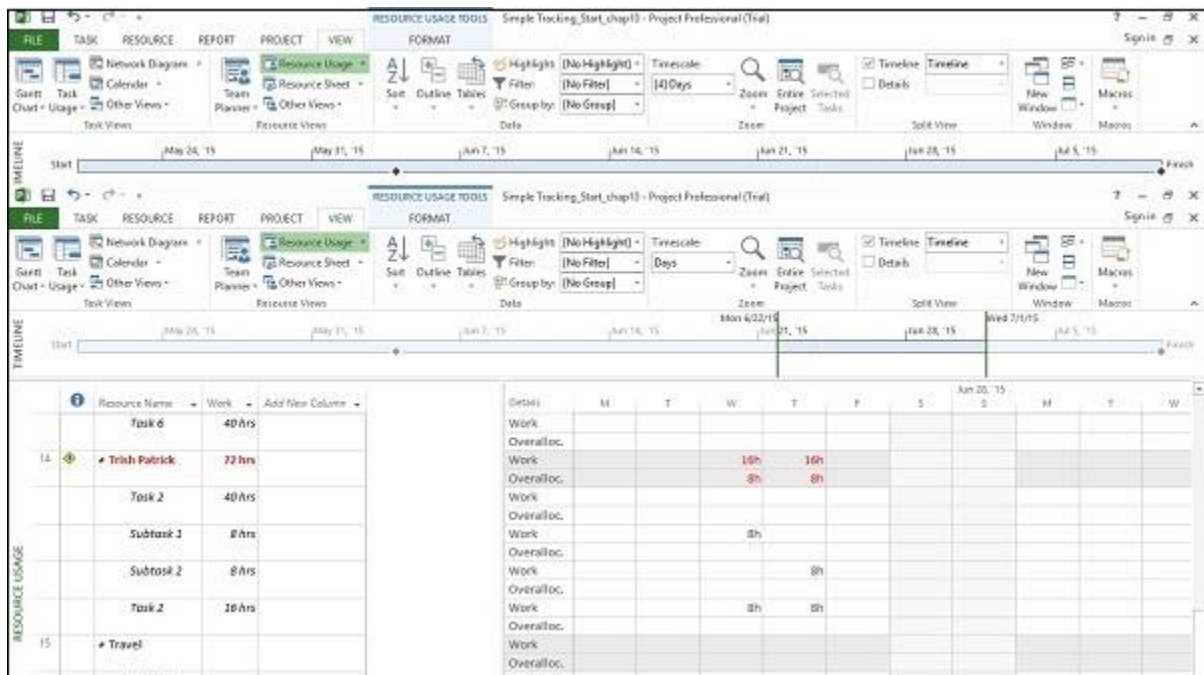
Click View Tab → Resource Views group → Resource Usage view.

In the following example, Trish Patrick is over allocated, the Resource Name and Work appear in red.

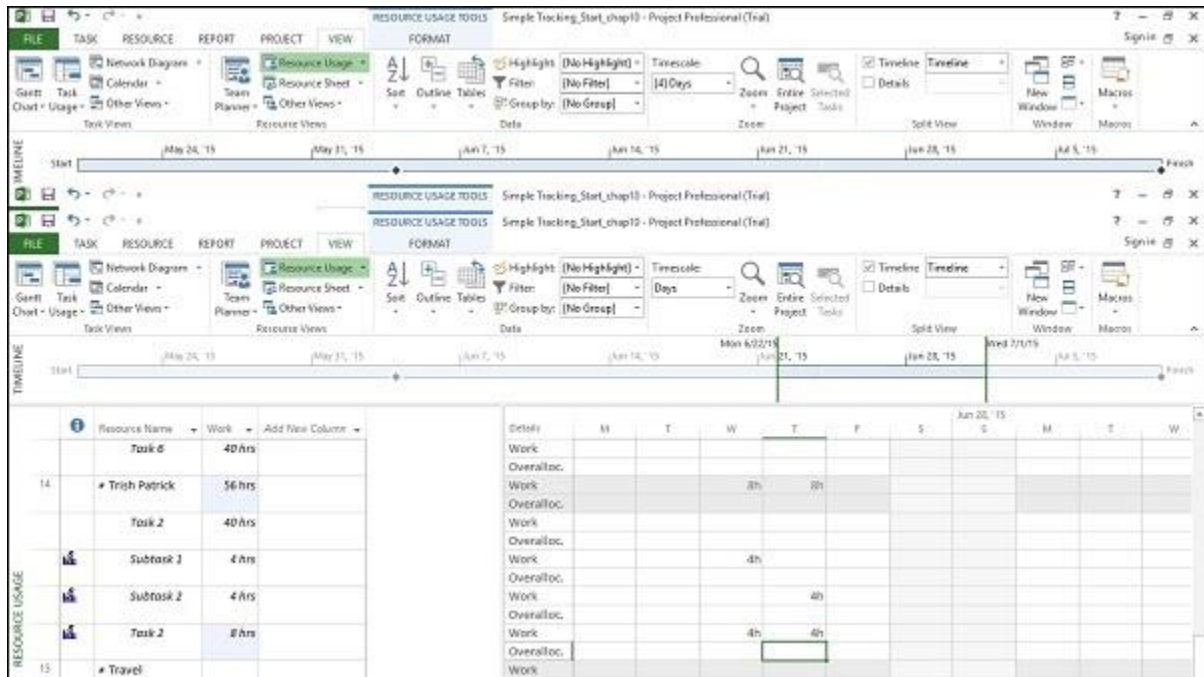


On View tab → Zoom group → Timescale box → Days.

You can also right-click on the Time-phased grid in the right hand side window to display amount of overallocation by switching on overallocation.



Now you can reduce the assigned hours. In the following example, 8-hour assignment is reduced to 4-hour assignments. Not only is Trish Patrick's work reduced but total work in the plan has changed. You will also notice a new icon in the indicator column to let you know that the assignment work has been edited.



Decrease Task Duration

You can decrease task duration (if no actual work has been entered) to reduce the amount of work required of the resource, who is assigned to complete the task. If actual work has been recorded, you must manually reduce the remaining work on the task.

Remove Over Allocated Resource

You can just remove a resource assignment from an overallocated resource.



Lab 16 & 17

Task Constraints

Introduction: Advanced Scheduling with MS Project 2013

Problem Statement:

- What is meant by task constraints
- Apply different tasks constraints and what is meant by each



Lead and Lag

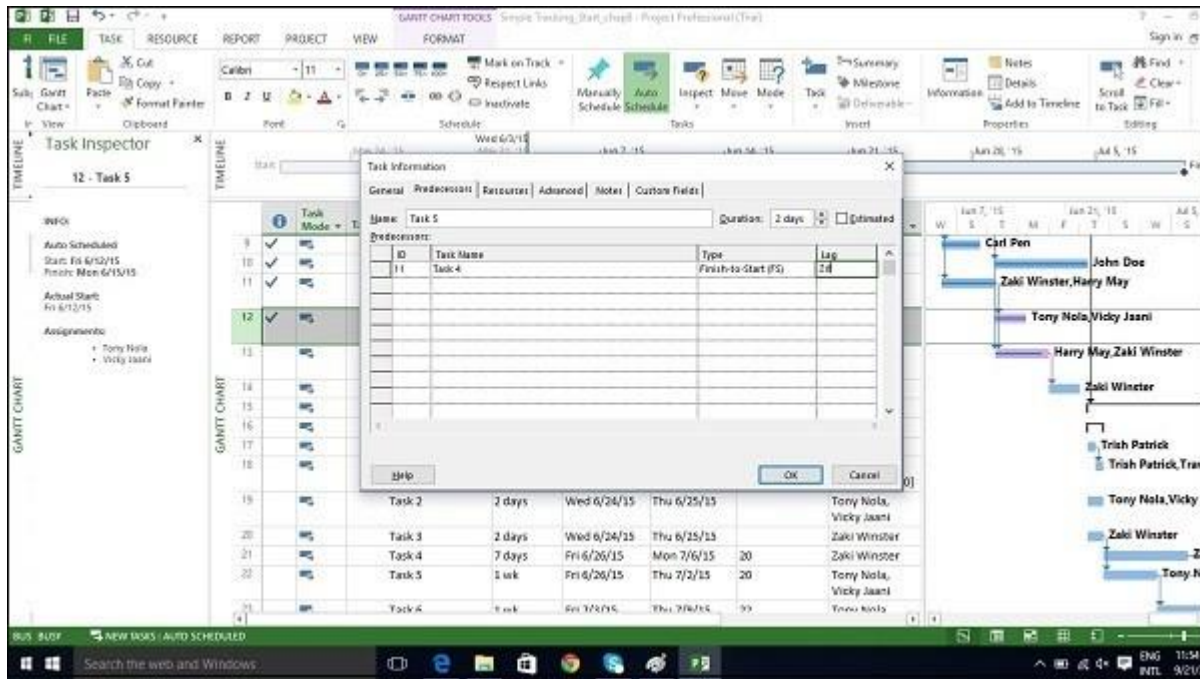
By default when you link tasks they are assigned a “Finish to Start” relationship. In this relationship,

- ♦ Lead – Lead time causes successor task to begin before its predecessor task ends.
- ♦ Lag – Lag time causes successor task to start after its predecessor task ends.

Click Task Tab → double-click the required Task under Task Name column → Task Information dialog box opens → Predecessors Tab.

Under Lag heading column, enter the lag in terms of hours, days, weeks, or years. You can also apply lag or lead as a percentage. If you enter 50% for the selected Task which is 6 days long, the task is delayed by 3 days after the predecessor ends.

Lag is entered as positive units and lead in negative units (example, -3d or -50%).



Apply Task Constraints

- In MS Project 2013 by default each task is constrained as “As Soon As Possible” when Automatic Scheduling is turned ON.
- As Soon As Possible means the task starts as soon as the project starts, if there are no dependencies that would delay it.
- No fixed start or end dates are imposed by this constraint type, but of course predecessor and successor dependencies are maintained.

When MS Project 2013 performs calculations to save you time in a project that's running late, constraint settings are enforced.

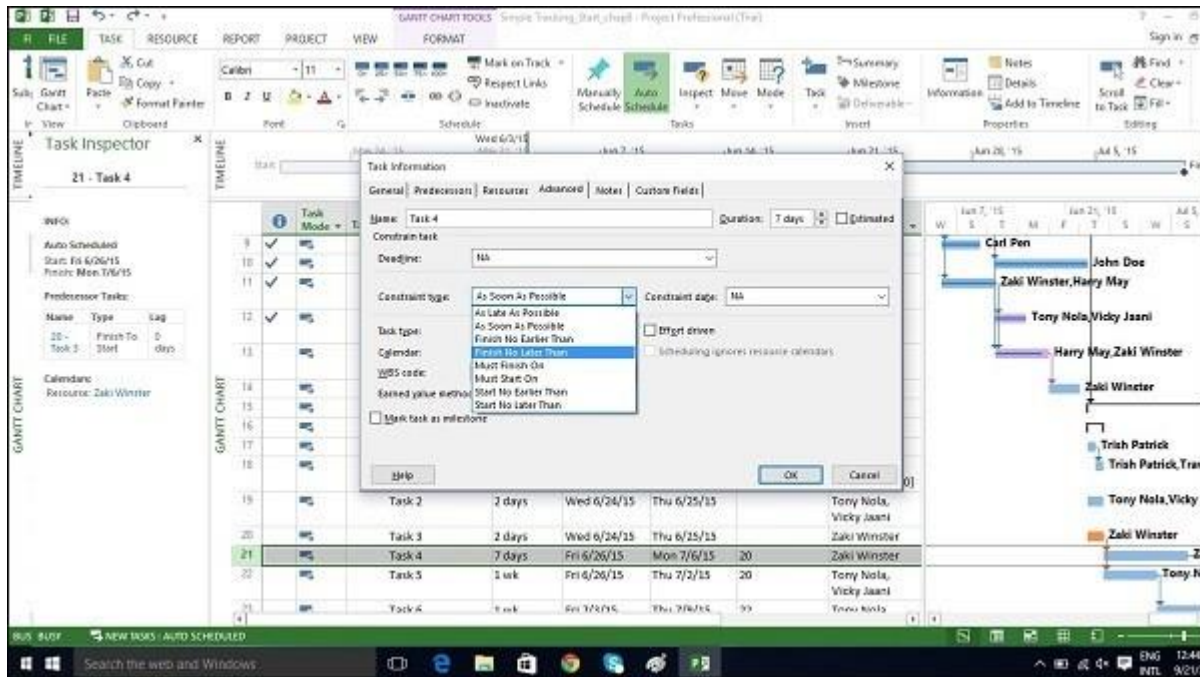
There are 8 Task Constraints.



Constraint type	Constraint name	Description
Flexible	As Late As Possible (ALAP)	Task is scheduled as late as possible with the task ending before the project ends and without delaying subsequent tasks. Default constraint when you schedule from the project finish date. Do not enter a task start or finish date with this constraint.
	As Soon As Possible (ASAP)	Task is scheduled to begin as early as possible. Default constraint when you schedule from the project start date. Do not enter a start or finish date with this constraint.
Semi-Flexible	Start No Earlier Than (SNET)	Task is scheduled to start on or after a specified date.
	Finish No Earlier Than (FNET)	Task is scheduled to finish on or after a specified date.
	Start No Later Than (SNLT)	Task is scheduled to start on or before a specified date.
	Finish No Later Than (FNLT)	Task is scheduled to finish on or before a specified date.
Inflexible	Must Finish On (MFO)	Task is scheduled to finish on a specified date.
	Must Start On (MSO)	Task is scheduled to start on a specified date.

Click Task Tab → double-click the required Task under Task Name column → Task Information dialog box opens → Advanced Tab.

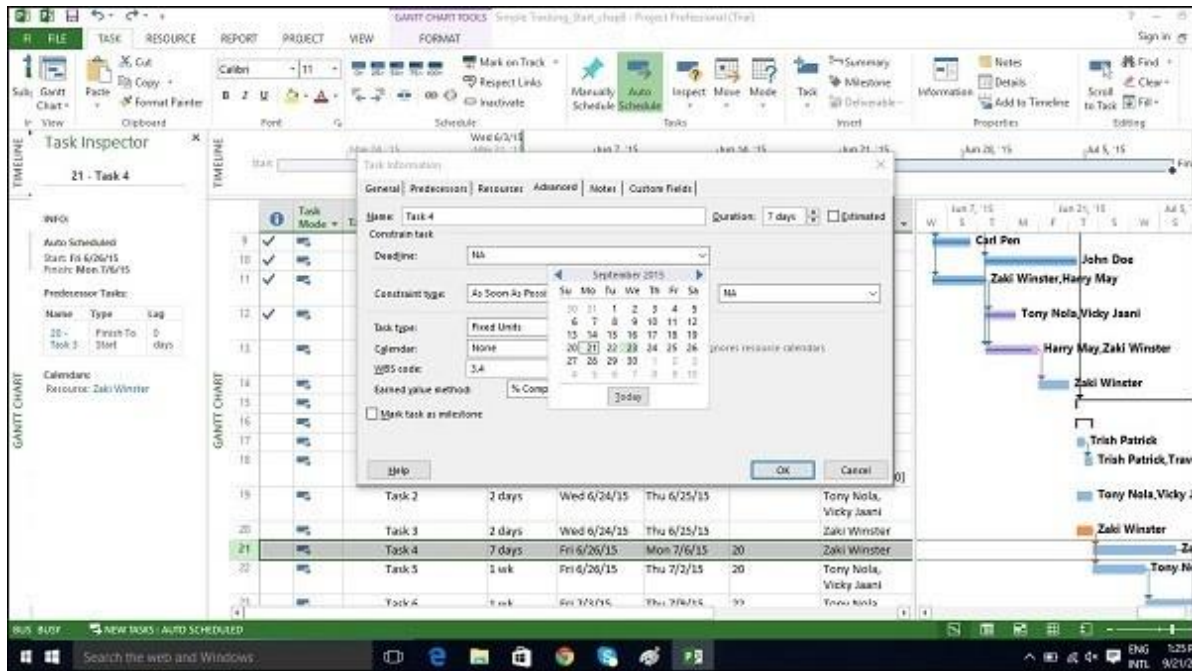
Click dropdown box for Constraint type. Choose the constraint you would like to apply



Enter Deadline Date

If you use Task Constraints, you limit your scheduling flexibility, where MS Project 2013 will fix a particular start or finish date of the task according to the constraint. It is a better idea to use a Deadline Date which has no effect on the scheduling of a task or summary task. MS Project will alert you with a red exclamation symbol in the indicators column, if the scheduled completion of the task exceeds its deadline date.

Click Task Tab → double-click the required Task under Task Name column → Task Information dialog box opens → Advanced Tab.



Lab Task 15

- Create additional columns for “free slack, total slack, Early start, Early finish, late start and late finish” for your FYP Task list
- Create critical path
- Apply all 8 task constraints to your FYP task list



Lab 18

Resource Over Allocation

Introduction: Advanced Scheduling with MS Project 2013

Problem Statement:

- Learn resource allocations
- Resolve resource over allocation

Lab Task

Open the “Lab 16 practice task.mpp” file and check the resource allocations. See if there is any resource over allocations. Apply the steps learnt in lab 14 to resolve resource over allocation.



Lab 19& 18

Levelling

Introduction: Level Over Allocated Resources

Problem Statement:

- Learn what is meant by leveling
- Learn to set priorities
- Learn to level over allocated resource
- Check plan's cost



Level Over Allocated Resources

If resources are over allocated you can use resource-leveling feature in MS Project 2013. It works by either splitting tasks or by adding delay to tasks to ensure the resource is not overloaded. Leveling can delay the individual task finish dates and even the project finish date.

Project first delays tasks to use up any available slack. Once the slack becomes zero, MS Project 2013 makes changes according to priorities, dependency relationships and task constraints (such as a Finish No Later Than constraint).

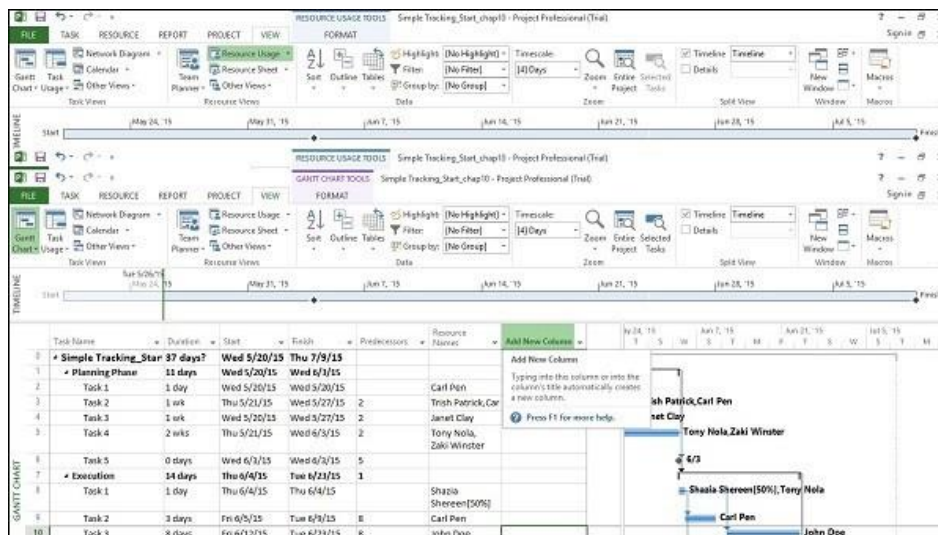
Set Priorities

It is always better to set task priorities (this is a measure of a task's importance/availability for leveling). You can enter value between 1 and 1000, according to the amount of control you like in the leveling process. A priority level of 1000 will ensure MS Project does not level a particular task. By default, priority is set at 500 or a medium level of control. Tasks that have lower priority are delayed or split before those that have higher priority.

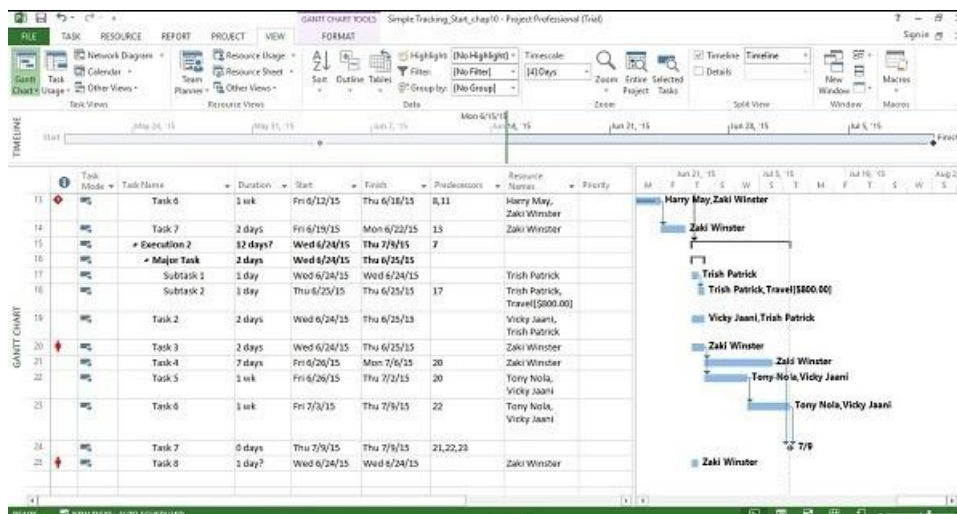
Click View Tab → Task Views → Gantt chart View.

In the Gantt chart table area, scroll to the right to see Add New Column.

Click on the dropdown box and select Priority.



Now you can add priority to each task as required.



Levelling

Steps in the Levelling process are only a few, but it is important to understand what each option does. The steps are as follows –

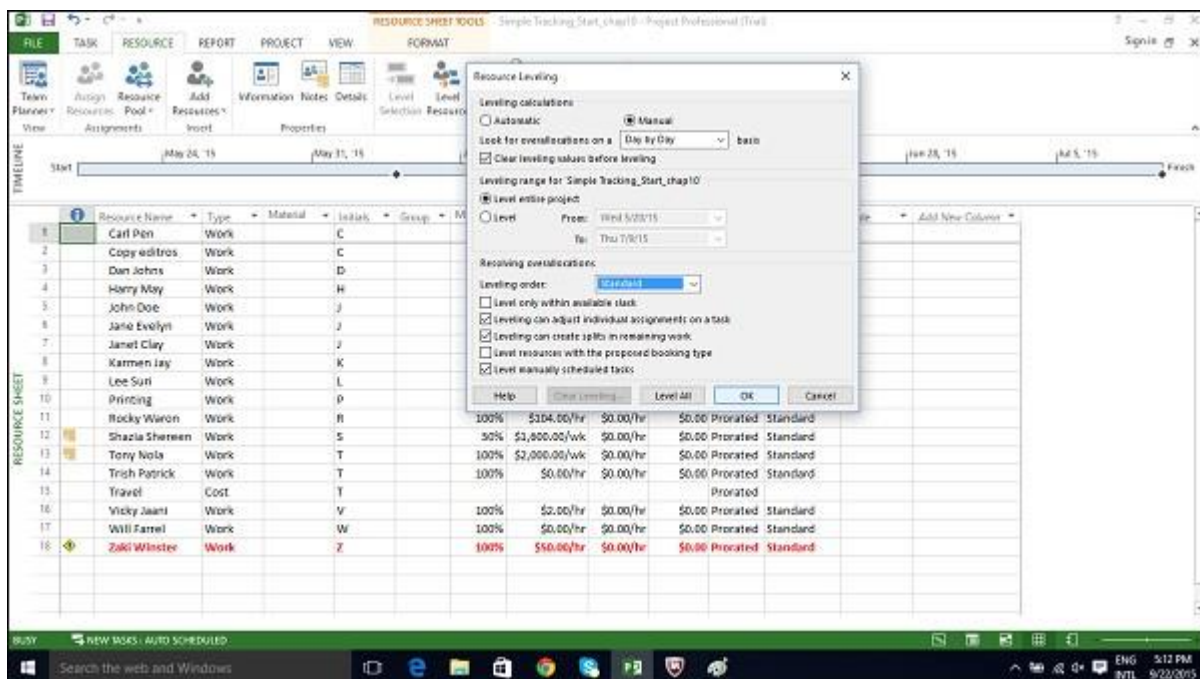
Click on View Tab → Resource View group → Resource Sheet.

Click Resource tab → Level group → Levelling Options → Level All.

Project does levelling and over allocated indicators are removed (If levelling is done completely, sometimes this might not happen).

In the following section, we will look at Levelling Options in detail –

Click Resource tab → Level group → Levelling Options.





Step 1: In Resource Leveling dialog box, under Level calculations, try to use Manual more often. This will ensure MS Project 2013 does the leveling process only when you ask it to, and not as soon as a resource becomes over allocated even if you don't want it to (when you choose Automatic option). For examples, if a resource is over allocated, for say half an hour more in a week, from 40 hours to

40.5 hours, you wouldn't want this to inconvenience you by getting automatically leveled.

Step 2: In Resource Leveling dialog box, under Level calculations, choose Day by Day basis for "Look for over allocations on a" option. Doing so will not level resources, but it will determine when Project displays over allocation indicators next to resourcenames.

Step 3: In Resource Leveling dialog box, under Level calculations, use the clear leveling values before leveling checkbox is selected. Doing so will ensure Project removes any existing leveling delays from all tasks and assignments before leveling. And if you previously leveled the plan and then added more assignments, you might want checkbox to be unchecked to ensure you don't lose the previous leveling results.

Step 4: In Resource Leveling dialog box, under **Leveling range for "....."**, you can choose Level **entire project**. Here you choose to level either the entire plan or only assignments that fall within a date range you specify.

Step5: In Resource Leveling dialog box, under Resolving over allocations, **leveling order** dropdown box you can choose Standard. You have 3 options here –

- **ID only** option delays tasks only according to their ID numbers. *Numerically higher ID numbers (for example, 10) will be delayed before numerically lower ID numbers. You might want to use this option when your plan has no task relationships or constraints.*
- **Standard option** delays tasks according to *predecessor relationships, start dates, task constraints, slack, priority, and IDs.*
- **Priority, standard option** looks at the task priority value before the other standard criteria (Task priority is a numeric ranking between 0 and 1000).

Step6: In Resource Leveling dialog box, under Resolving over allocations, you have several options that you can select. These are explained as follows –

- **Level only within available slack.** Selecting this checkbox would prevent Project from extending the plan's finish date. MS Project will use only the free slack within the existing schedule, which could mean that resource over allocations might not be fully resolved.
- **Leveling can adjust individual assignments.** Selecting this checkbox allows Project to add a leveling delay (or split work on assignments if Leveling Can Create Splits in Remaining Work is also selected) independently of any other resources assigned to the same task. This might cause resources to start and finish work on a task at different times.
- **Leveling can create splits in remaining work checkbox.** This allows Project to split work on a task (or on an assignment if Leveling Can Adjust Individual Assignments on a Task is also selected) as a way of resolving over allocation.



- **Level manually scheduled tasks.** Selecting this allows Project to level a manually scheduled task just as it would an automatically scheduled task

Lab 20 & 22

Introduction: Learn how to track progress by saving baselines

Problem Statement:

- Learn what is a baseline
- Learn how to set a baseline-table
- Learn how to set a baseline
- Learn what is interim plan



MS Project - Track Progress

Once your project plan is ready in MS Project, it becomes essential for a project manager to measure the actuals (in terms of work completed, resources used and costs incurred) and to revise and change information about tasks and resources due to any changes to the plans.

Save a Baseline

To evaluate project performance, you need to create a baseline against which you will compare the progress. One needs to save the baseline, once a plan is fully developed. Of course, one can always add new tasks, resources, constraints and costs to the plan.

Also note, it makes sense to save the baseline before entering any actual values such as percentage of task completion.

Note – With MS Project 2013, you can save up to 11 Baselines in a Single plan. These multiple baselines seem contrary to the definition of baseline. You can use this flexibility when –

- You have a baseline plan for the external customer and another for the internal team.
- You are preparing for a risk event. You want to develop separate baseline plans for risk response and recovery.
- You are accommodating a big change request; you might still want to keep the original plan for future reference when communicating with a stakeholder.

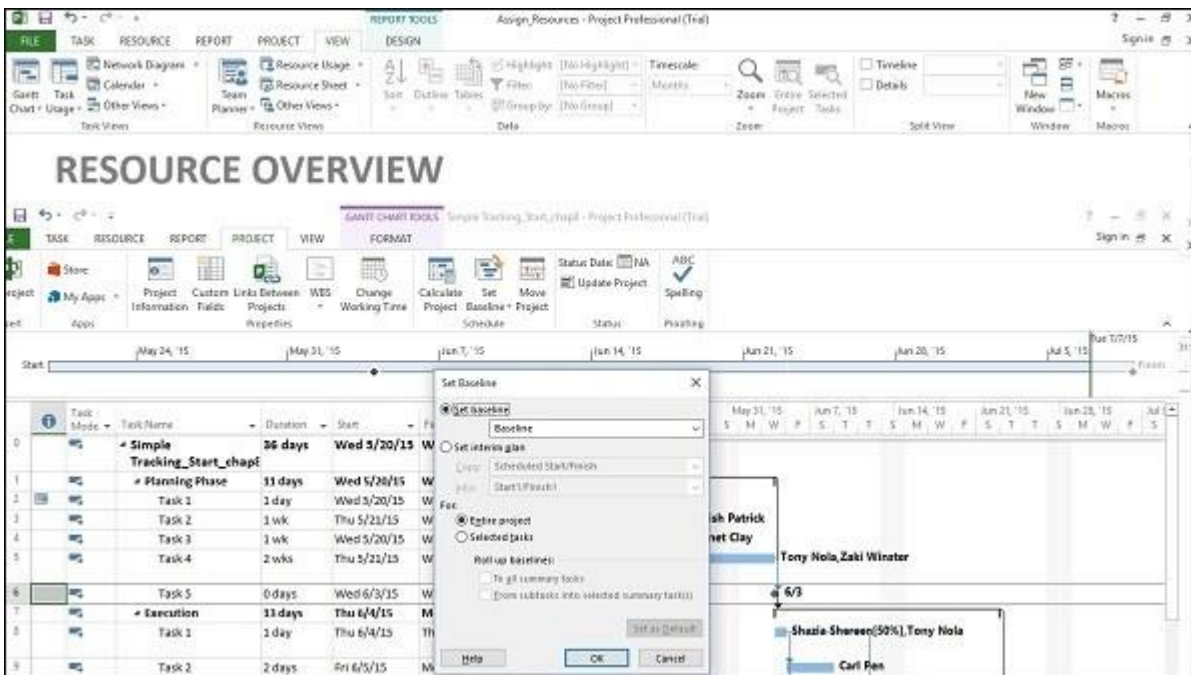
Create a Baseline table

Click View → table → more tables → baseline → OK.

Note: Now all the table entries are zero as no baseline is set yet.

Create a Baseline

Click Project Tab → Schedule group → Set Baseline → OK.



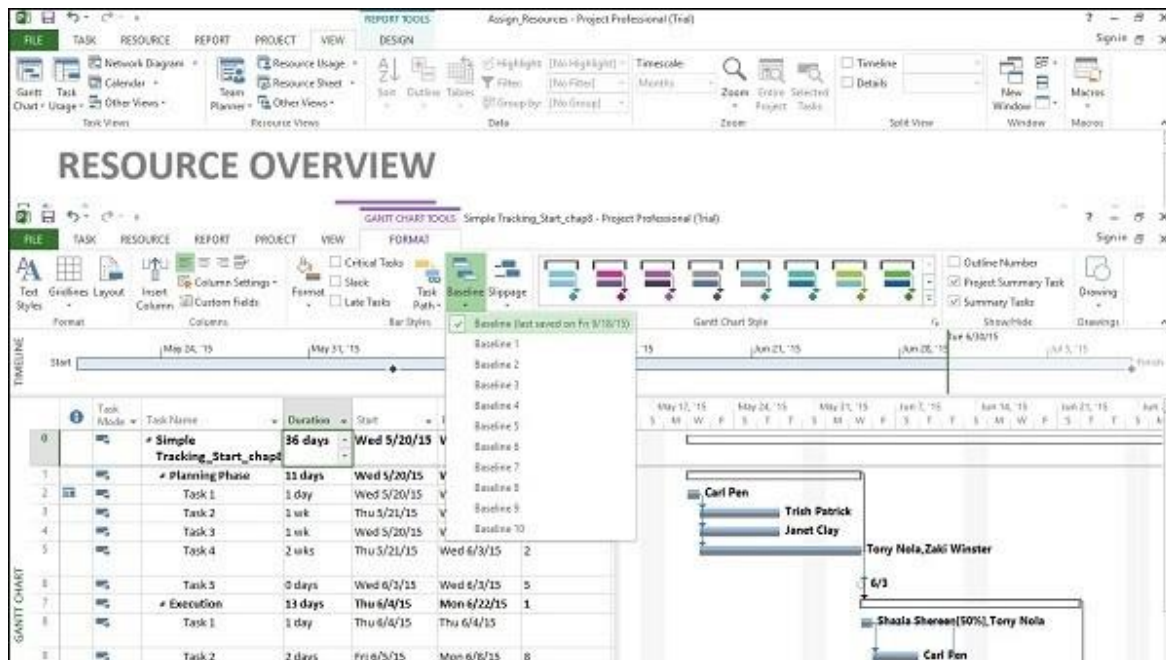
The screenshot displays the Microsoft Project Professional interface. The 'PROJECT' tab is active, and the 'Set Baseline' dialog box is open in the center. The dialog has two main options: 'Set baseline for' and 'Set interval for'. Under 'Set baseline for', the 'Entire project' radio button is selected. Under 'Set interval for', the 'Selected tasks' radio button is selected. The 'Set as default' button is visible at the bottom right of the dialog. The background shows the 'Resource Overview' view with a Gantt chart and a task list. The task list includes tasks such as 'Simple Tracking_Start_chapt', 'Planning Phase', 'Task 1', 'Task 2', 'Task 3', 'Task 4', 'Task 5', 'Execution', 'Task 1', and 'Task 2'.

View Baseline on Gantt Chart

Click View Tab → Task Views group → Gantt Chart.

Click Format Tab → Bars and Styles group → Baseline (that you want to display).

You will see Baseline Gantt bars displayed together with the current Gantt bars.



Update a Baseline

As time and work progresses on a project, you might need to change the baseline as well. You have several options for the same –

- Update the baseline.
- Update the baseline for selected tasks. Save
- multiple baselines.



Update the Baseline for the Entire Project

This simply replaces the original baseline values with the currently scheduled values.

Click Project Tab → Schedule group → Set Baseline → OK.

Update the Baseline for Selected Tasks

This does not affect the baseline values for other tasks or resource baseline values in the plan.

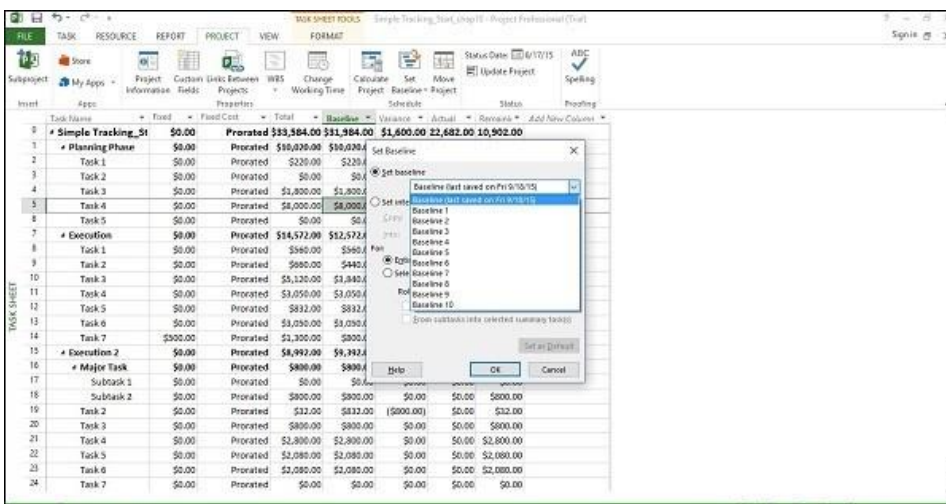
Click Project Tab → Schedule group → Set Baseline → For select Selected tasks → OK.

Save Multiple Baselines

You can save up to 11 baselines in a single plan. The first one is called Baseline, and the rest are Baseline 1 through Baseline 10.

Click Project Tab → Schedule group → Set Baseline → click the dropdown box to save any baseline you like.

Click OK.

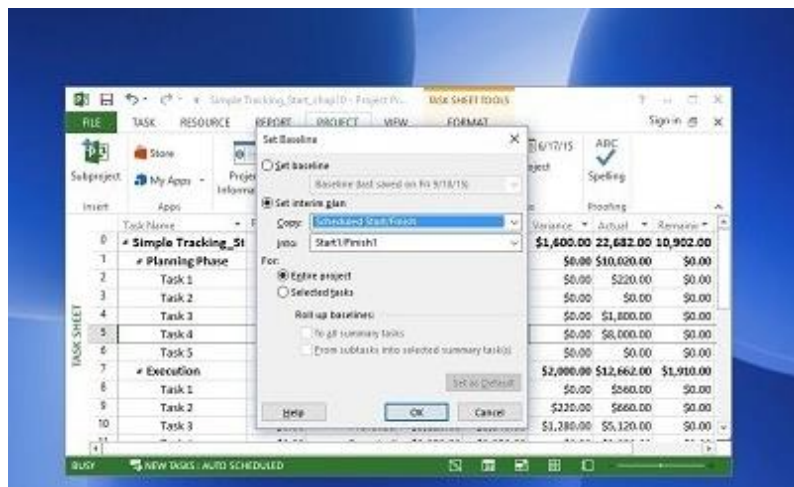


Interim Plans

An interim plan saves only two kinds of information for each task – Current start dates and Current finish dates.

It can be used as a project marker. It is visually easy to see how off-track or on-track the project progress is. Because it only specifies dates, it is simple, clear and easy information.

Click Project Tab → Schedule group → Set Baseline → Set interim plan → OK.



Lab task

Save a base line for your FYP project plan. After doing changes in durations save another baseline and explore the differences.



Lab 23 & 24

Task Slippage

Introduction: Advanced Scheduling with MS Project 2013

Problem Statement:

- Learn what is task slippage
- How to view tasks with variance?

After creating a project plan and baselines, the project begins. At this stage, the project manager would be focusing on collecting, monitoring, analyzing project performance, and updating project status by communicating with the stakeholders.

When there is a difference between what is planned and the actual project performance, it is called a **Variance**. Variance is mostly measured in terms of Time and Cost.

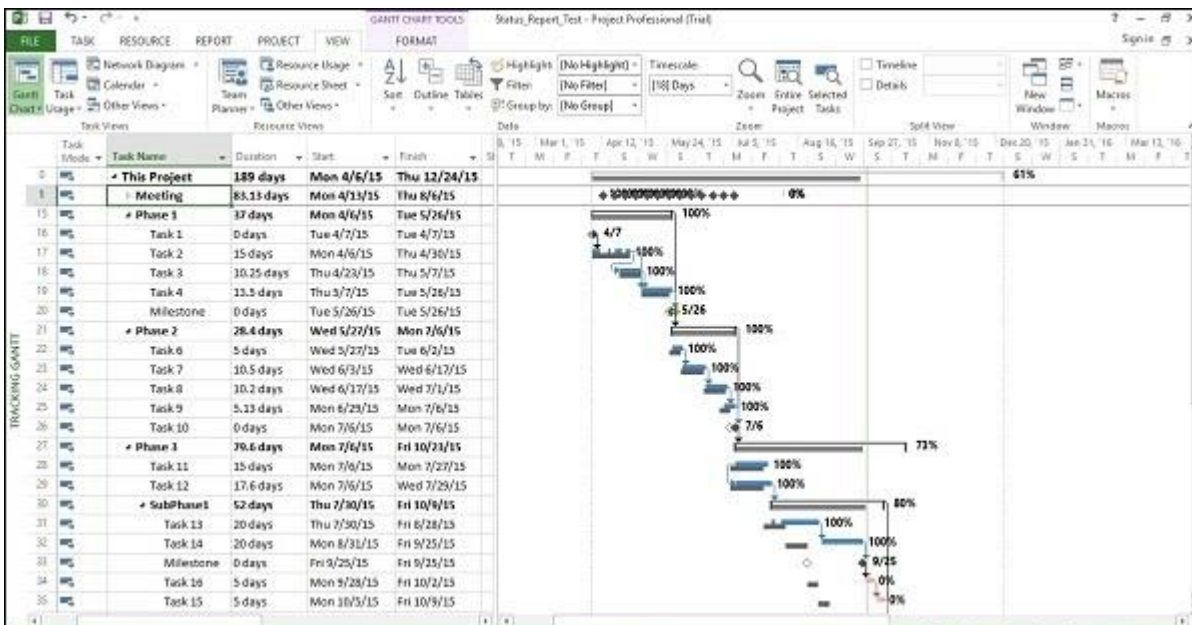
Task Slippage

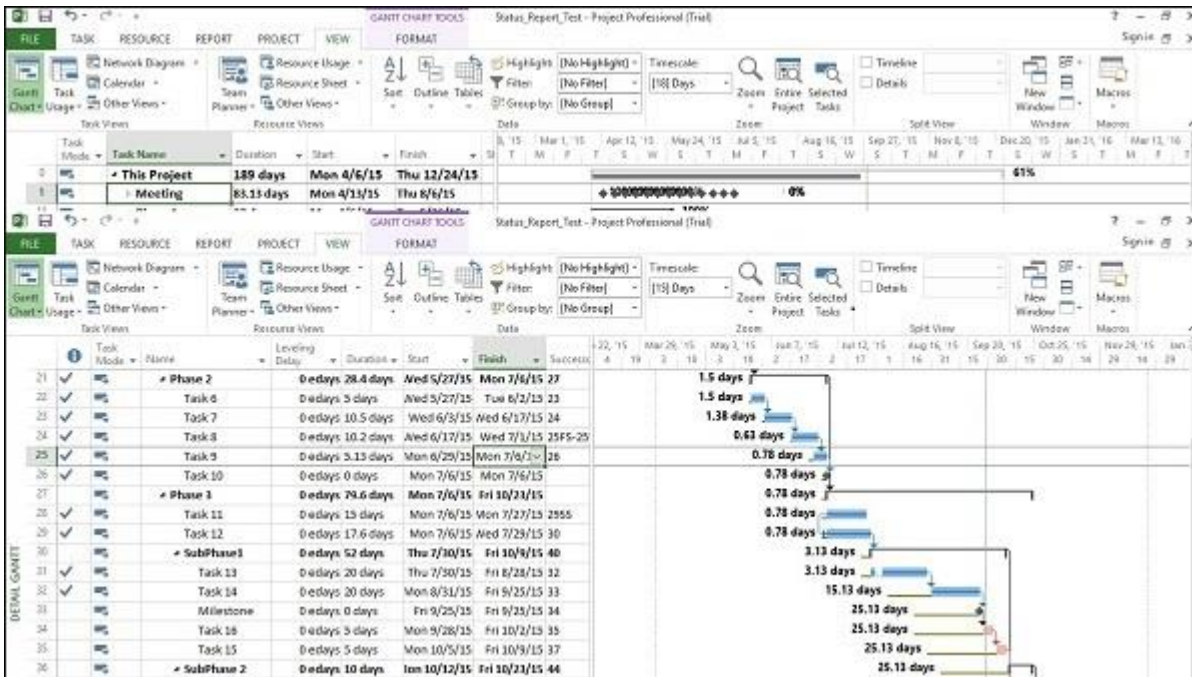
There are several ways to view task with variance.

Method 1: Graphical View by Tracking Gantt

Click View tab → Task Views group → Gantt Chart dropdown → Tracking Gantt.

By comparing the currently scheduled Gantt bars with baseline Gantt bars, you can see what tasks started later than planned or took longer to complete.





Method 3: Variance Table

Click View tab → Data group → Tables → Variance.

Method 4: Filters

Click View tab → Data group → Filters → More
Filters → choose filter as Late tasks, Slipping task, etc.

MS Project 2013 will filter the task list to show only the tasks filtered in this process. So if you select Slipping Task, you will view only incomplete tasks. Any task that is already completed will not show up.

Lab Task



Lab 25

Introduction: Advanced Scheduling with MS Project 2013

Problem Statement:

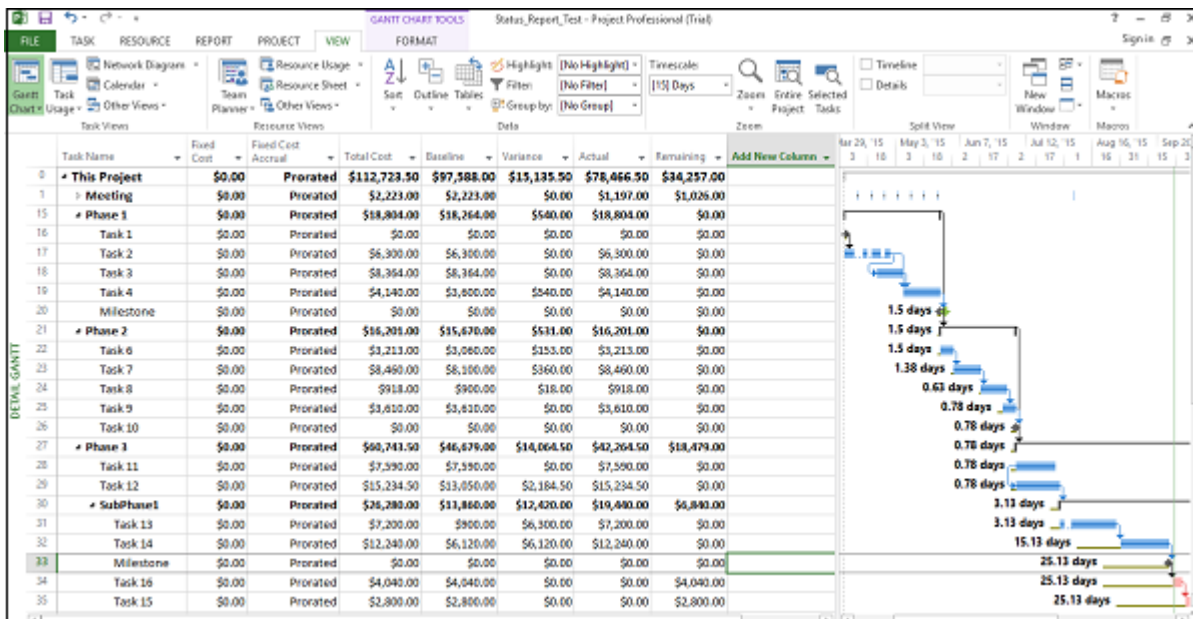
- Learn about task costs
- Learn about resource costs
- Learn about project reports

Task Costs

To examine cost in a project life cycle, you should be aware of these terms and what they mean in MS Project 2013 –

- **Baseline costs** – All planned costs as saved in baseline plan.
- **Actual costs** – Costs that have been incurred for tasks, resources, or assignments.
- **Remaining costs** – Difference between baseline/current costs and actual costs.
- **Current costs** – When plans are changed due to assigning or removing resources, or adding or subtracting tasks, MS Project 2013 will recalculate all costs. This will appear under the fields labeled Cost or Total Cost. If you have started to track actual cost, it will include actual cost+ remaining cost (uncompleted task) per task.
- **Variance** – Difference between Baseline Cost and the Total Cost (current or scheduled cost).

Click View Tab → Data group → Tables → Cost.



You will be able to view all relevant information. You can also use filters to see tasks that have run over budget.

Click View tab → Data group → Filters → More Filters → Cost Overbudget → Apply.

Resource Cost

For some organizations, resources costs are primary costs, and sometimes the only cost, so these need to be closely watched.

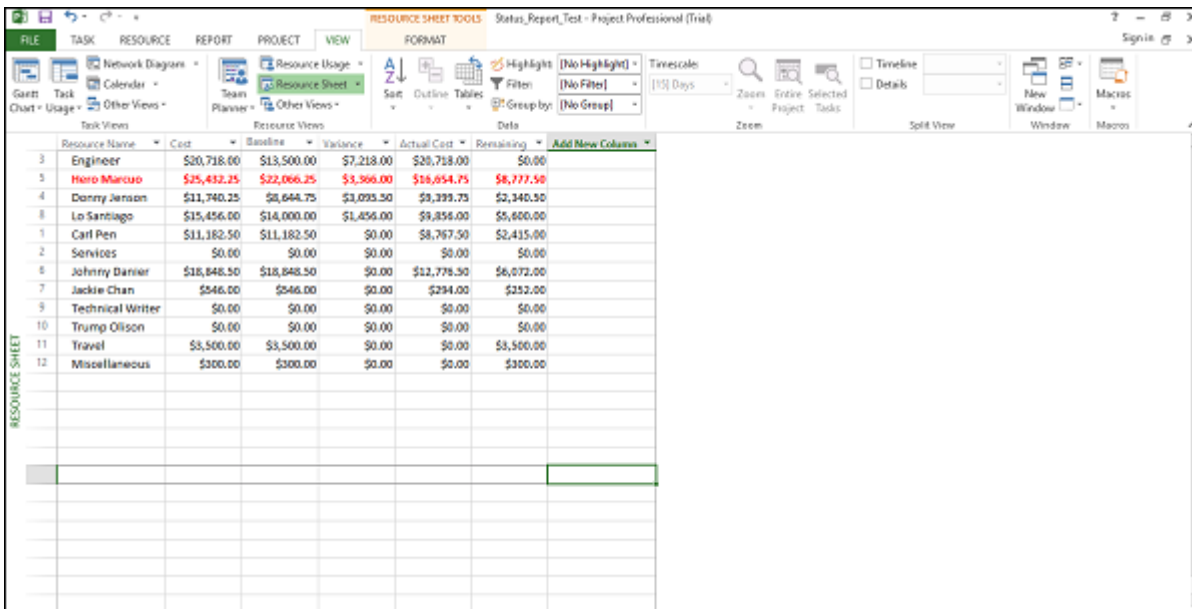
Click View tab → Resource Views group → Resource Sheet.

Click View tab → Data group → Tables → Cost.

We can sort the Cost column to see which resources are the most and least costly.

Click the AutoFilter arrow in Cost column heading, when the drop-down menu appears, click on Sort Largest to Smallest.

You can use the AutoFilter feature for each of the columns, By sorting Variance column, you will be able to see the variance pattern.



The screenshot shows the 'Resource Sheet' in Microsoft Project 2013. The 'VIEW' tab is active, and the 'Resource Sheet' is selected. The table displays resource costs and variances. The 'Cost' column is sorted in descending order, with 'Hero Marcio' having the highest cost at \$25,432.25. The 'Variance' column shows the difference between the actual cost and the baseline cost.

Resource Name	Cost	Baseline	Variance	Actual Cost	Remaining
Engineer	\$20,718.00	\$13,500.00	\$7,218.00	\$20,718.00	\$0.00
Hero Marcio	\$25,432.25	\$22,066.25	\$3,366.00	\$18,654.75	\$6,777.50
Denny Jensen	\$11,740.25	\$8,644.75	\$3,095.50	\$8,399.75	\$2,340.50
Lo Santiago	\$15,456.00	\$14,000.00	\$1,456.00	\$9,856.00	\$5,600.00
Carl Pen	\$11,182.50	\$11,182.50	\$0.00	\$8,767.50	\$2,415.00
Services	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Johnny Danier	\$18,848.50	\$18,848.50	\$0.00	\$12,778.30	\$6,070.20
Jackie Chan	\$546.00	\$546.00	\$0.00	\$294.00	\$252.00
Technical Writer	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Trump Olison	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Travel	\$3,500.00	\$3,500.00	\$0.00	\$0.00	\$3,500.00
Miscellaneous	\$300.00	\$300.00	\$0.00	\$0.00	\$300.00

Project Report

Project 2013 comes with a set of predefined reports and dashboards. You'll find all of these on the Report tab. You can create and customize graphical reports for your project as well.

Dashboard Reports

Click Report → View Reports group → Dashboards.

Resource Reports

Click Report → View Reports group → Resources.



Cost Reports

Click Report → View Reports group → Costs.

Progress Reports

Click Report → View Reports group → In Progress.

Custom Reports

Click Report → View Reports group → New Report.

There are four options.

- **Blank** – Creates a blank canvas. Use the Report Tools - Design tab to add charts, tables, text, and images.
- **Chart** – Creates a chart comparing Actual Work, Remaining Work, and Work by default. Use the Field List pane to pick different fields to compare. The look of the chart can be changed by clicking on Chart Tools tabs, Design, and Layout tabs.
- **Table** – Creates a table. Use the Field List pane to choose what fields to display in the table (Name, Start, Finish, and % Complete appear by default). Outline level box lets you select how many levels in the project outline the table should show. The look of the table can be changed by clicking on Table Tools tabs, Design, and Layout tabs.
- **Comparison** – Creates two charts side-by-side. Charts will have the same data at first. You can click one of the charts and pick the data you want in the Field List pane to begin differentiating them.



Lab 26 & 27

Network Diagram

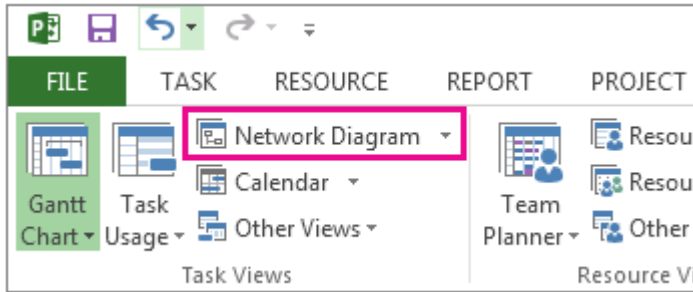
Introduction: Advanced Scheduling with MS Project 2013

Problem Statement:

- Creating Network diagram from the scratch
- How to represent non critical tasks, critical tasks, summary tasks, milestones.
- How to add predecessor, resources, indentations

Network Diagram

To find the Network diagram view, choose **View > Network Diagram**.



Add a legend

Choose **File > Print > Page Setup**.

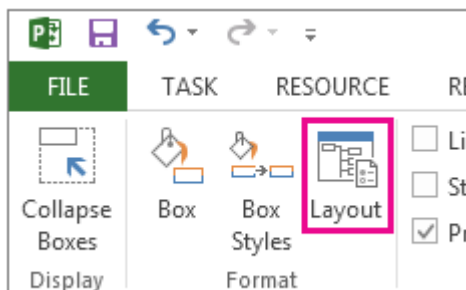
On the Legend tab, decide how you want your legend to look, which pages it should show up on, and then labels you want.

Choose OK.

Automatically change the way the boxes are laid out

Choose **View > Network Diagram**.

Choose **Format > Layout**.

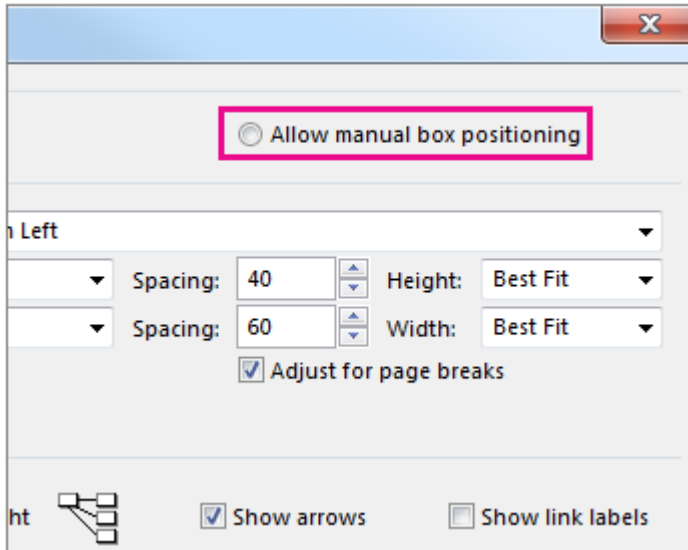


Under Box Layout, choose the box arrangement, alignment, spacing, height, and width that work best for you. To space boxes evenly, select Fixed in the Height and Width boxes.

Keep in mind that grouped tasks are positioned automatically. You'll need to undo grouping if you want to change them.

Manually change the way boxes are laid out

If you've gotten this far and still don't like how your boxes are positioned, click **Format > Layout**, select **Allow manual box positioning**, choose **OK**, and then drag the boxes to the spot you want.



If you manually reposition a task, you can change the layout of any linked tasks or subtasks associated with it by right-clicking on the task and choosing **Layout Related Tasks Now**.

Change the line style between boxes

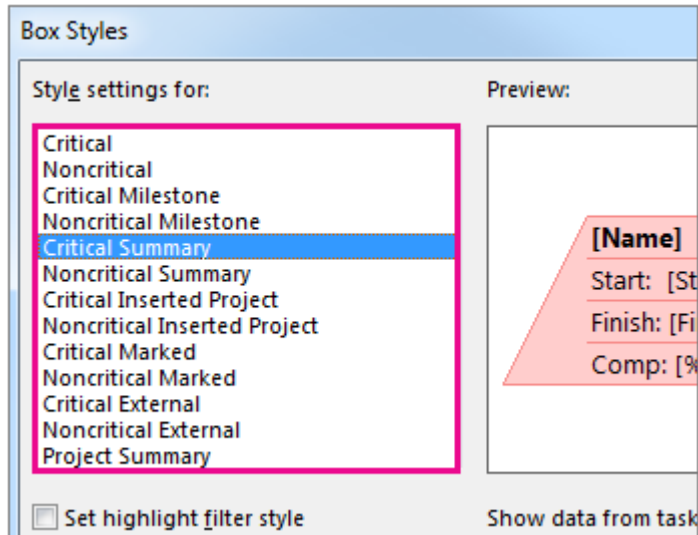
If you have a lot of tasks that you've linked to predecessor or successor tasks, the links between boxes can be really hard to follow. Try changing the line style, and then arranging them in way that's easier to see.

- Choose View > Network Diagram.
- Choose Format > Layout.
- Under **Link style**, select **Rectilinear** or **Straight**. Rectilinear links look like this  , and straight links look like this .
- Select Show arrows to add arrows that point to predecessor and successor tasks. Select Show link labels to add dependency and lead or lag time to the link line.

Choose what kind of task information to show

- If things are looking cluttered (or you start to experience information overload), try changing the task information in each box so you only see what's most important.
- Choose View > Network Diagram.
- Choose Format > Box Styles.

In the Style settings for list, select the task that you want to change.



- Under **Border**, choose the shape, color, width, and gridline options to create the look you want.
- Select a name under **Data template** to apply your changes to an existing template. To create a new template that will use your changes, choose **More Templates**, and then choose **New** (to create a new template), **Copy** (to base the new template on an existing one), **Edit** (to change a template), or **Import** (to import a template from another project).
- Choose **OK**.

Lab Task

The SDLC has phases:

- Initiation
- Planning and requirement gathering
- Designing
- Development
- Testing
- Deployment

Create a network diagram of these phases showing properly:

1. milestones,
2. predecessors,
3. summary tasks and
4. resources.



Lab 28 & 29

Introduction to Asana

Introduction: Getting started with asana

Problem Statement:

- Getting started with asana
- Creating profile, adding team members
- Starting a new project
- Assigning tasks, subtasks, deadlines
- Adding attachments



What is Asana?

Asana is an Online Project Management App where you can Organize:

- Projects
- Tasks
- Team members

With Asana you can set:

- Project milestones
- Assign tasks to teams
- See current status
- Assign tasks with due dates
- See current projects
- And real time update when any changes are made

It makes it easier for people to work together as a **team**.

Asana works with PC, iOS, and Android



Asana for iOS, iPad, and Android

Made by Asana

[MORE INTEGRATIONS](#)

Download Asana on your iPhone, iPad, or Android device to plan your day, share ideas, and get team updates on the go.

Get Asana for your iOS or Android device

iPhone and iPad

1. Download our Asana app from [the App Store](#)
2. For help getting up to speed, read our [iOS Quick Start Guide](#)

Android

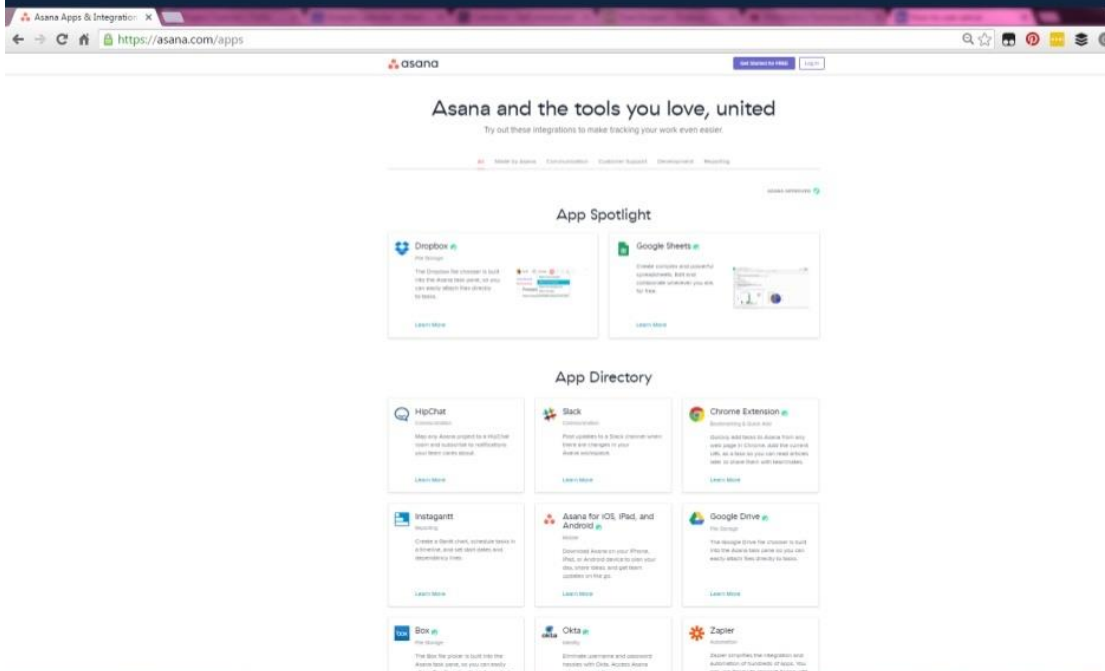
1. Download our Asana app from [the Google Play Store](#)
2. For help getting up to speed, read our [Android Quick Start Guide](#)



Feedback & Support

[Asana Support](#)

and integrates with other Apps too



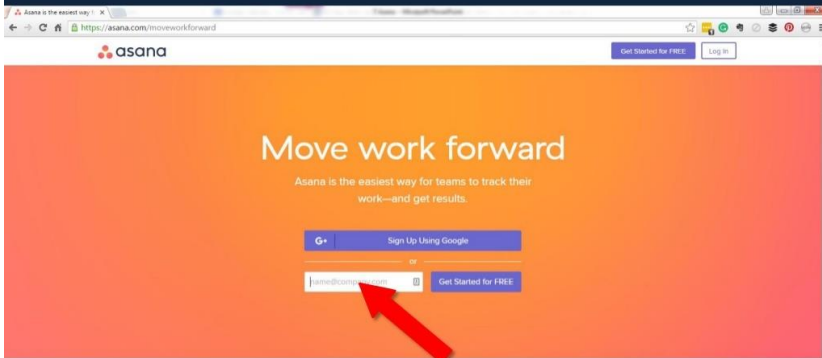
The screenshot shows the Asana website's 'Apps & Integration' page. At the top, a dark blue banner reads 'and integrates with other Apps too'. Below this, the page title is 'Asana and the tools you love, united' with the subtitle 'Try out these integrations to make tracking your work even easier'. The page is divided into two main sections: 'App Spotlight' and 'App Directory'. The 'App Spotlight' section features two featured integrations: 'Dropbox' and 'Google Sheets'. The 'App Directory' section is a grid of various other integrations including 'HipChat', 'Slack', 'Chrome Extension', 'Instagram', 'Asana for iOS, iPad, and Android', 'Google Drive', 'Box', 'Clio', and 'Zapier'. Each integration card includes a logo, a brief description, and a 'Learn More' link.

How to create an account on asana?

Type “asana.com” and press “Enter”



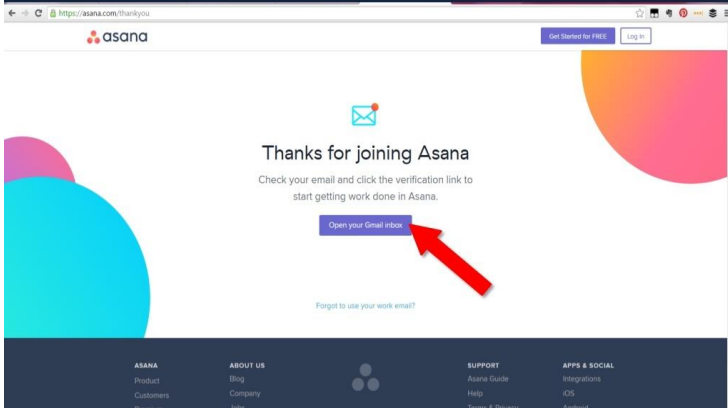
Type your email here



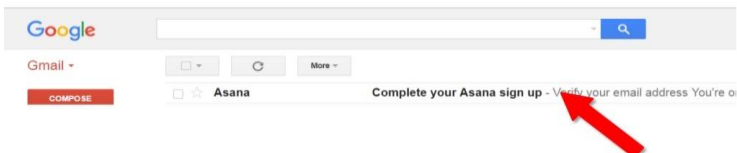
And click “get started for free”



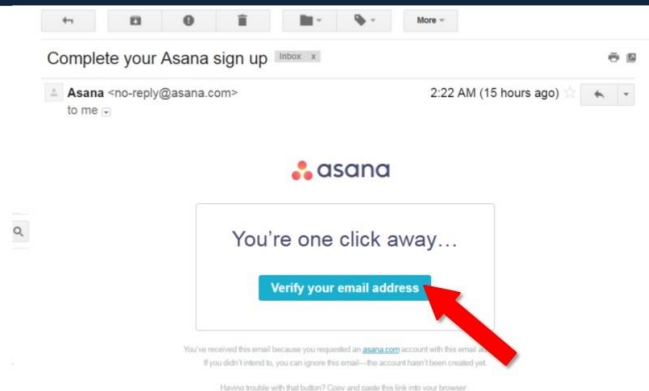
Click this to go to your email inbox



Click this to open email from Asana

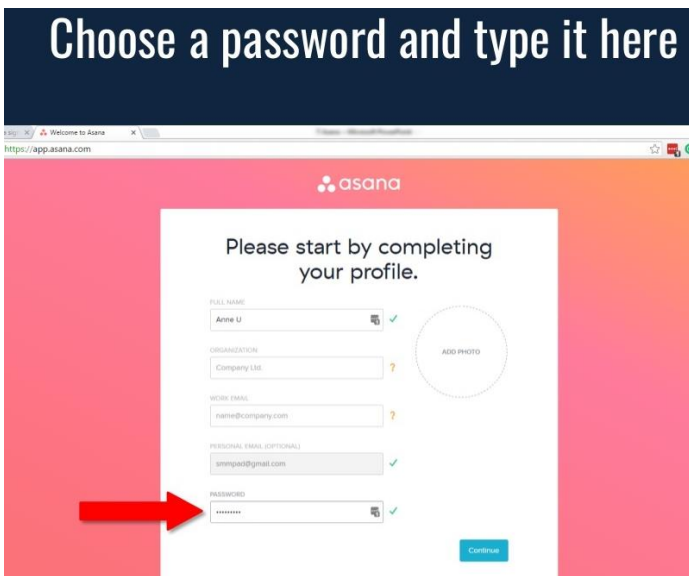


Click this to verify your email address

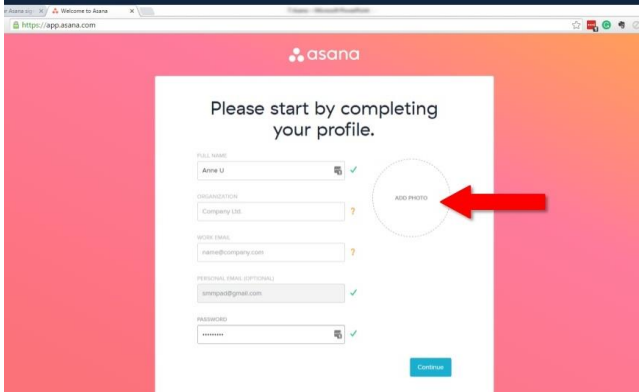




Type your name, company's name and work email.



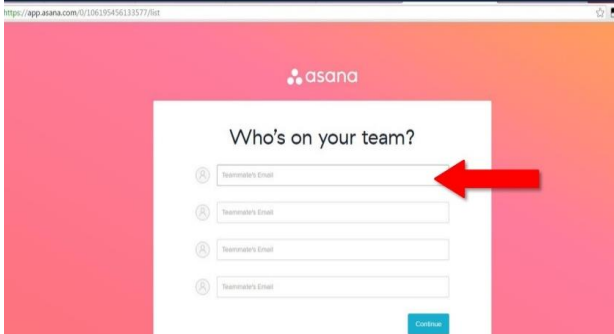
Click this if you want to add your photo
(or just skip this)



The screenshot shows the Asana profile completion page. The title is "Please start by completing your profile." There are several input fields: "FULL NAME" (filled with "Anne U"), "ORGANIZATION" (filled with "Company Ltd"), "WORK EMAIL" (filled with "name@company.com"), "PERSONAL EMAIL (OPTIONAL)" (filled with "annepaul@gmail.com"), and "PASSWORD" (filled with "password"). To the right of the "FULL NAME" field is a circular "ADD PHOTO" button, which is highlighted by a red arrow. A "Continue" button is at the bottom right.

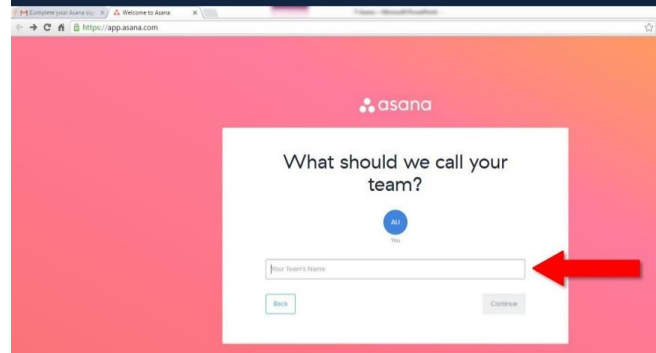
And click “continue”

Type your teammate's email here
(or skip this and add them later)



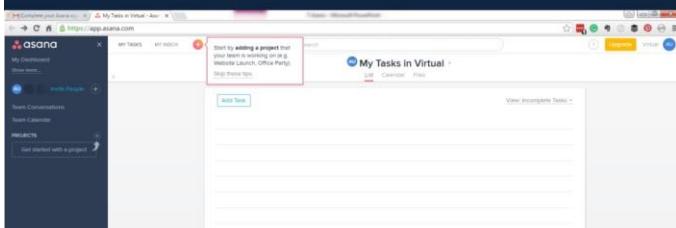
The screenshot shows the "Who's on your team?" screen. It has four input fields, each with a placeholder "Teammate's Email". A red arrow points to the first input field. A "Continue" button is at the bottom right.

Type your Team's name



The screenshot shows the "What should we call your team?" screen. It has a single input field with a placeholder "Your Team's Name". A red arrow points to this input field. There are "Back" and "Continue" buttons at the bottom.

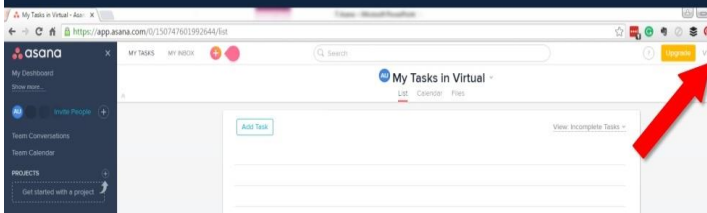
Welcome to your Asana account



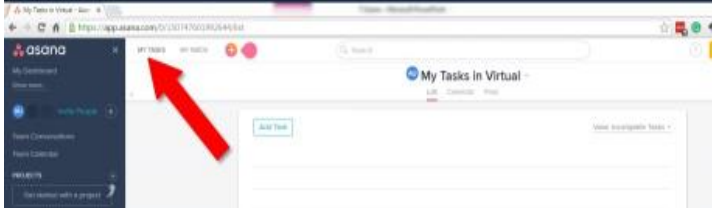
The screenshot shows the "Welcome to your Asana account" screen. It has a sidebar on the left with "My Dashboard", "My Tasks", and "My Projects". The main area has a "Start by adding a project" button and a "My Tasks in Virtual" section. A red box highlights the "Start by adding a project" button.

Quick tour to your Asana work space

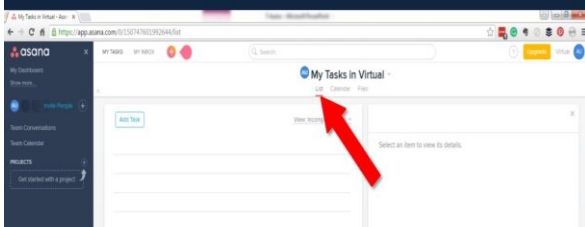
On the top-right corner is your Team's name



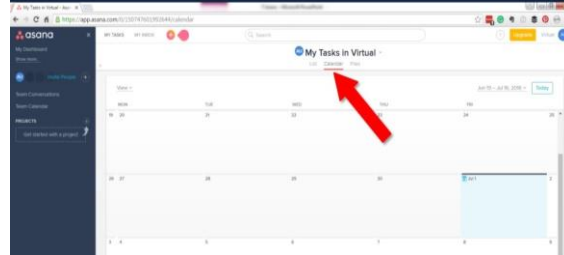
You are currently viewing "My Tasks"



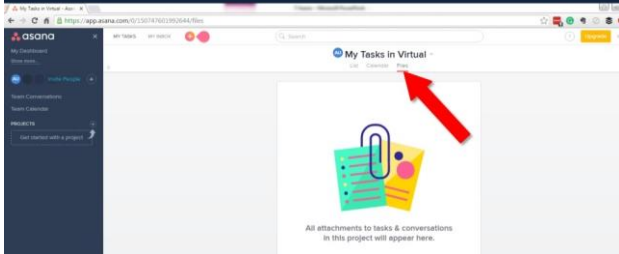
Click "List" to go to List view



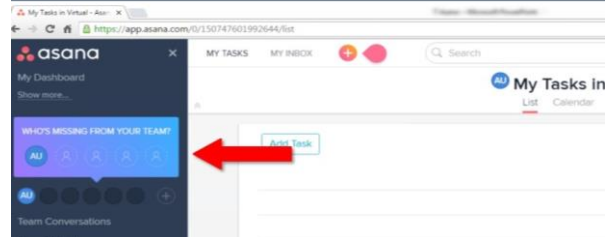
Click "Calendar" to go to Calendar view



Click "Files" to see your file attachments

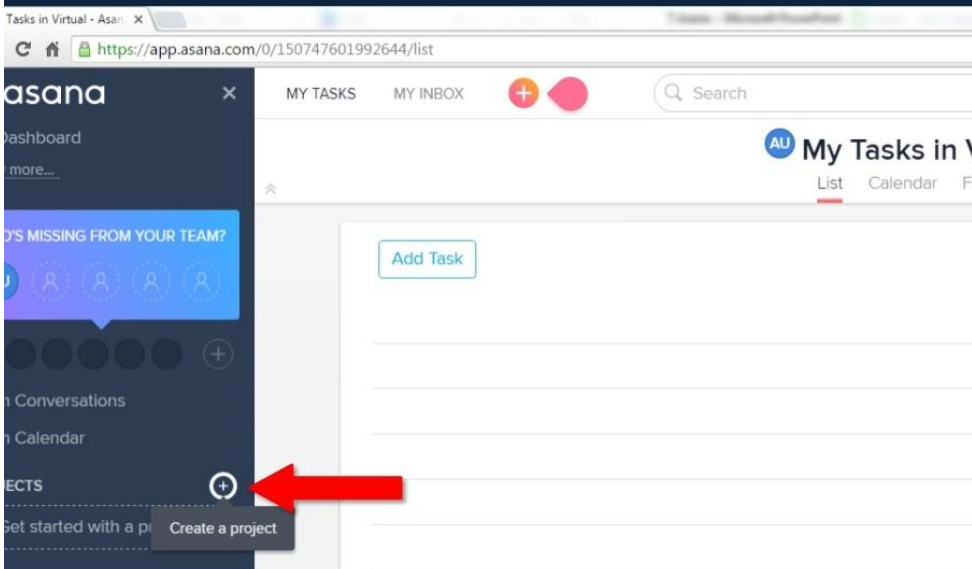


where you can click and see your Team Members



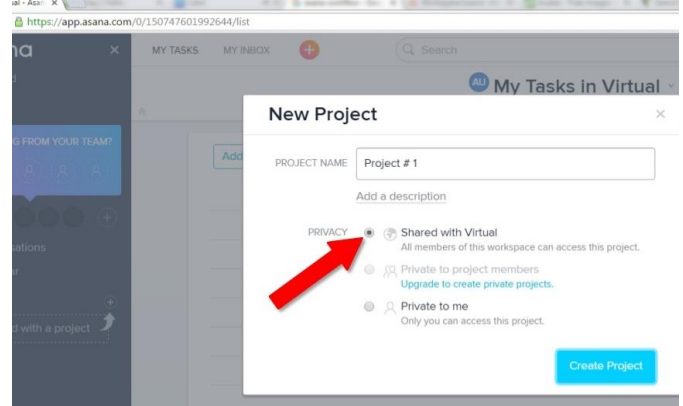
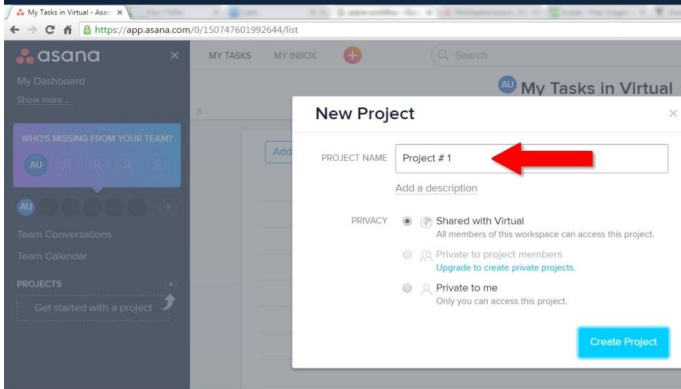
Asana Projects

Click this to create a project

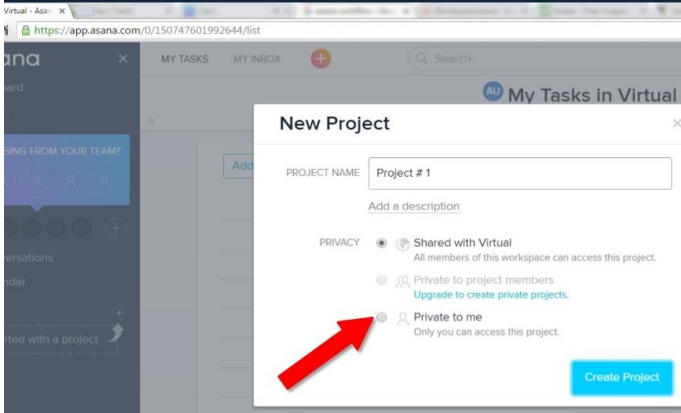


Type the name of your project

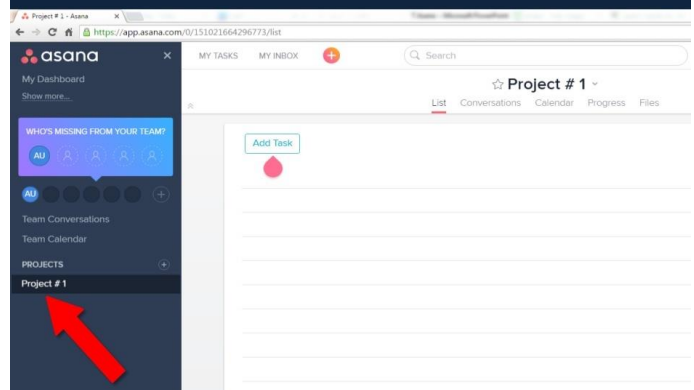
Click this if it's a Team Project



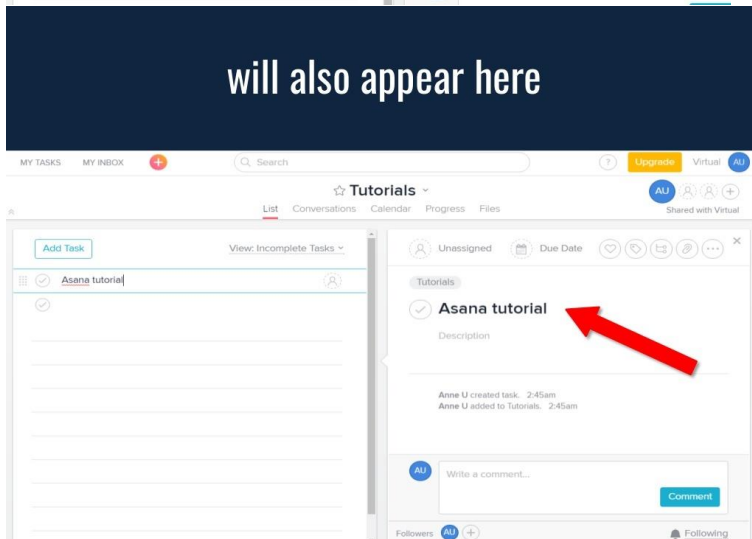
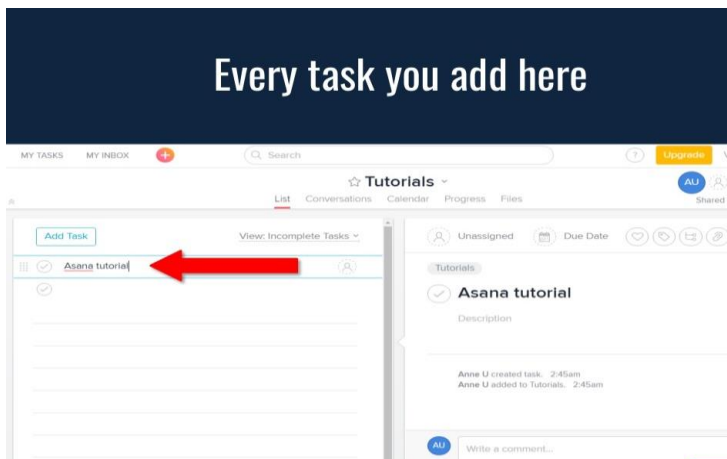
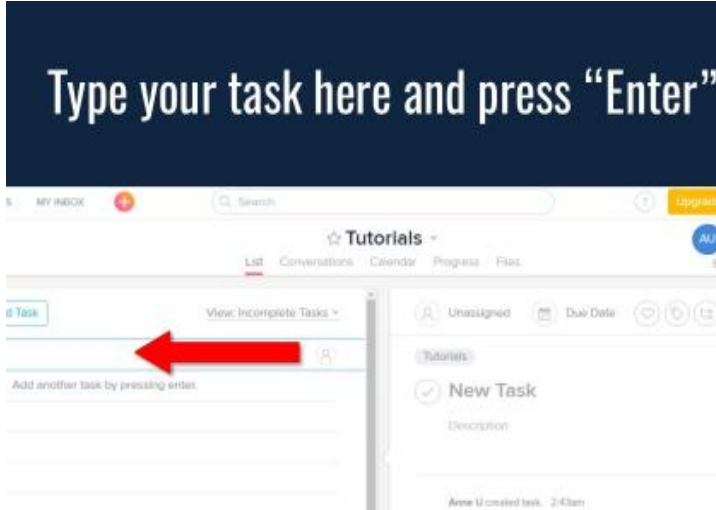
Or this if it's a private project



Here is your new project under the Dashboard

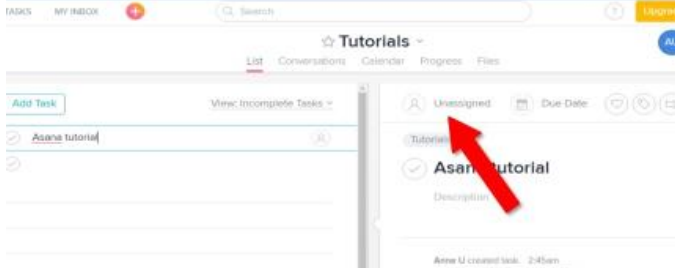


Add tasks in Asana

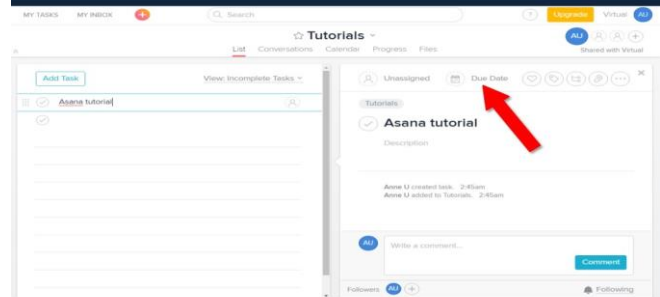




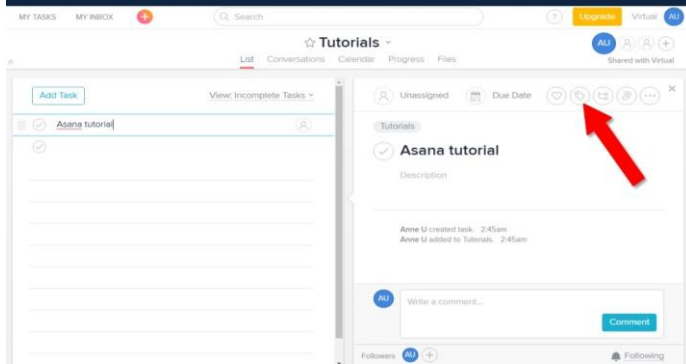
Click this to assign task to a team member or to yourself



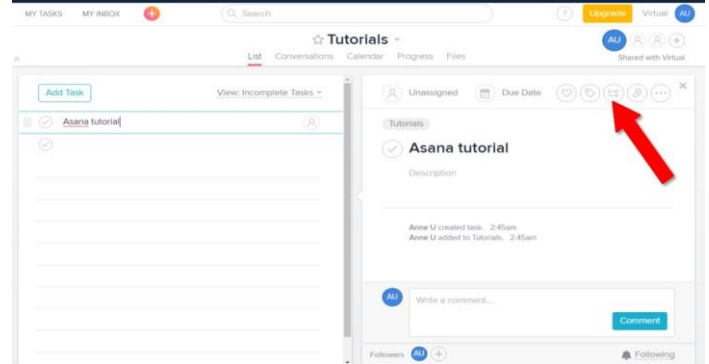
Click this to assign due date for the task



Click this to create "Tags" for the task

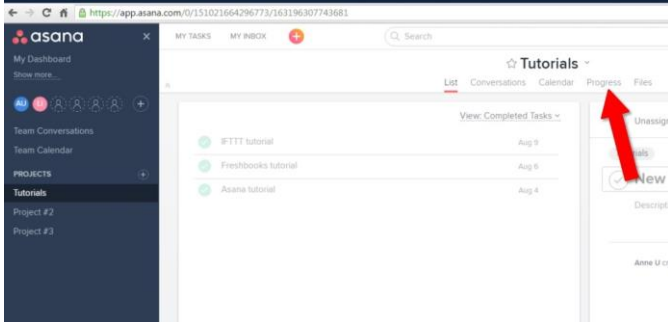


Click this to add subtasks

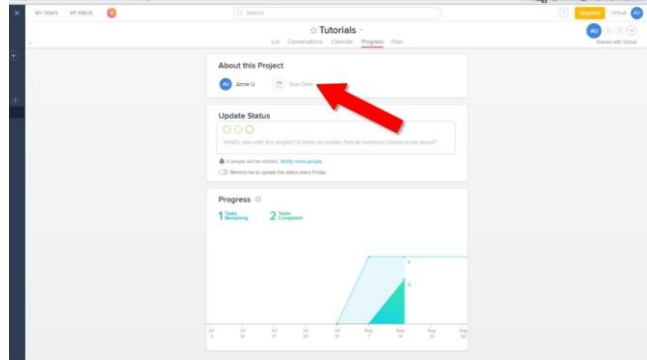


See Your Project Progress

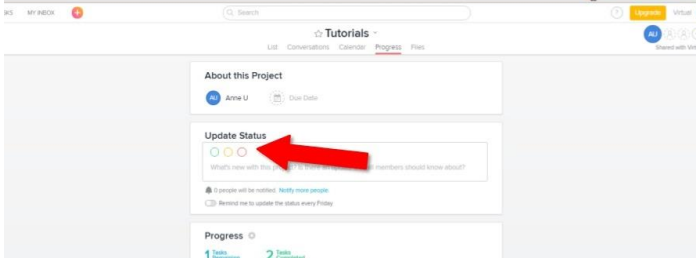
Click "Progress"



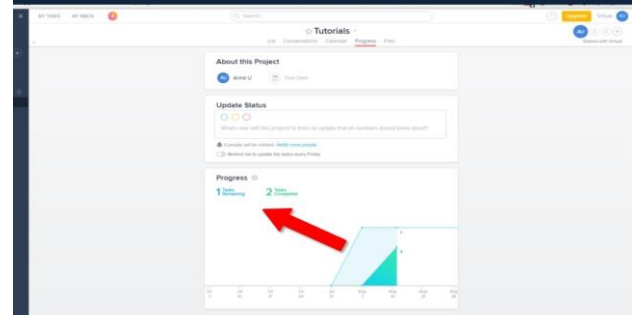
You can add a due date



Click on 1 of these circles to choose the current status of the project

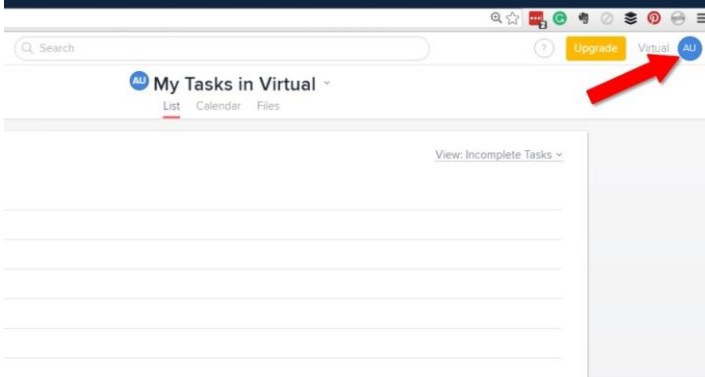


And see the complete and incomplete tasks

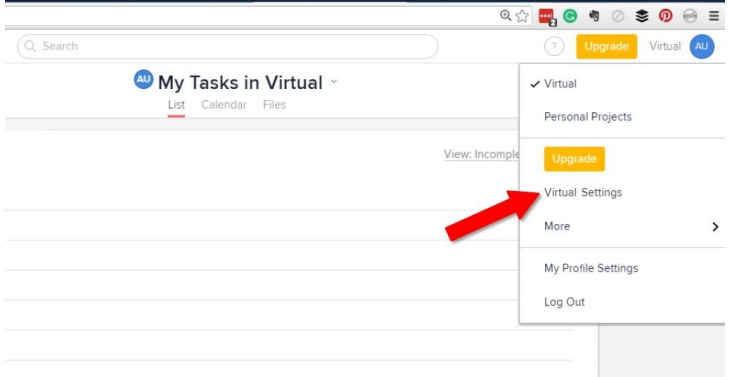




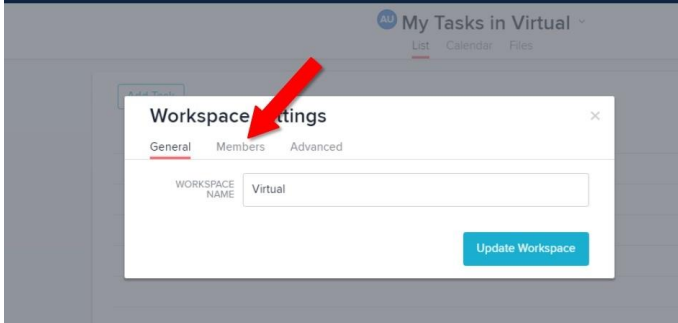
On the top-left corner, click your initials



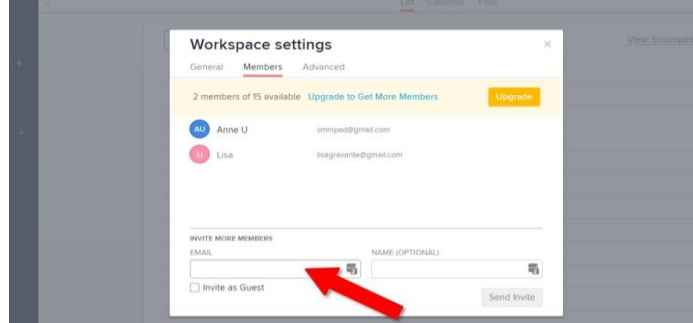
Click "Virtual Settings"



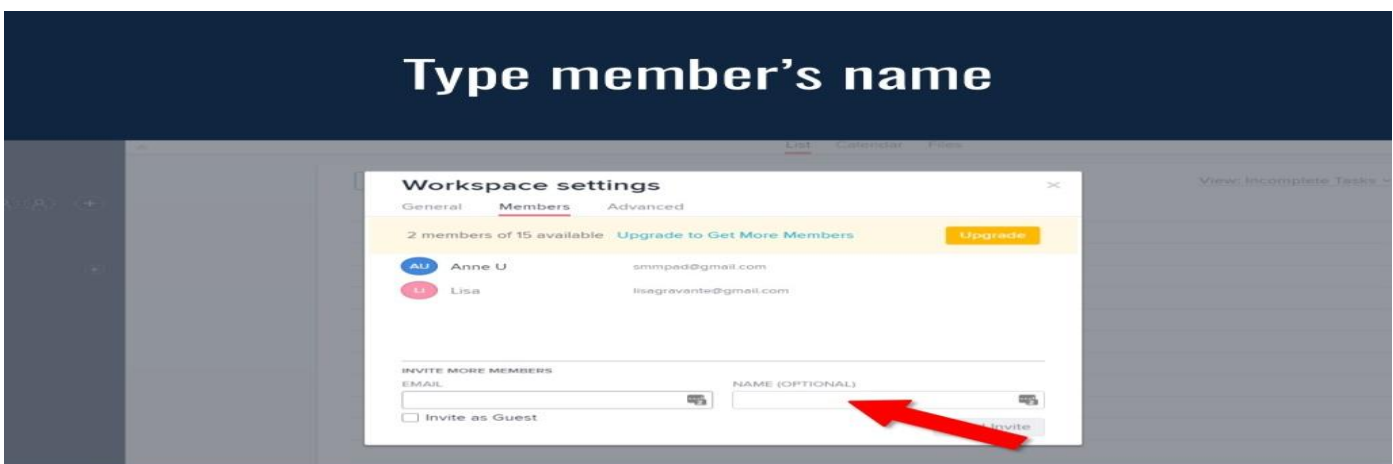
Click "Members"



To add another member, type email here



Type member's name



And send invite



